

Radiation Gauge Reconstruction Notes

Peter Zimmerman*

* zimmerator@protonmail.com

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I. INTRODUCTION

A. Corrector Tensor

The corrector tensor $x_{\mu\nu}$ is determined by the source T_{ab} through a hierarchy of equations descending from the l^a -sector (IRG variant) of Einstein's equations $l^a(\mathcal{E}_{ab}[x] = T_{ab})$. In GHP form, these equations read [1]

$$\rho^2 \mathbf{P} \left(\frac{\bar{\rho}}{\rho^3} \mathbf{P} \left[\frac{\rho}{\bar{\rho}} x_{m\bar{n}} \right] \right) = T_{l\bar{l}} \quad (\text{I.1})$$

for $x_{m\bar{n}}$,

$$\begin{aligned} & \frac{\rho}{2(\rho + \bar{\rho})} \mathbf{P} \left((\rho + \bar{\rho})^2 \mathbf{P} \frac{x_{nm}}{\rho(\rho + \bar{\rho})} \right) \\ &= T_{lm} - \frac{1}{2} \{ (\mathbf{P} + \rho - \bar{\rho})(\delta + \bar{\tau}' - \tau) + 2\bar{\tau}'(\mathbf{P} - 2\rho) - (\delta - \tau - \bar{\tau}')\bar{\rho} + 2\rho\tau \} x_{m\bar{n}}. \end{aligned} \quad (\text{I.2})$$

for x_{nm} , and

$$\begin{aligned} & \frac{1}{2}(\rho + \bar{\rho})^2 \mathbf{P} \left(\frac{1}{\rho + \bar{\rho}} x_{nn} \right) \\ &= T_{ln} - \text{Re}\{ \delta' \delta + (\bar{\tau}' - \tau) \delta' + [\delta(\bar{\tau} + \tau')] - 2\tau' \tau - 2\Psi_2 - 2\bar{\rho}\rho' + 2(\rho' \mathbf{P} + \rho \mathbf{P}') - \mathbf{P}' \mathbf{P} \} x_{m\bar{n}} \\ & \quad - \text{Re}\{ [(\mathbf{P} - 2\rho)(\delta' - \bar{\tau}) + (\tau' + \bar{\tau})(\mathbf{P} + \bar{\rho}) - 2(\delta' - \tau')\rho - 2\bar{\tau} \mathbf{P}] x_{nm} \} \end{aligned} \quad (\text{I.3})$$

for x_{nn} .

In this note, we give explicit solutions to equations (I.1), (I.2), and (I.3) for a point particle (IV.1) in terms of readily computable coefficients written in terms of the source.

We work in mostly outgoing Kerr-Newman coordinates wherein equations (I.1), (I.2), and (I.3) reduce to a sequence of three ODEs along the outgoing null-generators $l^\alpha = \partial_r$ of increasing r the affine parameter labeling cuts of $\mathcal{I}^+$, or in BL coordinates. For initial data, we assume trivial conditions on $x_{\alpha\beta}$ at $\mathcal{H}^-$, consistent with a bound orbit supported away from the horizon for all time.

Throughout the note, we adopt the mostly minus metric “west coast” signature $(+, -, -, -)$ and geometrized units $G = c = 1$. Greek letters denote spacetime indices, lowercase Latin letters spatial indices, capital Latin letters denote quantities on the $\mathbb{S}^2$.

To avoid tracking factors of π , we adopt the convention wherein the perturbed EFE reads

$$\mathcal{E}_{\mu\nu} = T_{\mu\nu}. \quad (\text{I.4})$$

Unless stated otherwise, all other conventions and notation will be chosen to match GHZ.

II. PRELIMINARIES

III. CALCULATION OF $x_{\mu\nu}$

A. General solution in the source-free region

The solution to (I.1) in the vacuum region $r > r_{\max}$ is given by

$$x_{m\bar{m}}^> = a^\circ \left(\frac{\bar{\rho}}{\rho} + \frac{\rho}{\bar{\rho}} \right) + b^\circ(\rho + \bar{\rho}), \quad r > r_{\max}, \quad (\text{III.1})$$

where $\rho/\bar{\rho} + \bar{\rho}/\rho$ and $\rho + \bar{\rho}$ are homogeneous solutions. The integration constants a° and b° are determined by the source and will be given in a later section.

We next turn to the nm -component, as determined by (I.2). To integrate this equation, we write it first in Held form

$$\frac{\rho}{2(\rho + \bar{\rho})} \mathbb{P} \left((\rho + \bar{\rho})^2 \mathbb{P} \frac{x_{nm}}{\rho(\rho + \bar{\rho})} \right) = T_{lm} - \frac{1}{2} \left\{ \bar{\rho}(\mathbb{P} + \rho - \bar{\rho})\tilde{\mathfrak{d}} - \bar{\rho}(3\bar{\rho} + \rho)\tau^\circ \mathbb{P} + 4\rho\bar{\rho}^2\tau^\circ \right\} x_{m\bar{m}}, \quad (\text{III.2})$$

wherein we have used Eqs. (C.7), (C.8), and (C.9). Integrating the result give

$$x_{nm}^> = \bar{\rho}\Omega^\circ\tilde{\mathfrak{d}}a^\circ + \frac{(\rho + \bar{\rho})^2}{\rho}\tau^\circ a^\circ + \bar{\rho}\tilde{\mathfrak{d}}b^\circ + \bar{\rho}(\rho + \bar{\rho})\tau^\circ b^\circ + \frac{1 - \Omega^{\circ 2}\bar{\rho}^2}{\bar{\rho}}c^\circ + \rho(\rho + \bar{\rho})e^\circ, \quad r > r_{\max}. \quad (\text{III.3})$$

We begin the calculation of x_{nm} by recasting Eq. (I.3) in Held form

$$\frac{1}{2}(\rho + \bar{\rho})^2 \mathbb{P} \left(\frac{x_{nn}}{\rho + \bar{\rho}} \right) = T_{ln} + \text{Re}\mathcal{U}x_{m\bar{m}} + \text{Re}[\mathcal{V}x_{nm}], \quad (\text{III.4})$$

where

$$\begin{aligned} \mathcal{U} = & \rho\bar{\rho} \left[-\tilde{\mathfrak{d}}'\tilde{\mathfrak{d}} - \bar{\tau}^\circ(\rho + \bar{\rho})\tilde{\mathfrak{d}} + \bar{\tau}^\circ\tilde{\mathfrak{d}}\mathbb{P} + \tau^\circ(\bar{\rho} - \rho)\tilde{\mathfrak{d}}' + \tau^\circ\tilde{\mathfrak{d}}'\mathbb{P} \right] + \tilde{\mathbb{P}}'\mathbb{P} \\ & + \frac{1}{2} \left[\Psi^\circ(\rho^2 - \bar{\rho}^2) - 2\rho\bar{\rho}(\bar{\rho} - \rho)\tau^\circ\bar{\tau}^\circ - 4\bar{\rho}\rho'^\circ(1 + \rho\Omega^\circ) \right] \mathbb{P} - 2\rho\tilde{\mathbb{P}}' \\ & + \rho^2\Psi^\circ(2\rho - \bar{\rho}) + 2\bar{\rho}^2(\rho'^\circ - \rho(2\rho - \bar{\rho})\tau^\circ\bar{\tau}^\circ) - \rho\bar{\rho}(\rho - 3\bar{\rho})\rho'^\circ\Omega^\circ, \end{aligned} \quad (\text{III.5})$$

and

$$\mathcal{V} = -\rho\{-3\rho\tilde{\mathfrak{d}}' + [\tilde{\mathfrak{d}}' - \bar{\tau}^\circ(2\bar{\rho} + \rho)]\mathbb{P} + 2\rho^2\bar{\tau}^\circ\} \quad (\text{III.6})$$

The solution is given by

$$\begin{aligned}
x_{nn}^> = & 2 \operatorname{Re} \left\{ \frac{1}{2} \frac{\rho + \bar{\rho}}{\rho^2} \tilde{\mathbf{P}}' a^\circ + (1 + \rho \bar{\rho} \Omega^\circ) \tilde{\delta}' \tilde{\delta} a^\circ + 2 (\rho^2 + 2\rho \bar{\rho}) \tau^\circ \bar{\tau}^\circ a^\circ \right. \\
& + \left(\frac{\rho^2}{\bar{\rho}} + \frac{3}{2} \rho \bar{\rho} \Omega^\circ \right) \Psi^\circ a^\circ - 2\rho' a^\circ + \tilde{\mathbf{P}}' b^\circ + 2\rho \bar{\rho} \tau^\circ \tilde{\delta} b^\circ + 2\rho^2 \bar{\rho} \tau^\circ \bar{\tau}^\circ b^\circ + \rho^2 \Psi^\circ b^\circ \\
& \left. + \rho^2 \tilde{\delta}' e^\circ + 2\rho^2 \bar{\rho} \bar{\tau}^\circ e^\circ - \frac{\bar{\rho}}{\rho} \left(4\Omega^\circ + \frac{1}{\rho} \right) \tilde{\delta}' c^\circ + 4 \frac{\bar{\rho}}{\rho} \bar{\tau}^\circ c^\circ + (\rho + \bar{\rho}) f^\circ \right\} \quad (\text{III.7})
\end{aligned}$$

IV. POINT-PARTICLE SOURCE

Let us now take the perturbation to be sourced by a particle of mass μ on a worldline γ parametrized by proper time here denoted τ . We further denote the position and four velocity of the particle by $x_p^\mu(\tau)$ and $v^\mu = dx_p^\mu/d\tau$, respectively. It carries the point particle stress energy distribution

$$T_{\mu\nu} = \mu \int_\gamma v_\mu v_\nu \delta_4(x, x_p(\lambda)) d\lambda. \quad (\text{IV.1})$$

We consider a bound orbit with radii between $r_{\min} < r_p < r_{\max}$ and denote the coordinate location of the particle in retarded coordinates as $x_p^\mu = (u_p, r_p, z_p, \varphi_p)$.

A. Held Decomposition of the source

With retarded KN time $-\infty < u_p < \infty$ taken as the time parameter, we now rewrite the source as

$$T_{\mu\nu} = \int_{-\infty}^{\infty} \left[\mu \frac{v_\mu v_\nu}{\Sigma_p v^u} \delta_p^2(\Omega) \right] \delta_p(u) \delta_p(r) du_p, \quad (\text{IV.2})$$

$$= T_{\mu\nu}^\circ(u, z, \varphi; r_p) \delta_p(r), \quad (\text{IV.3})$$

$$= J_{\mu\nu}^\circ(u; r_p, z_p, \varphi_p) \delta_p^2(\Omega) \delta_p(r), \quad (\text{IV.4})$$

where in general $x^p(u)$ after doing the u_p integration, and where (in the outgoing Kerr chart)

$$T_{\mu\nu}^\circ := \mu \frac{v_\mu v_\nu}{\Sigma_p \gamma_p} \delta_p^2(\Omega), \quad J_{\mu\nu}^\circ := \mu \frac{\gamma_p x'_\mu x'_\nu}{\Sigma_p}, \quad (\text{IV.5})$$

with the coordinate four-velocity

$$x'_\mu := g_{\mu\nu} \frac{dx^\nu}{d\tau} \frac{d\tau}{du} = \frac{v_\mu}{\gamma_p} \quad (\text{IV.6})$$

and $\delta_p^2(\Omega) = \delta(z - z_p)\delta(\varphi - \varphi_p)$ and where $v^\alpha \partial_\alpha u = \gamma_p$. Here, $J_{\mu\nu}^\circ$ and $T_{\mu\nu}^\circ$ are taken at $r_p(u)$, $\Sigma_p = r_p^2 + a^2 z_p^2$ and it is understood that $r_p = r_p(\tau(u_p))$ etc. We also introduce the quantities T_μ° and J_μ° by

$$T_{\mu\nu} l^\nu = T_\mu^\circ(u, z, \varphi) \delta_p(r) = J_\mu^\circ(u; z_p, \varphi_p) \delta_p(r) \delta_p^2(\Omega) \quad (\text{IV.7})$$

B. Action angle representation for general orbits

Bound geodesic motion in Kerr spacetime¹ is completely integrable and therefore admits a natural description in terms of action–angle variables. In this description the orbits wrap invariant tori labeled by a set of conserved quantities

$$\mathcal{P} = \{E, L_z, Q\} \leftrightarrow \{p, e, \iota\} \quad (\text{IV.10})$$

associated to the Killing symmetries $E = -p_\mu t^\mu$, $L_z = p_\mu \phi^\mu$, $Q = p_\mu p_\nu Q^{\mu\nu}$.

We follow [2] to decompose the source into modes conjugate to the natural averaged Mino and Boyer-Lindquist frequencies of the orbit,

$$\Upsilon_\alpha = \left\langle \frac{d\psi_\alpha}{d\lambda} \right\rangle_\lambda, \quad \Omega_\alpha = \frac{\Upsilon_\alpha}{\Upsilon_t}. \quad (\text{IV.11})$$

respectively.

$$\omega_{m\mathbf{k}} = k_z \Omega_z + k_r \Omega_r + m \Omega_\phi \quad (\text{IV.12})$$

with associated angles

$$\vartheta_\alpha = \Omega_\alpha t + \vartheta_\alpha^0, \quad (\text{IV.13})$$

where ϑ_α^0 are chosen to be zero except for $\vartheta_z^0 = -\Omega_z t_z^0$ where t_z^0 is the first value of t at which $z = z_{\max}$.

The instantaneous phases ψ_α are given in terms of ϑ_α

$$\psi_\alpha(\vartheta_\beta) = \vartheta_\alpha + \Delta_{\vartheta} \psi_\alpha(\boldsymbol{\vartheta}) - \Delta_{\vartheta} \psi_\alpha(0) \quad (\text{IV.14})$$

¹ To keep the formulas consistent with [2], in this section we use λ as Mino time and define

$$\dot{x}_p^\mu \equiv \frac{dx_p^\mu}{d\lambda}, \quad d\tau = \Sigma d\lambda, \quad \Sigma = r^2 + a^2 z^2, \quad v^\mu = \frac{dx_p^\mu}{d\tau} = \frac{\dot{x}_p^\mu}{\Sigma}. \quad (\text{IV.8})$$

Thus

$$\dot{t} = \Sigma v^t, \quad \dot{\phi} = \Sigma v^\phi, \quad \dot{r} = \Sigma v^r. \quad (\text{IV.9})$$

where the oscillatory piece is given by the Fourier sum

$$\Delta_{\vartheta}\psi_{\alpha}(\boldsymbol{\vartheta}) = \sum_{\mathbf{k}} \frac{(\mathbf{f}_{\alpha}/\mathbf{f}_t)_{\mathbf{k}}}{-i\Omega_{\alpha,\mathbf{k}}} e^{-i\vartheta_{\mathbf{k}}} \quad (\text{IV.15})$$

where

$$\mathbf{f}_{\alpha} = \frac{d\psi_{\alpha}}{d\lambda} \quad (\text{IV.16})$$

are the instantaneous frequencies associated to the Keplerian phases ψ_r and ψ_z defined implicitly in terms of BL coordinates by

$$r = \frac{pM}{1 + e \cos \psi_r}, \quad z = z_{\max} \cos \psi_z \quad (\text{IV.17})$$

as given in Eqs. (216) and (217) of [2]. They have the Fourier decomposition given in Eq. (263) of [2]. Here we've used the vector notation $\boldsymbol{\vartheta} = (\vartheta_r, \vartheta_z)$, $\mathbf{k} = (k_r, k_z)$ etc and

$$\vartheta_{\mathbf{k}} = \vartheta_z k_z + \vartheta_r k_r \quad (\text{IV.18})$$

The discrete harmonics of the stress tensor is now presented.

$$T_{\mu\nu} = \sum_{\mathbf{k}, m} T_{\mu\nu}^{(m, \mathbf{k})}(\mathcal{P}; r, z) e^{im\phi - i\vartheta_{m\mathbf{k}}} \quad (\text{IV.19})$$

where

$$\vartheta_{m\mathbf{k}} = m\vartheta_{\phi} + k_r\vartheta_r + k_z\vartheta_z \quad (\text{IV.20})$$

and where

$$\frac{d\vartheta_{m\mathbf{k}}}{dt} = m\Omega_{\phi} + k_z\Omega_z + k_r\Omega_r + \varepsilon [\text{Flux terms}] + \dots, \quad (\text{IV.21})$$

$$T_{\mu\nu}^{(m, \mathbf{k})} = \frac{1}{(2\pi)^3 \Sigma^2} \frac{\Upsilon_r \Upsilon_z}{\Upsilon_t} \left[\Theta(r - r_p) - \Theta(r - r_a) \right] \Theta(z_{\max} - |z|) \sum_{\sigma_r = \pm, \sigma_z = \pm} \frac{\dot{x}_{\mu}(\boldsymbol{\psi}^{\sigma}) \dot{x}_{\nu}(\boldsymbol{\psi}^{\sigma})}{\dot{r}(\boldsymbol{\psi}_r^{\sigma_r}) \dot{z}(\boldsymbol{\psi}_z^{\sigma_z})} e^{i\vartheta_{\mathbf{k}}(\boldsymbol{\psi}^{\sigma}) - im\Delta_{\vartheta}\phi(\boldsymbol{\psi}^{\sigma})} \quad (\text{IV.22})$$

where $\dot{x}^{\mu} = g_{\mu\nu}(\mathbf{x})\dot{x}^{\nu}$, $\dot{x}^t = \mathbf{f}_t(\boldsymbol{\psi}, \mathcal{P}^i)$, $\dot{x}^{\phi} = \mathbf{f}_{\phi}(\boldsymbol{\psi}, \mathcal{P}^i)$ are the instantaneous frequencies

$$\mathbf{f}^t = \Sigma \frac{dt}{d\tau}, \quad \mathbf{f}^{\phi} = \Sigma \frac{d\phi}{d\tau}, \quad (\text{IV.23})$$

$\dot{x}_t = -E_{\vartheta}(p, e, \iota)\Sigma$, $\dot{x}_{\phi} = L_{\vartheta}(p, e, \iota)\Sigma$, and $\mathcal{P}(p^i; a, M) = \{p, e, \iota; a, M\}$ are system parameters.

The BL mean phases $\vartheta_{\mathbf{k}}$ are related to the mean Mino phases

$$q_a(\psi_a) = \Upsilon_a \int^{\psi_a} \frac{d\psi'_a}{\mathbf{f}_a(\psi'_a)}, \quad a = r, z. \quad (\text{IV.24})$$

by

$$\vartheta_{\mathbf{k}}(\boldsymbol{\psi}) = q_{\mathbf{k}}(\boldsymbol{\psi}) + \Omega_{\mathbf{k}} (\delta t_r(\psi_r) + \delta t_z(\psi_z)), \quad (\text{IV.25})$$

where $\delta t_a(\boldsymbol{\psi}) = \Delta t_a[q_a(\boldsymbol{\psi})] - \Delta t_a(0) = \int^{\psi_a} d\psi'_a (T_a(\psi'_a) - \langle T_a \rangle_\lambda) / \mathfrak{f}_a(\psi'_a)$. Here we have used $T_r = (r^2 + a^2)P(r)/\Delta$ and $T_z = -a^2 E(1 - z^2)$.

Here we take the choice of origin with Eq. (264) of [2].

V. CCK-ORI POINT PARTICLE Φ^S

Following Ori [3] and the presentation in [4],² (Ori Eq. above 26) we solve the inversion relation

$$\mathbf{P}^4 \bar{\Phi} = -4\psi_0 \quad (\text{V.1})$$

via for mode decomposed $\{0, -4\}$ scalar $\bar{\Phi}$ in the Boyer-Lindquist chart

$$t = u + r_*, \quad \phi = \varphi - \phi^\sharp \quad (\text{V.2})$$

where

$$r_*(r) = \int \frac{r^2 + a^2}{\Delta} dr, \quad \phi^\sharp(r) = a \int \frac{dr}{\Delta}. \quad (\text{V.3})$$

This entails using a phase

$$H_{\text{outgoing}}^{\ell m \omega} = H_{\text{BL}}^{\ell m \omega} \exp \left\{ i\omega(r_* - r_{*p}) + im(\phi^\sharp - \phi_p^\sharp) \right\} \quad (\text{V.4})$$

relating Ori's BL radial kernel $H^{\ell m \omega}(r, r')$ to the IRG (conjugated) Hertz potential in the outgoing Kerr chart. The Kinnersley tetrad in these coordinates takes the form

$$\begin{aligned} l^\alpha \partial_\alpha &= \frac{1}{\Delta} (\partial_t + \Delta \partial_r + a \partial_\phi), \\ n^\alpha \partial_\alpha &= \frac{1}{2\Sigma} ((r^2 + a^2) \partial_t - \Delta \partial_r + a \partial_\phi), \\ m^\alpha \partial_\alpha &= \frac{1}{\sqrt{2}(r + ia z)} \left(ia \sqrt{1 - z^2} \partial_t - \sqrt{1 - z^2} \partial_z + \frac{ia}{\sqrt{1 - z^2}} \partial_\phi \right) \end{aligned}$$

To determine the radial junction conditions at $r = r_p$, we use that ψ_0 for a point particle $x_p^\mu = (t_p, r_p, \theta_p, \phi_p)$ has the form

$$\psi_0(t, r, \theta, \phi) = \psi_0^p(t, z, \phi) \delta_p(r) + \psi_0^+(t, r, \theta, \phi) \Theta_p^+(r) + \psi_0^-(t, r, \theta, \phi) \Theta_p^-(r) \quad (\text{V.5})$$

² The naming conventions in these two references differ. Ori using the conventions $C^+ \equiv C^{(out)} \Leftrightarrow C_{\ell m \omega}^{\text{up}}$, $C^- \equiv C^{(down)} \Leftrightarrow C_{\ell m \omega}^{\text{in}}$, $C^{(in)} \Leftrightarrow C_{\ell m \omega}^{\text{down}}$, $C^{(up)} \Leftrightarrow C_{\ell m \omega}^{\text{out}}$ and ori uses the notation $p = 16(D_{\ell m}(a\omega)^2 + 9M^2\omega^2)$ = (Ori Eq. 21) and $Q = \prod_{s=2}^{-1} [4(\omega - m\Omega_+)Mr_+ + is(r_+ - r_-)]$

where $\psi^\pm$ are homogeneous solutions to the Teukolsky equation to the right and left of the particle, and where

$$\psi_0^p = \frac{\Sigma_p}{\Delta_p} T_{mm}^\circ \quad (\text{V.6})$$

results from distribution analysis of the $s = 2$ Teukolsky equation with point particle source. These conditions give $[\partial_r^n \Phi] = 0$ for $n < 3$ and

$$[\partial_r^3 \Phi] = -\frac{4\Sigma_p}{\Delta_p} \delta_p^2(\Omega) J_{\bar{m}\bar{m}}^\circ.$$

The CCK Hertz potential decomposes into a string piece Φ^S , which is a third order polynomial in r and no-string Φ^N piece:

$$\Phi = \Phi^N + \Phi^S \Theta_p^+ \quad (\text{V.7})$$

where

$$\Phi^N = \Phi^+ \Theta_p^+ + \Phi^- \Theta_p^- \quad (\text{V.8})$$

and

$$\bar{\Phi}^S = \sum_{j=0}^3 \hat{\mathcal{B}}_j^\circ \frac{(r - r_p)^j}{j!} = \sum_{j=0}^3 \mathcal{B}_j^\circ r^j \quad (\text{V.9})$$

where

$$r = -\frac{\rho + \bar{\rho}}{2\rho\bar{\rho}} \quad (\text{V.10})$$

is a homogeneous solution and that

$$\mathcal{B}_j^\circ \doteq \{i, i-1\}. \quad (\text{V.11})$$

For a point particle source, the junction conditions $[\bar{\Phi}] = 0$, $[\mathbf{p}\bar{\Phi}] = 0$, $[\mathbf{p}^2\bar{\Phi}] = 0$, $[\mathbf{p}^3\bar{\Phi}] = -4\psi_0^\circ$, together with the definition of ψ_0 and the field equations can be used to derive an explicit formula for $\mathcal{B}_j^\circ$. Starting from

$$\bar{\Phi}^S = \int dV' \mathcal{H}^S(x, x') {}_{+2}T(x') \quad (\text{V.12})$$

where ${}_{+2}T(x') = \mathcal{S}^{\alpha\beta} T_{\alpha\beta} = \mathcal{S}^{\alpha\beta} \mu \int d\lambda v_\alpha v_\beta \delta_4(x, x_p(\lambda))$ and

$$\mathcal{H}^S(x, x') = \sum_{\ell m} \int d\omega e^{-i\omega(u-u')} {}_{+2}S_{\ell m\omega}(z) {}_{+2}\bar{S}_{\ell m\omega}(z_p) \sum_{j=0}^3 H_j^{\ell m\omega}(r, r_p) \frac{(r - r_p)^j}{j!}. \quad (\text{V.13})$$

Integrating by parts and throwing away the boundary term gives

$$\bar{\Phi}^S = \mu \int d\lambda v^\alpha v^\beta \mathcal{S}_{\alpha\beta}^\dagger[\mathcal{H}^S] \quad (\text{V.14})$$

from which we can compare with the general solution (V.7) and (V.9) to arrive at

$$\mathcal{B}_j^\circ = \mu \int_\gamma d\lambda v^\alpha v^\beta \mathcal{S}_{\alpha\beta}^\dagger[\mathcal{H}_j] \quad (\text{V.15})$$

where

$$\begin{aligned} \mathcal{S}_{\alpha\beta}^\dagger = & -l_\alpha l_\beta (\eth - \tau)(\eth + 3\tau) - m_\alpha m_\beta (\mathbb{P} - \rho)(\mathbb{P} + 3\rho) \\ & + l_{(\alpha} m_{\beta)} [(\mathbb{P} - \rho + \bar{\rho})(\eth + 3\tau) + (\eth - \tau + \bar{\tau}')(\mathbb{P} + 3\rho)]. \end{aligned} \quad (\text{V.16})$$

taken at the particle location and where

$$\mathcal{H}_j = \sum_{\ell m} \int_{-\infty+i\epsilon}^{\infty+i\epsilon} d\omega e^{-i\omega(t-t_p)+im(\phi-\phi_p)} h_j^{\ell m\omega}(r, r_p) {}_{+2}S_{\ell m\omega}(z) {}_{+2}\bar{S}_{\ell m\omega}(z_p). \quad (\text{V.17})$$

The quantity $h_j^{\ell m\omega}$ is the radial ‘‘Ori kernel’’

$$\begin{aligned} h_j^{\ell m\omega}(r, r_p) = & \frac{\Delta_p^2}{{}_{+2}W(\omega)} \exp\left(i\omega(r_* - r_{*p}) + im(\phi^\sharp - \phi_p^\sharp)\right) \Theta(r - r_p) \\ & \times \left(C^{\text{in}} {}_{+2}R_{\ell m\omega}^{\text{up}}(r_p) \mathcal{D}_{\omega m-2}^j R_{\ell m\omega}^{\text{in}}(r_p) - C^{\text{up}} {}_{+2}R_{\ell m\omega}^{\text{in}}(r_p) D_{\omega m-2}^j R_{\ell m\omega}^{\text{up}}(r_p)\right) \end{aligned} \quad (\text{V.18})$$

Here ${}_{+2}W(\omega)$ is related to the Wronskian on ${}_{+2}R$ solutions

$${}_{+2}W(\omega) = \Delta^3 \left({}_{+2}R^{\text{up}} \frac{d}{dr} {}_{+2}R^{\text{in}} - {}_{+2}R^{\text{in}} \frac{d}{dr} {}_{+2}R^{\text{up}} \right) \quad (\text{V.19})$$

and ${}_s\mathcal{D}_{m\omega}$ Chandrasehkar’s boost raising operator on modes³

$${}_{+2}R_{\ell m\omega}(r) = \left(\prod_{s=-2}^1 {}_s\mathcal{D}_{\omega m} \right) R_{\ell m\omega}(r) \quad (\text{V.20})$$

$${}_s\mathcal{D}_{\omega m} = \partial_r - \frac{iK_{m\omega}}{\Delta}, \quad K_{m\omega} = \omega(r^2 + a^2) - am \quad (\text{V.21})$$

and constants C^{in} and C^{up} are given by⁴

$$C^{\text{in}} = \frac{1}{\prod_{s=2}^{-1} [4(\omega - m\Omega_+)Mr_+ + is(r_+ - r_-)]}, \quad (\text{V.22})$$

$$C^{\text{up}} = \frac{16\omega^4}{\mathfrak{C}_{2,\ell m}(a\omega)}. \quad (\text{V.23})$$

³ The boost raising operator is just the GHP $\mathbb{P}$ operator Kinnersley’s tetrad acting on a $\{\ell m\omega\}$ mode function expressed in the BL chart. The boost lowering $\mathbb{P}'$ maps to ${}_s\mathcal{D}_{m\omega}^\dagger = \partial_r + iK/\Delta$

⁴ Ori calls these $C^{(\text{down})}$ and $C^{(\text{out})}$

where $\mathfrak{C}_{2,\ell m}(\omega) = \mathfrak{B}_{2,\ell m}(a\omega) + 144M^2\omega^2$ is the real $s = 2$ radial Teukolsky–Starobinsky constant in the Casals–Teixeira da Costa convention[5].⁵

To derive (V.17), we note that the inversion relation

$$-\frac{1}{4}\mathbb{P}^4\bar{\Phi} = \psi_0 \quad (\text{V.24})$$

provides a relation between

$$+2g_{\ell m\omega}(r, r') = A_{\ell m\omega}^+(r') +2R_{\ell m\omega}^+(r)\Theta(+[r - r']) + A_{\ell m\omega}^-(r') -2R_{\ell m\omega}^-(r)\Theta(-[r - r']) \quad (\text{V.25})$$

$$A^\pm(r') = \frac{1}{\Delta} \frac{+2R^\mp(r')}{W[+2R^-, +2R^+]}, \quad (\text{V.26})$$

the Green's function for the radial -2 -equation,⁶

and

$$-2h_{\ell m\omega}(r, r') = h_{\ell m\omega}^+(r, r')\Theta(r - r') + h_{\ell m\omega}^-(r, r')\Theta(r' - r) \quad (\text{V.27})$$

$$-\frac{1}{4}\mathcal{D}_{m\omega-2}^4 h_{\ell m\omega} = +2g_{\ell m\omega} \quad (\text{V.28})$$

the radial kernel for the radial modes of the (complex conjugated) Hertz potential $\bar{\Phi}$.

The Teukolsky-Starobinsky Identities relate the two sets of homogenous solutions

$$+2R_{\ell m\omega}^\pm = C_{\ell m\omega}^\pm \mathcal{D}_{m\omega-2}^4 R_{\ell m\omega}^\pm \quad (\text{V.29})$$

where

$$C^+ = C^{\text{up}}, \quad C^- = C^{\text{in}} \quad (\text{V.30})$$

as above. Continuity of R across $r = r'$ ensures that

$$\mathcal{D}_{m\omega}^n[-2h_{\ell m\omega}^+] = \mathcal{D}_{m\omega}^n[-2h_{\ell m\omega}^-], \quad n = 0, \dots, 3 \quad (\text{V.31})$$

and the condition

$$\mathcal{D}_{m\omega}^4[-2h_{\ell m\omega}^\pm(r, r')] = A_{\ell m\omega}^\pm(r') +2R_{\ell m\omega}^\pm(r) \quad (\text{V.32})$$

requires the kernel is proportional to $-2R_{\ell m\omega}^\pm(r)$

$$-2h_{\ell m\omega}^\pm(r, r') = C_{\ell m\omega}^\pm A_{\ell m\omega}^\pm(r') -2R_{\ell m\omega}^\pm(r) \quad (\text{V.33})$$

which is just the TS identity we've seen already (V.29). Using the property

$$\mathcal{D}_{m\omega}(f_{\ell m\omega}(r)e^{-i(m\phi_\sharp - \omega r_*)}) = f'_{\ell m\omega}(r)e^{-i(m\phi_\sharp - \omega r_*)} \quad (\text{V.34})$$

⁵ When $a = 0$ this denominator becomes $\mathfrak{C}_{2,\ell}(\omega) = (\ell - 1)^2\ell^2(\ell + 1)^2(\ell + 2)^2 + 144M^2\omega^2$.

⁶ $R^+ \equiv R^{\text{up}}$ and $R^- \equiv R^{\text{in}}$

The radial Green's function h is determined up to a homogeneous function $\mathfrak{h}(r, r')$ in the kernel $\mathcal{D}^4$ to fix boundary conditions and ensure the junctions across r_p are satisfied

$$-{}_2h_{\ell m \omega}^+(r, r') = C_{\ell m \omega}^+ A_{\ell m \omega}^+(r') - {}_2R_{\ell m \omega}^+(r) + e^{-i(m\phi_\# - \omega r_*)} \mathfrak{h}_{\ell m \omega}^+(r, r'), \quad (\text{V.35})$$

$$\mathfrak{h}_{\ell m \omega}^+(r, r') = \sum_{j=0}^3 \mathfrak{h}_{\ell m \omega, j}^+(r') (\Delta r)^j. \quad (\text{V.36})$$

Plugging (V.35) into (V.28) and demanding continuity of the four derivatives fixes the coefficients and yields (V.18). After using the Teukolsky equation to successively reduce derivatives one finds

$$\begin{aligned} \mathfrak{h}_{\ell m \omega, j}^+ = & C_{\ell m \omega}^- A_{\ell m \omega}^-(r') \left(\hat{f}_j^{\ell m \omega} - {}_2R_{\ell m \omega}^- + \hat{g}_j^{\ell m \omega} \frac{d}{dr} - {}_2R_{\ell m \omega}^- \right) \Big|_{r=r'} \\ & - C_{\ell m \omega}^+ A_{\ell m \omega}^+(r') \left(\hat{f}_j^{\ell m \omega} - {}_2R_{\ell m \omega}^+ + \hat{g}_j^{\ell m \omega} \frac{d}{dr} - {}_2R_{\ell m \omega}^+ \right) \Big|_{r=r'} \end{aligned} \quad (\text{V.37})$$

where the $\hat{f}_j$ and $\hat{g}_j$ are analytic functions of r given on pp. 18 of [3].

A. Asymptotic Teukolsky amplitudes and the Teukolsky–Starobinsky identities

It is useful to relate the leading asymptotic coefficient of the spin-+2 Teukolsky field, ψ_0 , to the more commonly used spin-−2 amplitude associated with ψ_4 . This relation is particularly useful when comparing numerical large- r fits against the amplitudes returned directly by the Black Hole Perturbation Toolkit.

We use the usual separated form

$$\Psi_s = \sum_{\ell m \omega} R_{s\ell m \omega}(r) {}_sS_{\ell m \omega}(\theta) e^{im\phi - i\omega t}, \quad (\text{V.38})$$

where, for gravitational perturbations,

$$\Psi_{+2} = \psi_0, \quad \Psi_{-2} = \zeta^4 \psi_4, \quad \zeta = r - ia \cos \theta. \quad (\text{V.39})$$

At future null infinity the purely outgoing radial solutions behave as

$$R_{+2, \ell m \omega}^{\text{Up}} \sim r^{-5} e^{i\omega r_*}, \quad (\text{V.40})$$

$$R_{-2, \ell m \omega}^{\text{Up}} \sim r^3 e^{i\omega r_*}. \quad (\text{V.41})$$

Thus, introducing retarded time

$$u = t - r_*, \quad (\text{V.42})$$

the corresponding Weyl scalars have the asymptotic form

$$\psi_0 \sim \sum_{\ell m \omega} \frac{a_{0,\ell m \omega}}{r^5} {}_{+2}S_{\ell m \omega}(\theta) e^{im\phi - i\omega u}, \quad (\text{V.43})$$

$$\psi_4 \sim \sum_{\ell m \omega} \frac{Z_{\ell m \omega}^\infty}{r} {}_{-2}S_{\ell m \omega}(\theta) e^{im\phi - i\omega u}. \quad (\text{V.44})$$

1. Point-particle amplitudes from the BHPToolkit

For a point-particle source the Black Hole Perturbation Toolkit returns the amplitudes multiplying the ingoing and outgoing homogeneous solutions. In particular, outside the radial support of the source,

$$R_{s\ell m \omega}^{\text{ret}}(r) = Z_{s\ell m \omega}^{\mathcal{J}} R_{s\ell m \omega}^{\text{Up}}(r). \quad (\text{V.45})$$

The homogeneous Up solution is normalized to unit transmission amplitude at infinity. Consequently,

$$Z_{-2,\ell m \omega}^{\mathcal{J}} = Z_{\ell m \omega}^\infty, \quad (\text{V.46})$$

$$Z_{+2,\ell m \omega}^{\mathcal{J}} = a_{0,\ell m \omega}. \quad (\text{V.47})$$

Thus, for the Mathematica definitions

```
Cplus[-Weylspin, l, m] =
  TeukolskyPointParticleMode[
    -Weylspin, l, m, 0, 0, orbit] ["Amplitudes"] ["\[ScriptCapitalI]"];

Cplus[Weylspin, l, m] =
  TeukolskyPointParticleMode[
    Weylspin, l, m, 0, 0, orbit] ["Amplitudes"] ["\[ScriptCapitalI]"];
```

with

$$\text{Weylspin} = 2, \quad (\text{V.48})$$

one has, for a nonstatic mode,

$$\boxed{C_+[-2, \ell, m] = Z_{\ell m \omega}^\infty, \quad C_+[+2, \ell, m] = a_{0,\ell m \omega}.} \quad (\text{V.49})$$

Therefore, if the only quantity required is the leading coefficient in

$$\psi_{0,\ell m} = \frac{a_{0,\ell m}}{r^5} + O(r^{-6}) \quad (\text{V.50})$$

on an outgoing $u = \text{const.}$ slicing, it is not necessary to perform a numerical extrapolation of $r^5\psi_0$. The coefficient can be read directly from the spin-+2 point-particle amplitude,

$$\boxed{a_{0,\ell m\omega} = C_+[+2, \ell, m].} \quad (\text{V.51})$$

2. Teukolsky–Starobinsky relation

The spin-+2 and spin-−2 descriptions are not independent. In vacuum they are related by the Teukolsky–Starobinsky identities. The identity relevant at future null infinity has the schematic form

$$\Delta^2 \mathcal{D}^4 (\Delta^2 \psi_0) = \frac{1}{4} \mathcal{L}^4 (\zeta^4 \psi_4) + 3M \partial_t (\zeta^4 \overline{\psi_4}), \quad (\text{V.52})$$

where $\mathcal{D}$ denotes the appropriate radial differential operator and $\mathcal{L}$ the corresponding angular operator.

For an outgoing Fourier mode,

$$e^{im\phi - i\omega u}, \quad (\text{V.53})$$

the radial operator contributes, at leading order,

$$\Delta^2 \mathcal{D}^4 (\Delta^2 \psi_0) \sim \omega^4 a_{0,\ell m\omega} r^3 {}_{+2}S_{\ell m\omega} e^{im\phi - i\omega u}. \quad (\text{V.54})$$

Define the angular Teukolsky–Starobinsky coefficient $\mathfrak{D}'_{\ell m\omega}$ through

$$\mathcal{L}^4 {}_{-2}S_{\ell m\omega} = \mathfrak{D}'_{\ell m\omega+2} S_{\ell m\omega}. \quad (\text{V.55})$$

The asymptotic identity therefore gives

$$\boxed{4\omega^4 a_{0,\ell m\omega} = \mathfrak{D}'_{\ell m\omega} Z_{\ell m\omega}^\infty - 12iM\omega \eta_{\ell m\omega} \overline{Z_{\ell, -m, -\omega}^\infty}} \quad (\text{V.56})$$

where $\eta_{\ell m\omega}$ denotes the phase associated with the chosen conjugation convention for the spin-weighted spheroidal harmonics.

For a physical pair of modes satisfying a definite equatorial-parity relation,

$$\eta_{\ell m\omega} \overline{Z_{\ell, -m, -\omega}^\infty} = p_{\ell m\omega} Z_{\ell m\omega}^\infty, \quad p_{\ell m\omega} = \pm 1, \quad (\text{V.57})$$

the relation becomes diagonal,

$$a_{0,\ell m\omega} = \frac{\mathfrak{D}'_{\ell m\omega} - 12iM\omega p_{\ell m\omega}}{4\omega^4} Z_{-2,\ell m\omega}^\infty. \quad (\text{V.58})$$

Equivalently, in terms of the BHPToolkit amplitudes,

$$C_+[+2, \ell, m] = \frac{\mathfrak{D}'_{\ell m\omega} - 12iM\omega p_{\ell m\omega}}{4\omega^4} C_+[-2, \ell, m], \quad (\text{V.59})$$

provided that the phase convention used for $p_{\ell m\omega}$ agrees with that of the spheroidal harmonics in the Toolkit.

3. *Kerr Starobinsky constant*

Let

$$\lambda = {}_{+2}\lambda_{\ell m\omega} \quad (\text{V.60})$$

denote the spin-+2 angular Teukolsky separation constant. The normalization-independent angular Starobinsky quantity is

$$\begin{aligned} D_{\ell m\omega} = & (\lambda + 4)^2(\lambda + 6)^2 \\ & + 8a\omega(m - a\omega)(\lambda + 4)(5\lambda + 26) \\ & + 48a^2\omega^2 [2(\lambda + 4) + 3(m - a\omega)^2]. \end{aligned} \quad (\text{V.61})$$

The complete gravitational Starobinsky quantity appearing in the radial identity is then

$$\mathfrak{C}_{2,\ell m}(\omega) = D_{\ell m}(\omega) + 144M^2\omega^2. \quad (\text{V.62})$$

Here $\mathfrak{C}_{\ell m}(\omega)$ is real and positive for the real frequencies relevant to bound geodesic motion. Different conventions in the literature may call either $\mathfrak{C}_{\ell m}(\omega)$ or $\mathcal{C}_{\ell m\omega}^2$ the “Starobinsky constant.”

Taking the modulus squared of Eq. (V.58) gives the convention-independent relation

$$|a_{0,\ell m\omega}|^2 = \frac{\mathfrak{C}_{2,\ell m}(\omega)}{16\omega^8} |Z_{-2,\ell m\omega}^\infty|^2. \quad (\text{V.63})$$

Hence the two amplitudes returned independently by the BHPToolkit must satisfy

$$|C_+[+2, \ell, m]|^2 = \frac{\mathfrak{C}_{\ell m}(\omega)}{16\omega^8} |C_+[-2, \ell, m]|^2. \quad (\text{V.64})$$

This provides a particularly clean numerical check because it is independent of the relative phase convention for the spin- ± 2 spheroidal harmonics.

Indeed, the BHPToolkit computes the infinity flux from the spin- -2 amplitude as

$$\dot{E}^\infty = \frac{|Z_{-2,\ell m \omega}^\infty|^2}{4\pi\omega^2}, \quad (\text{V.65})$$

whereas when the same physical flux is computed from the spin- $+2$ amplitude it uses

$$\dot{E}^\infty = \frac{16\omega^6}{4\pi\mathfrak{C}_{2,\ell m}(\omega)} |Z_{+2,\ell m \omega}^\infty|^2. \quad (\text{V.66})$$

Equating the two expressions reproduces Eq. (V.64).

4. *Schwarzschild limit*

For $a = 0$,

$$\lambda = \ell(\ell + 1) - 6, \quad (\text{V.67})$$

and hence

$$(\lambda + 4)(\lambda + 6) = (\ell - 1)\ell(\ell + 1)(\ell + 2). \quad (\text{V.68})$$

Defining

$$\Lambda_\ell \equiv (\ell - 1)\ell(\ell + 1)(\ell + 2), \quad (\text{V.69})$$

we obtain

$$D_{\ell m \omega} \longrightarrow \Lambda_\ell^2 \quad (\text{V.70})$$

and

$$\boxed{\mathfrak{C}_{2,\ell m}(\omega) = \Lambda_\ell^2 + (12M\omega)^2.} \quad (\text{V.71})$$

The amplitude relation therefore reduces to

$$\boxed{a_{0,\ell m \omega} = \frac{\Lambda_\ell - 12iM\omega p_{\ell m}}{4\omega^4} Z_{\ell m \omega}^\infty.} \quad (\text{V.72})$$

For a circular equatorial orbit, the mode frequency is

$$\omega = m\Omega_\phi. \quad (\text{V.73})$$

With a convention in which the parity factor is $p_{\ell m} = (-1)^{\ell+m}$, this becomes

$$\boxed{a_{0,\ell m} = \frac{(\ell - 1)\ell(\ell + 1)(\ell + 2) - 12iM(m\Omega_\phi)(-1)^{\ell+m}}{4(m\Omega_\phi)^4} Z_{\ell m}^\infty.} \quad (\text{V.74})$$

5. Practical consequence

Suppose a numerical calculation of ψ_0 gives the large- r expansion

$$\psi_{0,\ell m}^{[u]}(r) = \frac{a_{0,\ell m}}{r^5} + \frac{a_{1,\ell m}}{r^6} + \frac{a_{2,\ell m}}{r^7} + \dots \quad (\text{V.75})$$

The leading coefficient may be obtained in three equivalent ways:

$$a_{0,\ell m \omega} = \lim_{r \rightarrow \infty} r^5 \psi_{0,\ell m}^{[u]}(r) = Z_{+2,\ell m \omega}^{\mathcal{J}} = C_+[+2, \ell, m]. \quad (\text{V.76})$$

Alternatively, it can be reconstructed from the spin- -2 amplitude using the Teukolsky–Starobinsky identity.

For numerical purposes, Eq. (V.76) is preferable: the point-particle solver already performs the source convolution and returns the coefficient multiplying the unit-normalized outgoing solution. A direct polynomial fit of $r^5 \psi_0$ is therefore useful primarily as an independent check rather than as the preferred method for obtaining a_0 .

A simple convention-independent diagnostic is

$$\frac{16\omega^8 |C_+[+2, \ell, m]|^2}{\mathfrak{C}_{2,\ell m}(\omega) |C_+[-2, \ell, m]|^2} = 1. \quad (\text{V.77})$$

For nonstatic modes this ratio should agree with unity to the numerical accuracy of the Teukolsky calculation.

Finally, the static sector $\omega = 0$ must be treated separately. The formulas above contain explicit inverse powers of ω and apply only to radiative, nonstatic modes.

B. Hertz potential from the asymptotic spin-+2 Teukolsky amplitude

Once the leading asymptotic coefficient of the spin-+2 Teukolsky field is known, the corresponding spin- -2 Hertz potential in the exterior vacuum region can be obtained algebraically using the Teukolsky–Starobinsky identities.

We write the outgoing spin-+2 radial Teukolsky mode as

$$R_{+2,\ell m \omega}^{\text{ret}}(r) = a_{0,\ell m \omega} R_{+2,\ell m \omega}^{\text{Up}}(r), \quad (\text{V.78})$$

where the homogeneous up solution is normalized at future null infinity according to

$$R_{+2,\ell m \omega}^{\text{Up}} \sim r^{-5} e^{i\omega r_*}. \quad (\text{V.79})$$

Thus the corresponding Weyl scalar has the asymptotic behavior

$$\psi_{0,\ell m\omega} \sim \frac{a_{0,\ell m\omega}}{r^5} {}_{+2}S_{\ell m\omega}(\theta) e^{im\phi - i\omega u}, \quad u = t - r_*. \quad (\text{V.80})$$

The Hertz potential relevant for ingoing-radiation-gauge metric reconstruction satisfies the spin-2 Teukolsky equation. Hence, in the exterior vacuum region, its radial dependence is proportional to the spin-2 up solution,

$$R_{-2,\ell m\omega}^{\text{Up}} \sim r^3 e^{i\omega r_*}. \quad (\text{V.81})$$

Strictly speaking, the CCK inversion relation acts on the complex conjugate Hertz potential. Schematically,

$$\psi_0 \propto \mathbb{P}^4 \bar{\Phi}, \quad (\text{V.82})$$

where the precise overall numerical factor depends on the convention used for the Hertz potential and the Newman-Penrose tetrad. In the conventions commonly used in modern GHZ/CCK calculations one has

$$\mathbb{P}^4 \bar{\Phi} = 2\psi_0. \quad (\text{V.83})$$

The Teukolsky-Starobinsky identities allow this fourth-order inversion to be replaced, mode by mode, by an algebraic relation between the normalization constants of the spin-+2 and spin-2 homogeneous solutions. For an outgoing mode one may write

$$\bar{\Phi}_{\ell m\omega}^+ = H_{\ell m\omega}^\infty R_{-2,\ell m\omega}^{\text{Up}}(r) {}_{-2}S_{\ell m\omega}(\theta) e^{im\phi - i\omega t}, \quad (\text{V.84})$$

where the Hertz amplitude is

$$H_{\ell m\omega}^\infty = \frac{16\omega^4}{\mathfrak{C}_{2,\ell m}(\omega)} a_{0,\ell m\omega}. \quad (\text{V.85})$$

Here $\mathcal{C}_{\ell m\omega}^2$ denotes the gravitational Kerr Starobinsky quantity.

Thus the complete exterior radial Hertz mode is

$$\bar{\Phi}_{\ell m\omega}^+(r) = \frac{16\omega^4}{\mathfrak{C}_{2,\ell m}(\omega)} a_{0,\ell m\omega} R_{-2,\ell m\omega}^{\text{Up}}(r). \quad (\text{V.86})$$

1. Relation to the BHPToolkit amplitudes

For a point-particle source, the exterior retarded spin-+2 mode is written by the Black Hole Perturbation Toolkit as

$$R_{+2,\ell m\omega}^{\text{ret}} = Z_{+2,\ell m\omega}^{\mathcal{J}} R_{+2,\ell m\omega}^{\text{Up}}. \quad (\text{V.87})$$

Since the up solution is normalized to unit outgoing amplitude at future null infinity,

$$Z_{+2,\ell m\omega}^{\mathcal{J}} = a_{0,\ell m\omega}. \quad (\text{V.88})$$

Thus, for the Mathematica definition

```
Cplus[Weylspin, 1, m] =  
  TeukolskyPointParticleMode[  
    Weylspin, 1, m, 0, 0, orbit  
  ]["Amplitudes"] ["\[ScriptCapitalI]"];
```

with

$$\text{Weylspin} = 2, \quad (\text{V.89})$$

one has

$$a_{0,\ell m\omega} = C_+[2, \ell, m]. \quad (\text{V.90})$$

The exterior Hertz-potential normalization can therefore be obtained directly from the point-particle amplitude:

$$H_{\ell m\omega}^{\infty} = \frac{16\omega^4}{\mathfrak{C}_{2,\ell m}(\omega)} C_+[2, \ell, m]. \quad (\text{V.91})$$

2. Relation to the spin-−2 radiative amplitude

The same Hertz amplitude may also be expressed in terms of the spin-−2 radiative amplitude

$$Z_{\ell m\omega}^{\infty} = Z_{-2,\ell m\omega}^{\mathcal{J}}. \quad (\text{V.92})$$

The spin-+2 and spin-−2 asymptotic amplitudes satisfy

$$|a_{0,\ell m\omega}|^2 = \frac{\mathfrak{C}_{2,\ell m}(\omega)}{16\omega^8} |Z_{\ell m\omega}^{\infty}|^2. \quad (\text{V.93})$$

Substituting this into Eq. (V.85) gives

$$|H_{\ell m\omega}^{\infty}|^2 = \frac{16}{\mathfrak{C}_{2,\ell m}(\omega)} |Z_{\ell m\omega}^{\infty}|^2. \quad (\text{V.94})$$

Thus the three exterior amplitudes

$$Z_{-2,\ell m\omega}^{\infty}, \quad a_{0,\ell m\omega} = Z_{+2,\ell m\omega}^{\infty}, \quad H_{\ell m\omega}^{\infty}, \quad (\text{V.95})$$

are all algebraically related by the Teukolsky–Starobinsky identities.

VI. POINT PARTICLE CORRECTOR

The Held scalars determining the IRG corrector

$$x_{\mu\nu} = x_{\mu\nu}^+ \Theta(r - r_p)$$

for a point particle source are given by $\llbracket$ and the relations (using the derivative property of Dirac distributions $f(x)\delta'(x) = f(0)\delta'(x) - f'(0)\delta(x)$ and that $\tilde{\delta}\tau^\circ = 0$, $\tilde{\delta}\rho = \rho^2\tau^\circ$ and $\tilde{\delta}'\rho = -\rho^2\bar{\tau}^\circ$). Using the notation

$$\delta_p(\Omega; u) \equiv \delta^2(\Omega - \Omega_p(u)), \quad (\text{VI.1})$$

we find

$$a^\circ = -\frac{1}{8} \left(\frac{1}{\rho_p} + \frac{1}{\bar{\rho}_p} \right) J_l^\circ \delta_p(\Omega; u) \quad (\text{VI.2a})$$

$$b^\circ = \frac{1}{8} \left(\frac{1}{\rho_p^2} + \frac{1}{\bar{\rho}_p^2} \right) J_l^\circ \delta_p(\Omega) \quad (\text{VI.2b})$$

$$\begin{aligned} c^\circ &= -\frac{1}{12} \frac{\bar{\rho}}{\rho} \tilde{\delta} T_l^\circ + \frac{1}{12} \frac{\bar{\rho}_p}{\rho_p} (3\rho_p + \bar{\rho}_p) \tau_p^\circ T_l^\circ - \frac{1}{6} \left(1 + \frac{\bar{\rho}_p}{\rho_p} \right) T_m^\circ, \\ &= -\frac{1}{12} \frac{\bar{\rho}_p}{\rho_p} \tilde{\delta} (J_l^\circ \delta_p(\Omega)) + \frac{\bar{\rho}_p(\rho_p + \bar{\rho}_p) \bar{\tau}_p^\circ}{12\rho_p} \delta_p(\Omega) J_l^\circ + \frac{1}{12} \frac{\bar{\rho}_p}{\rho_p} (3\rho_p + \bar{\rho}_p) \tau_p^\circ J_l^\circ \delta_p(\Omega) \\ &\quad - \frac{1}{6} \left(1 + \frac{\bar{\rho}_p}{\rho_p} \right) J_m^\circ \delta_p(\Omega), \end{aligned} \quad (\text{VI.2c})$$

$$\begin{aligned} e^\circ &= -\frac{1}{12} \frac{\bar{\rho}}{\rho^4} \tilde{\delta} T_l^\circ + \frac{1}{12} \frac{\bar{\rho}_p^2}{\rho_p^4} \tau_p^\circ T_l^\circ + \frac{1}{6} \frac{2\rho_p - \bar{\rho}_p}{\rho_p^4} T_m^\circ, \\ &= -\frac{\bar{\rho}_p}{12\rho_p^4} \tilde{\delta} [J_l^\circ \delta_p(\Omega)] + \frac{\rho_p(4\rho_p + \bar{\rho}_p) \bar{\tau}_p^\circ}{12\bar{\rho}_p^4} J_l^\circ \delta_p(\Omega) \\ &\quad + \frac{\bar{\rho}_p^2 \tau_p^\circ}{12\rho_p^4} J_l^\circ \delta_p(\Omega) + \frac{2\rho_p - \bar{\rho}_p}{6\rho_p^4} J_m^\circ \delta_p(\Omega), \end{aligned} \quad (\text{VI.2d})$$

$$\quad (\text{VI.2e})$$

$$\begin{aligned}
f^\circ = \text{Re} \left[\frac{1}{4} \frac{\rho^2}{\bar{\rho}^3(\rho + \bar{\rho})} \tilde{\partial}' \tilde{\partial} T_l^\circ - \frac{1}{4} \frac{\rho^6 + 2\rho^5 \bar{\rho} - 3\rho^4 \bar{\rho}^2 + \rho^2 \bar{\rho}^4 - 2\rho \bar{\rho}^5 - \bar{\rho}^6}{\rho^3 \bar{\rho}^3(\rho + \bar{\rho})} \bar{\tau}^\circ \tilde{\partial} T_l^\circ \right. \\
+ \frac{1}{16} \frac{\rho(4\rho^2 + 3\rho \bar{\rho} - 2\bar{\rho}^2)}{\bar{\rho}^3(\rho + \bar{\rho})} \Omega^\circ \tilde{\mathbf{P}}' T_l^\circ + \frac{1}{4} \frac{\bar{\rho}_p(3\rho_p + \bar{\rho}_p)(2\rho_p^2 - \bar{\rho}_p^2)}{\rho_p^3(\rho_p + \bar{\rho}_p)^2} \Omega^\circ \rho'^\circ T_l^\circ \\
+ \frac{1}{2} \frac{\bar{\rho}_p^2(2\rho_p^3 - 2\rho_p^2 \bar{\rho}_p - 3\rho_p \bar{\rho}_p^2 - \bar{\rho}_p^3)}{\rho_p^3(\rho_p + \bar{\rho}_p)^2} \tau^\circ \bar{\tau}^\circ T_l^\circ \\
+ \frac{1}{16} \frac{2\rho_p^7 + 6\rho_p^6 \bar{\rho}_p + \rho_p^5 \bar{\rho}_p^2 - 21\rho_p^4 \bar{\rho}_p^3 + 3\rho_p^3 \bar{\rho}_p^4 + \rho_p^2 \bar{\rho}_p^5 - 6\rho_p \bar{\rho}_p^6 - 2\bar{\rho}_p^7}{\rho_p^3 \bar{\rho}_p^3(\rho_p + \bar{\rho}_p)^2} \Psi^\circ T_l^\circ \\
\left. - \frac{1}{2} \frac{2\rho^2 - \bar{\rho}^2}{\rho^3(\rho + \bar{\rho})} \tilde{\partial}' T_m^\circ - \frac{1}{2} \frac{2\rho_p^3 - 2\rho_p \bar{\rho}_p^2 - \bar{\rho}_p^3}{\rho_p^3(\rho_p + \bar{\rho}_p)} \bar{\tau}^\circ T_m^\circ \right]_{r_p} + \frac{1}{(\rho_p + \bar{\rho}_p)^2} T_n^\circ \quad (\text{VI.2f})
\end{aligned}$$

Introducing the quantities

$$E := \frac{1}{16} \frac{\rho(4\rho^2 + 3\rho \bar{\rho} - 2\bar{\rho}^2)}{\bar{\rho}^3(\rho + \bar{\rho})} \Omega^\circ \Big|_{r_p}, \quad (\text{VI.3})$$

$$F := \frac{1}{4} \frac{\rho^6 + 2\rho^5 \bar{\rho} - 3\rho^4 \bar{\rho}^2 + \rho^2 \bar{\rho}^4 - 2\rho \bar{\rho}^5 - \bar{\rho}^6}{\rho^3 \bar{\rho}^3(\rho + \bar{\rho})} \bar{\tau}^\circ \Big|_{r_p}, \quad (\text{VI.4})$$

$$G := -\frac{1}{2} \frac{2\rho^2 - \bar{\rho}^2}{\rho^3(\rho + \bar{\rho})} \Big|_{r_p},$$

$$H := \frac{1}{4} \frac{\rho^2}{\bar{\rho}^3(\rho + \bar{\rho})} \Big|_{r_p} \quad (\text{VI.5})$$

and using the rule for derivatives of δ_p we find that f° is given by

$$\begin{aligned}
f^\circ = \text{Re} \Bigg[& \left(H_p \tilde{\delta}' \tilde{\delta} - H_p (\rho_p + 4\bar{\rho}_p) \tau_p^\circ \tilde{\delta}' + H_p (\rho_p + 4\bar{\rho}_p) \bar{\tau}_p^\circ \tilde{\delta} \right. \\
& + H_p \left((\rho_p + 4\bar{\rho}_p) (\tilde{\delta}' \tau^\circ)_p - 2(\rho_p^2 + 4\rho_p \bar{\rho}_p + 6\bar{\rho}_p^2) \tau_p^\circ \bar{\tau}_p^\circ \right) \Big) J_l^\circ \delta_p(\Omega) \\
& - \left(\frac{(\rho^7 + 3\rho^6 \bar{\rho} + 3\rho^5 \bar{\rho}^2 - 3\rho^4 \bar{\rho}^3 - \rho^3 \bar{\rho}^4 + 3\rho^2 \bar{\rho}^5 + 3\rho \bar{\rho}^6 + \bar{\rho}^7) (\bar{\tau}^\circ)^2}{2\rho^3 \bar{\rho}^3 (\rho + \bar{\rho})} + \right. \\
& \left. + \frac{\rho^6 + 2\rho^5 \bar{\rho} - 3\rho^4 \bar{\rho}^2 + \rho^2 \bar{\rho}^4 - 2\rho \bar{\rho}^5 - \bar{\rho}^6}{\rho^3 \bar{\rho}^3 (\rho + \bar{\rho})} \bar{\tau}^\circ \tilde{\delta} \right) \Bigg|_p J_l^\circ \delta_p(\Omega) \\
& - \left(\frac{1}{2} \frac{2\rho^2 - \bar{\rho}^2}{\rho^3 (\rho + \bar{\rho})} \tilde{\delta}' - \frac{(2\rho + \bar{\rho})(2\rho^2 - 2\rho \bar{\rho} - \bar{\rho}^2) \tau^\circ}{2\rho^2 (\rho + \bar{\rho})} \right) \Bigg|_p J_m^\circ \delta_p(\Omega) \\
& + \left(\frac{1}{16} \frac{\rho(4\rho^2 + 3\rho \bar{\rho} - 2\bar{\rho}^2)}{\bar{\rho}^3 (\rho + \bar{\rho})} \Omega \tilde{\mathbf{P}}' - \frac{8\rho^3 + 15\rho^2 \bar{\rho} + 6\rho \bar{\rho}^2 - 2\bar{\rho}^3}{\bar{\rho}^3 (\rho + \bar{\rho})^2} \Omega^\circ (\tilde{\mathbf{P}}' \rho) \right. \\
& \left. + \frac{\rho(12\rho^3 + 22\rho^2 \bar{\rho} + 7\rho \bar{\rho} - 4\bar{\rho}^3)}{\bar{\rho}^4 (\rho + \bar{\rho})^2} \Omega^\circ (\tilde{\mathbf{P}}' \bar{\rho}) \right) \Bigg|_{r_p} \delta_p(\Omega) J_l^\circ \\
& + \frac{1}{4} \frac{\bar{\rho}_p (3\rho_p + \bar{\rho}_p) (2\rho_p^2 - \bar{\rho}_p^2)}{\rho_p^3 (\rho_p + \bar{\rho}_p)^2} \Omega_p^\circ \rho_p' J_l^\circ \delta_p(\Omega) \\
& + \frac{1}{2} \frac{\bar{\rho}_p^2 (2\rho_p^3 - 2\rho_p^2 \bar{\rho}_p - 3\rho_p \bar{\rho}_p^2 - \bar{\rho}_p^3)}{\rho_p^3 (\rho_p + \bar{\rho}_p)^2} \tau_p^\circ \bar{\tau}_p^\circ J_l^\circ \delta_p(\Omega) \\
& + \frac{1}{16} \frac{2\rho_p^7 + 6\rho_p^6 \bar{\rho}_p + \rho_p^5 \bar{\rho}_p^2 - 21\rho_p^4 \bar{\rho}_p^3 + 3\rho_p^3 \bar{\rho}_p^4 + \rho_p^2 \bar{\rho}_p^5 - 6\rho_p \bar{\rho}_p^6 - 2\bar{\rho}_p^7}{\rho_p^3 \bar{\rho}_p^3 (\rho_p + \bar{\rho}_p)^2} \Psi^\circ J_l^\circ \delta_p(\Omega) \\
& - \frac{1}{2} \frac{2\rho_p^2 - 2\rho_p \bar{\rho}_p^2 - \bar{\rho}_p^3}{\rho_p^3 (\rho_p + \bar{\rho}_p)} \tau_p^\circ J_m^\circ \delta_p(\Omega) \Bigg] \\
& + \frac{1}{(\rho_p + \bar{\rho}_p)^2} J_n^\circ \delta_p(\Omega)
\end{aligned} \tag{VI.6}$$

where we've used $F \tilde{\delta} (J_l^\circ \delta_p(\Omega)) = (F J^\circ)_p \tilde{\delta} \delta_p(\Omega) - (J_l^\circ)_p (\tilde{\delta}' F)_p \delta_p(\Omega)$ etc. The ℓ, m modes of the corrector Held scalars are now given. Using the results of appendix ?? we find that

[PZ: FINISH FOR GENERIC ORBIT]

$$a_{\ell m}^{\circ} = -\frac{1}{8} \left(\frac{1}{\rho_p} + \frac{1}{\bar{\rho}_p} \right) J_{l,\ell m}^{\circ}(u)_0 \bar{Y}_{\ell,m}, \quad (\text{VI.7a})$$

$$b_{\ell m}^{\circ} = \frac{1}{8} \left(\frac{1}{\rho_p^2} + \frac{1}{\bar{\rho}_p^2} \right) J_{l,\ell m}^{\circ}(u)_0 \bar{Y}_{\ell,m}, \quad (\text{VI.7b})$$

$$\begin{aligned} c_{\ell m}^{\circ} = & \frac{1}{12} \frac{\bar{\rho}_p}{\rho_p} (3\rho_p + \bar{\rho}_p) \tau_p^{\circ} J_{l-1}^{\circ} \bar{Y}_{\ell,m}^p - \frac{1}{6} \left(1 + \frac{\bar{\rho}_p}{\rho_p} \right) J_m^{\circ} \bar{Y}_{\ell,m}^p \\ & + \frac{\bar{\rho}_p(\rho + \bar{\rho}_p) \bar{\tau}_p^{\circ}}{12\rho_p} J_{l-1} \bar{Y}_{\ell,m}^p - \frac{\bar{\rho}_p}{12\rho_p} \langle {}_1Y_{\ell,m} | \tilde{\delta} J_l^{\circ} \delta_p(\Omega) \rangle_{L^2}, \end{aligned} \quad (\text{VI.7c})$$

$$\begin{aligned} e_{\ell m}^{\circ} = & \frac{\rho_p(4\rho_p + \bar{\rho}_p) \bar{\tau}_p^{\circ}}{12\bar{\rho}_p^4} J_{l-1}^{\circ} \bar{Y}_{\ell,m}^p \\ & + \frac{\bar{\rho}_p^2 \tau_p^{\circ}}{12\rho_p^4} J_{l-1} \bar{Y}_{\ell,m}^p + \frac{2\rho_p - \bar{\rho}_p}{6\rho_p^4} J_m^{\circ} \bar{Y}_{\ell,m}^p - \frac{\bar{\rho}_p}{12\rho_p^4} \langle {}_1Y_{\ell,m} | \tilde{\delta} J_l^{\circ} \delta_p(\Omega) \rangle_{L^2}, \end{aligned} \quad (\text{VI.7d})$$

$$\langle {}_sY_{\ell m-s} \mathcal{L}_{\omega_{m\mathbf{k}}}^{\dagger} f(\Omega, u) \delta_p(\Omega; u) \rangle = f(\Omega_p(u)) \left(\frac{\lambda_{\ell,s-1}}{\sqrt{2}\lambda_{\ell}} {}_{s-1}\bar{Y}^p(u) + \frac{am\omega_{m\mathbf{k}}}{\sqrt{2}} \sin \theta_p(u)_s \bar{Y}^p(u) \right) \quad (\text{VI.8})$$

$$\langle {}_sY_{\ell m-s} \mathcal{L}_{\omega_{m\mathbf{k}}} f(\Omega, u) \delta_p(\Omega; u) \rangle = -f(\Omega_p(u)) \left(\frac{\lambda_{\ell,s}}{\sqrt{2}\lambda_{\ell,s+1}} {}_{s+1}\bar{Y}^p(u) + \frac{am\omega_{m\mathbf{k}}}{\sqrt{2}} \sin \theta_p(u)_s \bar{Y}^p(u) \right) \quad (\text{VI.9})$$

A. Static background

We now take the Schwarzschild limit, for which we have a hypersurface orthogonal time translational killing vector t^{α} , $\mathcal{L}_t g_{\alpha\beta} = 0$. In the limit $a \rightarrow 0$, we find that the coefficients in x read

$$a^{\circ} = \frac{J_l^{\circ}}{4r_p} \delta_p^2(\Omega), \quad (\text{VI.10a})$$

$$b^{\circ} = r_p a^{\circ}, \quad (\text{VI.10b})$$

$$c^{\circ} = -\frac{1}{3} J_m^{\circ} \delta_p^2(\Omega) - \frac{1}{3r_p^2} \tilde{\delta} b^{\circ}, \quad (\text{VI.10c})$$

$$e^{\circ} = -\frac{r_p}{6} \left(r_p^2 J_m^{\circ} \delta_p^2(\Omega) - 2\tilde{\delta} b^{\circ} \right) \quad (\text{VI.10d})$$

$$f^{\circ} = \frac{r_p^2}{16} \left(2J_l^{\circ} + 4J_n^{\circ} - 2\tilde{\delta} J_m^{\circ} - 2\tilde{\delta}' J_m^{\circ} + \left(\tilde{\delta}' \tilde{\delta} + \tilde{\delta} \tilde{\delta}' \right) J_l \right) \delta_p^2(\Omega) \quad (\text{VI.10e})$$

with operators acting on everything to the right.

VII. IRG CORRECTOR FOR VARIOUS ORBITS

A. Circular geodesic orbit

We consider for the main a particle on a circular orbit $r = r_p = \text{const}$ with coordinate (outgoing KN) four velocity

$$v^\mu \partial_\mu = \gamma_p \left(\frac{\partial}{\partial u} + \omega_p \frac{\partial}{\partial \varphi} \right) \quad (\text{VII.1})$$

where

$$\gamma_p = du/d\tau = \frac{r_p^{3/2} \pm M^{1/2}a}{\sqrt{r_p^3 - 3Mr_p^2 \pm 2aM^{1/2}r_p^{3/2}}}, \quad (\text{VII.2})$$

$$\omega_p = \frac{\pm M^{1/2}}{r_p^{3/2} \pm aM^{1/2}}, \quad (\text{VII.3})$$

are essentially the specific conserved energy and angular momentum, $\gamma_p = -v_\mu t^\mu$, $\gamma_p \omega_p = \varphi_\mu v^\mu$ respectively. In the above, the lower sign is taken for retrograde motion and the upper for prograde orbits wrt to the BH spin.

In outgoing KN coordinates and Kinnersley tetrad the tetrad components of the four-velocity are

$$v_l = \gamma_p A_p, \quad (\text{VII.4a})$$

$$v_n = \frac{\gamma_p \Delta_p}{2\Sigma_p} A_p, \quad (\text{VII.4b})$$

$$v_m = \frac{i\bar{\rho}_p \gamma_p}{\sqrt{2}} B_p \sqrt{1 - z_p^2} \quad (\text{VII.4c})$$

where

$$A_p := 1 - a\omega_p(1 - z_p^2), \quad (\text{VII.5a})$$

$$B_p := (r_p^2 + a^2)\omega_p - a \quad (\text{VII.5b})$$

[PZ: Fix me] The components of stress energy needed for d_i° read

$$T_{ll}^\circ = \frac{\mu\gamma_p}{\Sigma_p} A_p^2 \delta_p^2(\Omega), \quad (\text{VII.6a})$$

$$T_{ln} = \frac{\mu\gamma_p \Delta_p}{2\Sigma_p^2} A_p^2 \delta_p^2(\Omega) \quad (\text{VII.6b})$$

$$T_{lm}^\circ = \frac{i\mu\bar{\rho}_p \gamma_p}{\sqrt{2}\Sigma_p} A_p B_p \sqrt{1 - z_p^2} \delta_p^2(\Omega) \quad (\text{VII.6c})$$

$$T_{mm}^\circ = -\frac{\mu\gamma_p \bar{\rho}_p^2}{2\Sigma_p} B_p^2 (1 - z_p^2), \quad (\text{VII.6d})$$

where

$$\delta_p^2(\Omega) = \delta(z)\delta(\varphi - \omega_p u) \quad (\text{VII.7})$$

And finally, using $\dot{x}_p^\alpha = dx_p^\alpha/du = v^\alpha/\gamma_p$ with v_α given in (VII.4) and taking $z_p = 0$ we find

$$J_l^\circ = \frac{\mu\gamma_p}{r_p^2}(1 - a\omega_p)^2 \quad (\text{VII.8a})$$

$$J_n^\circ = \frac{\mu\gamma_p}{2r_p^4}(1 - a\omega_p)^2\Delta_p \quad (\text{VII.8b})$$

$$J_m^\circ = \frac{i\mu\tilde{\gamma}_p}{\sqrt{2}r_p^3}(1 - a\omega_p)(a - \omega_p(r_p^2 + a^2)). \quad (\text{VII.8c})$$

since $\Sigma_p = r_p^2 + a^2 z_p^2 = r_p^2$ for an equatorial orbit.

B. Corrector scalars

The Held scalars for the corrector for the equatorial orbit $x_p^\mu = (u, r_p, 0, \Omega_p u)$,

$$\rho_p = \bar{\rho}_p = -1/r_p, \quad \tau_p^\circ = -ia/\sqrt{2}, \quad \Omega_p^\circ = 0, \quad \rho_p^{\prime\circ} = -1/2, \quad (\text{circular equatorial}) \quad (\text{VII.9})$$

and that $\Omega_p^\circ = 0$, $(\mathbb{P}'\Omega^\circ)_p = 0$, in mixed Held/coordinate form we have

$$a^\circ = \frac{r_p}{4} J_l^\circ \delta_p^2(\Omega) \quad (\text{VII.10a})$$

$$b^\circ = r_p a^\circ, \quad (\text{VII.10b})$$

$$c^\circ = -\frac{1}{12} \tilde{\delta}(J_l^\circ \delta_p(\Omega_p)) + \frac{ia}{6\sqrt{2}r_p} J_l^\circ \delta_p(\Omega) - \frac{1}{3} J_m^\circ \delta_p(\Omega), \quad (\text{VII.10c})$$

$$e^\circ = \frac{r_p^3}{12} \tilde{\delta} J_l^\circ \delta_p(\Omega) + \frac{r_p^2}{3} \frac{ia}{\sqrt{2}} J_l^\circ \delta_p^2(\Omega) - \frac{r_p^3}{6} J_m^\circ \delta_p^2(\Omega) \quad (\text{VII.10d})$$

$$f^\circ = \text{Re} \left[\frac{15Mr_p}{64} J_l^\circ - \frac{r_p}{2} \frac{ia}{\sqrt{2}} J_m^\circ - \frac{a^2}{4} J_l^\circ + \frac{r_p^2}{4} J_n^\circ \right] \delta_p^2(\Omega), \\ - \text{Re} \left[\frac{r_p^2}{8} \tilde{\delta}' \tilde{\delta} + \frac{5iar_p}{8\sqrt{2}} \tilde{\delta} - \frac{5iar_p}{8\sqrt{2}} \tilde{\delta}' + \frac{11a^2}{8} \right] J_l^\circ \delta_p(\Omega) \quad (\text{VII.10e})$$

$$+ \text{Re} \left[-\frac{5a^2}{4} - \frac{iar_p}{4\sqrt{2}} \tilde{\delta}' \right] J_l^\circ \delta_p(\Omega), \\ + \text{Re} \left[-\frac{r_p^2}{4} \tilde{\delta} + \frac{3iar_p}{4\sqrt{2}} \right] \delta_p(\Omega) J_m^\circ \quad (\text{VII.10f})$$

$$a_{\ell m}^{\circ} = \Gamma_p \delta_{[\ell|2] \ 0} Y_{\ell}^p e^{-im\omega_p u}, \quad (\text{VII.11a})$$

$$b_{\ell m}^{\circ} = r_p \Gamma_p \delta_{[\ell|2] 0} Y_{\ell}^p e^{-im\omega_p u}, \quad (\text{VII.11b})$$

$$\begin{aligned} c_{\ell m}^{\circ} = & -\frac{\Gamma_p}{3\sqrt{2}r_p} (\lambda_{\ell,-1} \delta_{[\ell|2] 0} Y_{\ell}^p + am\omega_p \delta_{[\ell+1|2] 1} Y_{\ell}^p) e^{-im\omega_p u} \\ & + \frac{i}{3\sqrt{2}r_p^2} (2a\Gamma_p - 3r_p\Lambda_p) \delta_{[\ell+1|2] 1} Y_{\ell}^p e^{-im\omega_p u} \end{aligned} \quad (\text{VII.11c})$$

$$\begin{aligned} e_{\ell m}^{\circ} = & \frac{\Gamma_p r_p^2}{3\sqrt{2}} (\lambda_{\ell,-1} \delta_{[\ell|2] 0} Y_{\ell}^p + am\omega_p \delta_{[\ell+1|2] 1} Y_{\ell}^p) e^{-im\omega_p u} \\ & + \frac{ir_p}{6\sqrt{2}} (8a\Gamma_p - 3r_p\Lambda_p) \delta_{[\ell+1|2] 1} Y_{\ell}^p e^{-im\omega_p u}, \end{aligned} \quad (\text{VII.11d})$$

$$\begin{aligned} f_{\ell m}^{\circ} = & \left(\frac{15Mr_p}{64} - \frac{23a^2}{8} \right) J_l^{\circ} Y_{\ell}^p \delta_{[\ell|2]} e^{-im\omega_p u} + \frac{r_p^2}{4} J_{n0}^{\circ} Y_{\ell}^p e^{-im\omega_p u} \delta_{[\ell|2]} \\ & + \frac{iar_p}{4\sqrt{2}} J_m^{\circ} \delta_{[\ell|2] \ 0} Y_{\ell}^p e^{-im\omega_p u} \\ & + -\frac{3iar_p}{8\sqrt{2}} J^{\circ} \frac{1}{\sqrt{2}} (\lambda_{\ell,-1} \delta_{[\ell|2] 1} Y_{\ell}^p + am\omega_p \delta_{[\ell|2] 0} Y_{\ell}^p) e^{-im\omega_p u} \\ & - \left(-\frac{r_p^2}{4} J_m^{\circ} + \frac{5iar_p}{8\sqrt{2}} J_l^{\circ} \right) \frac{1}{\sqrt{2}} (\lambda_{\ell,-1} \delta_{[\ell+1|2] 0} Y_{\ell}^p + am\omega_p \delta_{[\ell|2] 0} Y_{\ell}^p) e^{-im\omega_p u} \\ & - \frac{r_p^2}{8} J_l^{\circ} \left[\frac{1}{2} (\lambda_{\ell,-1} \lambda_{\ell,-1} - a^2(m\omega_p)^2) \delta_{[\ell|2] 0} Y_{\ell}^p + \frac{am\omega_p}{2} \lambda_{\ell,-1} (-_1 Y_{\ell}^p + _1 Y_{\ell}^p) \delta_{[\ell+1|2]} \right] e^{-im\omega_p u} \end{aligned} \quad (\text{VII.11e})$$

where we've introduced the notation

$$\delta_{[k|p]} := \delta_{k \bmod p, 0}, \quad (\text{VII.11f})$$

the dimensionless quantities

$$\Gamma_p \equiv \frac{\mu\gamma_p}{4r_p} (1 - a\omega_p)^2, \quad \Lambda_p = \frac{\mu\gamma_p}{3r_p^2} (1 - a\omega_p) (a - (r_p^2 + a^2)\omega_p), \quad (\text{VII.12})$$

and the combinatorial factor [\[PZ: check this\]](#)

$${}_s Y_{\ell}^p = (-)^{s+\frac{\ell+s}{2}} \sqrt{\frac{2\ell+1}{4\pi}} \sqrt{\frac{(\ell-s)!}{(\ell+s)!}} \frac{(\ell+s-1)!!}{(\ell-s)!!}. \quad (\text{VII.13})$$

to get the mode projections for for $c_{\ell m}^{\circ}$ and $e_{\ell m}^{\circ}$ we've used the mode forms of the angular operators in KN coordinates

$$\tilde{\delta} \mapsto -{}_s \mathcal{L}_{\omega}^{\dagger} = \hat{\delta} + \frac{ia\omega \sin \theta}{\sqrt{2}}, \quad \tilde{\delta}' \mapsto {}_s \mathcal{L}_{\omega} = \bar{\bar{\delta}} - \frac{ia\omega \sin \theta}{\sqrt{2}}, \quad (\text{VII.14})$$

to arrive at

$$\int_{\mathbb{S}^2} {}_s\mathcal{L}_{\omega_p}^\dagger [\delta(z)\delta(\varphi - \omega_p u)] {}_s\bar{Y}_{\ell,m} d\Omega = \frac{\lambda_{\ell,s}}{\sqrt{2}\lambda_{\ell,s-1}} {}_{s-1}\bar{Y}_{\ell,m}^p(\frac{\pi}{2}, \omega_p u) + \frac{am\omega_p}{\sqrt{2}} {}_s\bar{Y}_{\ell,m}^p(\frac{\pi}{2}, \omega_p u) \quad (\text{VII.15})$$

and

$$\int_{\mathbb{S}^2} {}_s\mathcal{L}_{\omega_p} [\delta(z)\delta(\varphi - \omega_p u)] {}_s\bar{Y}_{\ell,m} d\Omega = -\frac{\lambda_{\ell,s}}{\sqrt{2}\lambda_{\ell,s+1}} {}_{s+1}\bar{Y}_{\ell,m}^p(\frac{\pi}{2}, \omega_p u) - \frac{am\omega_p}{\sqrt{2}} {}_s\bar{Y}_{\ell,m}^p(\frac{\pi}{2}, \omega_p u) \quad (\text{VII.16})$$

To obtain (VII.15) and (VII.16) we've used (C.60) and

$$\langle {}_sY_{\ell,m} | \hat{\delta}\delta_p(\Omega) J_l^\circ \rangle_{L^2} = \langle \hat{\delta}^\dagger {}_sY_{\ell,m} | \delta_p(\Omega) \rangle_{L^2} \quad (\text{VII.17})$$

$$= - \int_{\mathbb{S}^2} J_l^\circ \delta_p(\Omega) \overline{\hat{\delta}_s Y_{\ell,m}}, \quad (\text{VII.18})$$

$$= \frac{\lambda_{\ell,s}}{\sqrt{2}\lambda_{\ell,s-1}} J_l^\circ {}_{s-1}\bar{Y}_{\ell,m}^p \quad (\text{VII.19})$$

after using $\bar{\hat{\delta}}_s Y_{\ell,m} = -2^{-1/2} \lambda_{\ell,s}/\lambda_{\ell,s-1} \hat{\delta}_s Y_{\ell,m} = 2^{-1/2} \lambda_{\ell,s}/\lambda_{\ell,s+1}$.

a. Stationary sector To obtain the stationary pieces we required the time-averaged averaged angular delta function on an equatorial worldline.

Spin-weighted spherical harmonics satisfy the completeness relation

$${}_s\delta(\Omega, \Omega') = \sum_{\ell=|s|}^{\infty} \sum_{m=-\ell}^{\ell} {}_sY_{\ell m}(\Omega) {}_s\bar{Y}_{\ell m}(\Omega'), \quad \Omega = (\theta, \phi). \quad (\text{VII.20})$$

We evaluate the primed point on an equatorial circular worldline,

$$\Omega' = \left(\frac{\pi}{2}, \varphi' = \omega_p u\right). \quad (\text{VII.21})$$

Using the azimuthal dependence of spin-weighted harmonics,

$${}_s\bar{Y}_{\ell, m}(\theta, \phi) = {}_s\bar{Y}_{\ell, m}(\theta) e^{-im\phi}, \quad (\text{VII.22})$$

the delta function becomes for an equatorial circular orbit

$${}_s\delta(\Omega, \Omega_p(u)) = \sum_{\ell, m} {}_sY_{\ell, m}(\Omega) {}_s\bar{Y}_{\ell, m}\left(\frac{\pi}{2}, 0\right) e^{-im\omega_p u}, \quad (\text{VII.23})$$

$$= \sum_{\ell m} {}_sY_{\ell, m}(\Omega) {}_sY_{\ell}^p \delta_{[\ell+s|2]} e^{-im\omega_p u} \quad (\text{VII.24})$$

and also

$$\tilde{\delta}_s \delta(\Omega, \Omega_p) \mapsto -{}_s\mathcal{L}_{m\omega_p}^{\dagger}, \quad \tilde{\delta}'_s \delta(\Omega, \Omega_p) \mapsto {}_s\mathcal{L}_{m\omega_p} \quad (\text{VII.25})$$

under mode decomposition. Define an average over one orbital period,

$$\langle f(u) \rangle = \frac{\omega_p}{2\pi} \int_0^{2\pi/\omega_p} du f(u). \quad (\text{VII.26})$$

Then

$$\langle e^{-im\omega_p u} \rangle = \delta_{m0}, \quad (\text{VII.27})$$

so that only axisymmetric modes survive:

$$\langle {}_s\bar{Y}_{\ell m}\left(\frac{\pi}{2}, \omega_p u\right) \rangle = {}_s\bar{Y}_{\ell 0}\left(\frac{\pi}{2}, 0\right) \delta_{m0}. \quad (\text{VII.28})$$

For $m = 0$, spin-weighted harmonics at the equator are real and obey the selection rule

$${}_sY_{\ell 0}\left(\frac{\pi}{2}, 0\right) \neq 0 \iff \ell + s \text{ even}. \quad (\text{VII.29})$$

Putting everything together, the (ℓ, m) harmonic coefficient of the orbit-averaged angular delta source is

$$\boxed{\langle \delta(\Omega - \Omega_p) \rangle_{\ell m} = {}_s Y_{\ell 0}(\frac{\pi}{2}) \delta_{m0} \delta_{(\ell+s) \bmod 2, 0}} \quad (\text{VII.30})$$

With the useful identities

$${}_0 Y_{\ell 0}(\frac{\pi}{2}, 0) = \sqrt{\frac{2\ell+1}{4\pi}} P_\ell(0) \delta_{\ell \bmod 2, 0} \quad (\text{VII.31})$$

$$P_{2k}(0) = (-1)^k \frac{(2k)!}{2^{2k} (k!)^2}, \quad k = 0, 1, 2, \dots \quad (\text{VII.32})$$

we find that

$${}_s \bar{Y}_{\ell 0}(\frac{\pi}{2}, 0) = (-1)^s {}_s Y_\ell^p \delta_{(\ell+s) \bmod 2, 0} = (-1)^s {}_s Y_\ell^p \delta_{[\ell+s|2]} \quad (\text{VII.33})$$

where

$${}_s Y_\ell^p = (-1)^{s+\frac{\ell+s}{2}} \sqrt{\frac{2\ell+1}{4\pi}} \sqrt{\frac{(\ell-s)!}{(\ell+s)!} \frac{(\ell+s-1)!!}{(\ell-s)!!}} \quad (\text{VII.34})$$

The **stationary pieces** are thus

$$\langle a^\circ \rangle_{\ell m} = \Gamma_p \delta_{[\ell|2]0} {}_0 Y_\ell^p \delta_{m,0}, \quad (\text{VII.35a})$$

$$\langle b^\circ \rangle_{\ell m} = r_p \langle a^\circ \rangle_{\ell m} \quad (\text{VII.35b})$$

$$\begin{aligned} \langle c^\circ \rangle_{\ell m} = & -\frac{\Gamma_p}{3\sqrt{2}r_p} \lambda_{\ell,-1} \delta_{[\ell|2]0} {}_0 Y_\ell^p \delta_{m,0} \\ & + \frac{i}{3\sqrt{2}r_p^2} (2a\Gamma_p - 3r_p\Lambda_p) \delta_{[\ell+1|2]1} {}_1 Y_\ell^p \delta_{m,0}, \end{aligned} \quad (\text{VII.35c})$$

$$\begin{aligned} \langle e^\circ \rangle_{\ell m} = & \frac{\Gamma_p r_p^2}{3\sqrt{2}} \lambda_{\ell,-1} \delta_{[\ell|2]0} {}_0 Y_\ell^p \delta_{m,0} \\ & + \frac{i r_p}{6\sqrt{2}} (8a\Gamma_p - 3r_p\Lambda_p) \delta_{[\ell+1|2]1} {}_1 Y_\ell^p \delta_{m,0} \end{aligned} \quad (\text{VII.35d})$$

where we've used

$$\langle {}_s \delta(\Omega - \Omega_p) \rangle_{\ell m} = {}_s Y_\ell^p \delta_{m,0} \delta_{(\ell+s) \bmod 2, 0} \quad (\text{VII.36})$$

$$\langle \hat{\delta}_s \delta(\Omega - \Omega_p) \rangle_{\ell m} = 2^{-s/2} {}_s Y_\ell^p \lambda_{\ell,-s} \delta_{m,0} \delta_{(\ell+s) \bmod 2, 0} \quad (\text{VII.37})$$

$$(\text{VII.38})$$

b. oscillator sector The **oscillatory** corrector scalars are

$$a_\omega^\circ = \frac{\mu\gamma_p}{8\pi r_p} (1 - a\omega_p)^2 \delta_p(z) \sum_m \delta(\omega - m\omega_p) e^{im\varphi}, \quad (\text{VII.39a})$$

$$b_\omega^\circ = r_p a_\omega^\circ, \quad (\text{VII.39b})$$

$$(\text{VII.39c})$$

Or, in terms of ℓ, m, ω modes

$$a_{\ell m \omega}^{\circ} = \frac{\mu \gamma_p}{8\pi r_p} (1 - a\omega_p)^2 \delta(\omega - m\omega_p) Y_{\ell}^p, \quad (\text{VII.40a})$$

$$b_{\ell m \omega}^{\circ} = r_p a_{\ell m \omega}^{\circ}, \quad (\text{VII.40b})$$

$$c_{\ell m}^{\circ} = -\frac{\Gamma_p}{3\sqrt{2}r_p} (\lambda_{\ell, -1} \delta_{[\ell|2]0} Y_{\ell}^p + am\omega_p \delta_{[\ell+1|2]1} Y_{\ell}^p) e\delta(\omega - m\omega_p) \\ + \frac{i}{3\sqrt{2}r_p^2} (2a\Gamma_p - 3r_p\Lambda_p) \delta_{[\ell+1|2]1} Y_{\ell}^p \delta(\omega - m\omega_p) \quad (\text{VII.40c})$$

$$e_{\ell m}^{\circ} = \frac{\Gamma_p r_p^2}{3\sqrt{2}} (\lambda_{\ell, -1} \delta_{[\ell|2]0} Y_{\ell}^p + am\omega_p \delta_{[\ell+1|2]1} Y_{\ell}^p) \delta(\omega - m\omega_p) \\ + \frac{ir_p}{6\sqrt{2}} (8a\Gamma_p - 3r_p\Lambda_p) \delta_{[\ell+1|2]1} Y_{\ell}^p \delta(\omega - m\omega_p) \quad (\text{VII.40d})$$

VIII. TRANSFORMATION TO NS GAUGE

CCK-Ori metric reconstruction puts the perturbation in either of the half-string forms

$$h_{\alpha\beta}^{\text{half}, \pm} = 2 \text{Re} \mathcal{S}_{\alpha\beta}^{\dagger} \Phi^{\pm} = h_{ab}^{\pm} + h_{\alpha\beta}^{\text{S}} \Theta_{\text{p}}^{\pm}, \quad (\text{VIII.1})$$

where $\Phi^{\pm}$ are Hertz potentials which are regular at the horizon (+) and at infinity (-). Each reconstructed solution contains a half-string singularity. A no-string metric is obtained by gluing together the regular halves ($h_{\alpha\beta}^{\text{half}, +}$ from r_p to the horizon and $h_{\alpha\beta}^{\text{half}, -}$ from r_p to infinity) across the particle,

$$h_{\alpha\beta}^{\text{N}} = h_{\alpha\beta}^{+} \Theta_{\text{p}}^{+} + h_{\alpha\beta}^{-} \Theta_{\text{p}}^{-} + D_{\alpha\beta} \delta_p(r), \quad \Theta_{\text{p}}^{\pm} \equiv \Theta(\pm(r - r_p)). \quad (\text{VIII.2})$$

Equivalently, one may begin from a half-string solution and perform a gauge transformation

$$h_{\alpha\beta}^{\text{no-string}} = h_{\alpha\beta}^{\text{half}} + \mathcal{L}_{\xi} g_{\alpha\beta}, \quad (\text{VIII.3})$$

with a gauge vector of the form

$$\xi^{\alpha} = \xi_{+}^{\alpha} \Theta_{\text{p}}^{+} + \xi_{-}^{\alpha} \Theta_{\text{p}}^{-} \quad (\text{VIII.4})$$

where $\xi_{\pm}^a$ are smooth at r_p .

Because the background metric is smooth, the only source of delta functions in $\mathcal{L}_{\xi} g_{\alpha\beta}$ arises from differentiating the Heaviside functions:

$$\nabla_{\beta} \xi_{\alpha} = (\nabla_{\beta} \xi_{\alpha}^{+}) \Theta_{\text{p}}^{+} + (\nabla_{\beta} \xi_{\alpha}^{-}) \Theta_{\text{p}}^{-} + (\nabla_{\beta} r) [\xi_{\alpha}]_{\text{p}} \delta(r - r_p), \quad (\text{VIII.5})$$

where

$$[\xi_\alpha]_{\text{p}} \equiv \xi_\alpha^+ - \xi_\alpha^-. \quad (\text{VIII.6})$$

The Lie derivative

$$\mathcal{L}_\xi g_{\alpha\beta} = \nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha \quad (\text{VIII.7})$$

therefore contains a delta-function piece

$$D_{\alpha\beta} \delta(r - r_{\text{p}}), \quad D_{\alpha\beta} = 2 (\nabla_{(\alpha} r) [\xi_{\beta)}]_{\text{p}}. \quad (\text{VIII.8})$$

A. $D_{\alpha\beta}$ algorithm

To compute the no-string gauge coefficient $D_{\alpha\beta}$ we follow the algorithm in [6].

1. Compute ξ_α° from the *AC-DC-Killing* prescription below

$$\xi_\alpha^\circ = \langle \xi_\alpha^\circ \rangle + Z_\alpha^\circ + \Xi_\alpha^\circ \quad (\text{VIII.9})$$

2. Build ξ_α^+ from ξ_α° using the Price-Whiting residual radiation gauge equations

$$\xi_l = \xi_l^\circ, \quad (\text{VIII.10a})$$

$$\begin{aligned} \xi_n = \xi_n^\circ + \frac{1}{2} \left(\frac{1}{\rho} + \frac{1}{\bar{\rho}} \right) \tilde{\mathcal{P}}' \xi_l^\circ + \rho \bar{\rho} \tau^\circ \bar{\tau}^\circ \xi_l^\circ + \frac{1}{2} (\Psi^\circ \rho + \bar{\Psi}^\circ \bar{\rho}) \xi_l^\circ + \rho \bar{\tau}^\circ \Omega^\circ \xi_m^\circ \\ - \bar{\rho} \tau^\circ \Omega^\circ \xi_{\bar{m}}^\circ - \rho \bar{\tau}^\circ \tilde{\mathcal{D}} \xi_l^\circ - \bar{\rho} \tau^\circ \tilde{\mathcal{D}}' \xi_l^\circ, \end{aligned} \quad (\text{VIII.10b})$$

$$\xi_m = \frac{1}{\bar{\rho}} \xi_m^\circ + \bar{\rho} \tau^\circ \xi_l^\circ - \tilde{\mathcal{D}} \xi_l^\circ. \quad (\text{VIII.10c})$$

3. Evaluate the jump at the particle:

$$[\xi_\alpha] = \xi_\alpha^+(r_{\text{p}}) - \xi_\alpha^-(r_{\text{p}}).$$

4. Form $D_{\alpha\beta} = 2\nabla_{(\alpha} r [\xi_{\beta)}]_{\text{p}}$ and project into NP form $D_{(a)(b)} = e_{(a)}^\alpha e_{(b)}^\beta D_{\alpha\beta}$

No differentiation of distributions is required. Only components involving the leg normal to $r = \text{const}$ are nonzero.

1. “AC-CD-Killing” decomposition

In Ref. we introduced the “AC-DC plus Killing” decomposition of the field

$$\xi_\alpha^\circ = \xi_\alpha^{S,\circ} + \Xi_\alpha^\circ = \langle \xi_\alpha^\circ \rangle + Z_\alpha^\circ + \Xi_\alpha^\circ \quad (\text{VIII.11})$$

where

$$\Xi^\alpha \partial_\alpha = u(\alpha \partial_u + \beta \partial_\varphi) \quad (\text{VIII.12})$$

with

$$\begin{aligned} \Xi_l^\circ &= 2\Xi_n^\circ = u(\alpha - a\beta \sin^2 \theta), \\ \Xi_m^\circ &= i \frac{u\beta \sin \theta}{\sqrt{2}} \end{aligned} \quad (\text{VIII.13a})$$

and where

$$\langle \xi^\circ \rangle = \lim_{U \rightarrow \infty} \frac{1}{2U} \int_{-U}^U \xi^\circ du \quad (\text{VIII.13b})$$

and

$$Z^\circ = \int_{-\infty}^{\infty} Z_\omega^\circ e^{-i\omega u} d\omega \quad (\text{VIII.13c})$$

for the various frame components. Here, α and β are determined by the $\ell = 0$ and the $\ell = 1, m = 0$ modes of Eqs. (102a) and (102e) or Ref. []

$$\alpha = \sqrt{\frac{1}{4\pi}} \langle a_{00}^\circ \rangle + a\beta, \quad \beta = -i\sqrt{\frac{3}{4\pi}} \langle c_{10}^\circ \rangle. \quad (\text{VIII.14})$$

The (independent) transformation equations to shadowless gauge divide into groups. The first group involves only the corrector through a° and b° and is independent of $\hat{h}^S$:

$$\begin{aligned} a^\circ &= \tilde{\mathbf{P}}' \xi_l^\circ + \frac{1}{2} \tilde{\delta} \xi_m^\circ + \frac{1}{2} \tilde{\delta}' \xi_m^\circ \\ b^\circ &= -\frac{1}{2} \left(\tilde{\delta} \tilde{\delta}' - \rho'^\circ - \bar{\rho}'^\circ \right) \xi_l^\circ + \xi_n^\circ + \frac{1}{2} \Omega^\circ (\tilde{\delta}' \xi_m^\circ - \tilde{\delta} \xi_{\bar{m}}^\circ) - \bar{\tau}^\circ \xi_m^\circ - \tau^\circ \xi_{\bar{m}}^\circ \end{aligned} \quad (\text{VIII.15a})$$

The second group is independent of the corrector and depends only on the coefficients d_1° and d_2° of $\hat{h}^S$:

$$\begin{aligned} \bar{d}_2^\circ &= \tilde{\delta} \xi_m^\circ \\ \bar{d}_1^\circ &= \tilde{\delta} \tilde{\delta} \xi_l^\circ + 2\tau^\circ \xi_m^\circ \end{aligned} \quad (\text{VIII.16a})$$

The last group involve both x_{ab} and the metric

$$\tilde{\delta}' \bar{d}_3^\circ + c^\circ = \tilde{\mathbf{P}}' \xi_m^\circ, \quad (\text{VIII.17a})$$

$$E^\circ = \Psi^\circ \tilde{\delta} \xi_l^\circ - \Omega^\circ \Psi^\circ \xi_m^\circ + g_a^\circ, \quad (\text{VIII.17b})$$

where

$$E^\circ \equiv -\tilde{\delta}' \bar{d}_0^\circ - \Omega^\circ \tilde{\delta}' \bar{d}_1^\circ - \Omega^{\circ 2} \tilde{\delta}' \bar{d}_2^\circ - \Omega^{\circ 3} \tilde{\delta}' \bar{d}_3^\circ + e^\circ. \quad (\text{VIII.18})$$

The four scalars d_i° appear in the ‘‘Held form’’ of the string potential

$$\Phi^S = d_0^\circ + \frac{d_1^\circ}{\rho} + \frac{d_2^\circ}{\rho^2} + \frac{d_3^\circ}{\rho^3} \quad (\text{VIII.19})$$

are related to the scalars $\mathcal{B}_i^\circ$ above in Eq.(??) via

$$\bar{d}_0^\circ = \mathcal{B}_0^\circ - \frac{1}{8} \Omega^\circ (4\mathcal{B}_1^\circ - 2\mathcal{B}_2^\circ \Omega^\circ + \mathcal{B}_3^\circ \Omega^{\circ 2}), \quad (\text{VIII.20a})$$

$$\bar{d}_1^\circ = -\mathcal{B}_1^\circ + \mathcal{B}_2^\circ \Omega^\circ - \frac{3}{4} \mathcal{B}_3^\circ \Omega^{\circ 2}, \quad (\text{VIII.20b})$$

$$\bar{d}_2^\circ = \mathcal{B}_2^\circ - \frac{3}{2} \mathcal{B}_3^\circ \Omega^\circ, \quad (\text{VIII.20c})$$

$$\bar{d}_3^\circ = -\mathcal{B}_3^\circ. \quad (\text{VIII.20d})$$

B. Stationary sector

We start by solving the stationary sector:

$$\langle a^\circ \rangle = \alpha - \frac{3}{2} a \beta \sin^2 \theta + \frac{1}{2} \left(\hat{\delta} \langle \xi_m^\circ \rangle + \hat{\delta}' \langle \xi_m^\circ \rangle \right), \quad (\text{VIII.21a})$$

$$\begin{aligned} \langle b^\circ \rangle = & -\frac{1}{2} \left(\hat{\delta} \hat{\delta}' + \hat{\delta}' \hat{\delta} + 1 \right) \langle \xi_l^\circ \rangle + \langle \xi_n^\circ \rangle \\ & - i a \cos \theta \left(\hat{\delta}' \langle \xi_m^\circ \rangle - \hat{\delta} \langle \bar{\xi}_m^\circ \rangle \right) + \frac{i a \sin \theta}{\sqrt{2}} \left(\langle \bar{\xi}_m^\circ \rangle - \langle \xi_m^\circ \rangle \right), \end{aligned} \quad (\text{VIII.21b})$$

$$\langle \bar{d}_2^\circ \rangle = \frac{1}{2} a \beta \sin^2 \theta + \hat{\delta} \langle \xi_m^\circ \rangle, \quad (\text{VIII.21c})$$

$$\langle \bar{d}_1^\circ \rangle = 2i a^2 \beta \sin^2 \theta \cos \theta - \hat{\delta} \hat{\delta} \langle \xi_l^\circ \rangle - i \sqrt{2} a \sin \theta \langle \xi_m^\circ \rangle, \quad (\text{VIII.21d})$$

$$\hat{\delta}' \langle \bar{d}_3^\circ \rangle + \langle c^\circ \rangle = \frac{i \beta \sin \theta}{\sqrt{2}}, \quad (\text{VIII.21e})$$

$$\begin{aligned} \langle E^\circ \rangle = & -\frac{i M a \sin \theta}{\sqrt{2}} (\alpha - a \beta \sin^2 \theta - \dot{a}/a) + M \hat{\delta} \langle \xi_l^\circ \rangle \\ & + 2i M a \cos \theta \langle \xi_m^\circ \rangle, \end{aligned} \quad (\text{VIII.21f})$$

$$\langle F^\circ \rangle = 3M(\alpha - a \beta \sin^2 \theta) + \dot{M}. \quad (\text{VIII.21g})$$

where $E^\circ = e^\circ - \tilde{\delta}' d_0^\circ - \Omega^\circ \tilde{\delta}' d_1^\circ - \Omega^{\circ 2} \tilde{\delta}' d_2^\circ - \Omega^{\circ 3} \tilde{\delta}' d_3^\circ$.

The trigonometric functions appearing in the stationary gauge-transform equations (VIII.21) admit the following decompositions into spin-weighted spherical harmonics:

$$\sin \theta = -\sqrt{\frac{8\pi}{3}} {}_{-1}Y_{10} = \sqrt{\frac{8\pi}{3}} {}_1Y_{10}, \quad (\text{VIII.22a})$$

$$\sin^2 \theta = 4\sqrt{\frac{2\pi}{15}} {}_2Y_{20} = -\frac{1}{3} \left(4\sqrt{\frac{\pi}{5}} Y_{20} - 2\sqrt{4\pi} Y_{00} \right), \quad (\text{VIII.22b})$$

$$\cos \theta = \sqrt{\frac{4\pi}{3}} {}_0Y_{10}, \quad (\text{VIII.22c})$$

$$\sin^2 \theta \cos \theta = 4\sqrt{\frac{2\pi}{105}} {}_2Y_{30}, \quad (\text{VIII.22d})$$

$$\sin^3 \theta = \frac{8}{5}\sqrt{\frac{2\pi}{3}} {}_1Y_{10} - \frac{8}{5}\sqrt{\frac{\pi}{21}} {}_1Y_{30}. \quad (\text{VIII.22e})$$

These factors multiply angular fields expanded as

$$X(\theta, \phi) = \sum_{\ell' m'} a_{\ell' m'} {}_{s_2}Y_{\ell' m'}(\theta, \phi). \quad (\text{VIII.23})$$

Multiplication by a trigonometric factor decomposed into ${}_s Y_{\ell 0}$ induces a *nearest-neighbor coupling* among angular modes. Each input mode (ℓ', m) contributes only to output modes with $\ell = \ell' - 1, \ell', \ell' + 1$, with coefficients given by Eq. (??).

To re-expand the product in spin-weighted harmonics, we use the angular-momentum decomposition (see appendix C 3 a

$${}_s Y_{\ell_1 0} {}_{s_2} Y_{\ell_2 m} = \sum_j C_{\ell_1 \ell_2 j; s_1 s_2 s}^m {}_s Y_{j m}, \quad s = s_1 + s_2, \quad (\text{VIII.24})$$

with coefficient

$$C_{\ell_1 \ell_2 j; s_1 s_2 s}^m = (-1)^{s+m} \sqrt{\frac{(2j+1)(2\ell_1+1)(2\ell_2+1)}{4\pi}} \begin{pmatrix} \ell_1 & \ell_2 & j \\ -s_1 & -s_2 & s \end{pmatrix} \begin{pmatrix} \ell_1 & \ell_2 & j \\ 0 & m & -m \end{pmatrix}, \quad (\text{VIII.25})$$

where $s = s_1 + s_2$ with the following nearest neighbor algorithm.

Projection Algorithm

Suppose we wish to extract the coefficient of ${}_s Y_{\ell m}$ from a product of the form

$$\sum_{\ell' m'} a_{\ell' m'} {}_{s_1} Y_{\ell_1 0} {}_{s_2} Y_{\ell' m'}. \quad (\text{VIII.26})$$

a. *Insert product decomposition.* Using Eq. (VIII.24),

$${}_{s_1}Y_{\ell_1 0} {}_{s_2}Y_{\ell' m'} = \sum_J C_{\ell_1 \ell' J; s_1 s_2 s}^{m'} {}_sY_{J m'}. \quad (\text{VIII.27})$$

b. *Project onto ${}_sY_{\ell m}$.* Multiply by ${}_sY_{\ell m}^*$ and integrate over the sphere:

$$\int_{\mathbb{S}^2} d\Omega {}_sY_{\ell m}^* {}_sY_{J m'} = \delta_{\ell J} \delta_{m m'}. \quad (\text{VIII.28})$$

c. *Collapse sums.* The projection enforces

$$J = \ell, \quad m' = m,$$

leaving

$$b_{\ell m} = \sum_{\ell'} a_{\ell' m} C_{\ell_1 \ell' \ell; s_1 s_2 s}^m. \quad (\text{VIII.29})$$

d. *Determine allowed ℓ' .* From the triangle inequality in the $3j$ symbols,

$$|\ell_1 - \ell'| \leq \ell \leq \ell_1 + \ell', \quad (\text{VIII.30})$$

so only a finite number of ℓ' contribute. For $\ell_1 = 1$ (as in $\sin \theta$ or $\cos \theta$),

$$\ell' = \ell - 1, \ell, \ell + 1. \quad (\text{VIII.31})$$

Result:

$$\langle a^\circ \rangle_{\ell m} = \sqrt{4\pi}(\alpha - a\beta)\delta_{\ell,0}\delta_{m,0} - \frac{\lambda_{\ell,-1}}{2\sqrt{2}} \left(\langle \xi_m^\circ \rangle_{\ell m} + (-1)^m \langle \bar{\xi}_m^\circ \rangle_{\ell, -m} \right) + \sqrt{\frac{4\pi}{5}} a\beta \delta_{\ell,2}\delta_{m,0}, \quad \forall \ell \geq 0, \quad (\text{VIII.32a})$$

as well as

$$\langle \xi_m^\circ \rangle_{\ell m} = \sqrt{2} \frac{\lambda_{\ell,-1}}{\lambda_{\ell,-2}} \left((-1)^{-m} \langle \bar{d}_2^\circ \rangle_{\ell, -m} - 2\sqrt{\frac{2\pi}{15}} a\beta \delta_{\ell,2}\delta_{m,0} \right), \quad \ell \geq 2, \quad (\text{VIII.32b})$$

$$\langle \xi_l^\circ \rangle_{\ell m} = 2\lambda_{\ell,2} \left[(-1)^{m+1} \langle \bar{d}_1^\circ \rangle_{\ell, -m} + 8ia^2\beta \sqrt{\frac{2\pi}{105}} \delta_{\ell,3}\delta_{m,0} \right. \quad (\text{VIII.32c})$$

$$\left. - 4ia\sqrt{\frac{\pi}{3}} \delta_{\ell,>1} \sum_{j=\ell-1}^{\ell+1} C_{2j\ell;112}^m \langle \xi_m^\circ \rangle_{jm} \right], \quad \ell \geq 2, \quad (\text{VIII.32d})$$

$$\begin{aligned} \langle \xi_n^\circ \rangle_{\ell m} &= \langle b^\circ \rangle_{\ell m} + \frac{1}{2}(1 - \lambda_{\ell,-1}^2) \langle \xi_l^\circ \rangle_{\ell m} - 2ia\sqrt{\frac{\pi}{6}} \delta_{\ell,>0} \sum_{j=\ell-1}^{\ell+1} C_{1j\ell;000}^m \lambda_{j,-1} \langle \xi_m^\circ \rangle_{jm} \\ &\quad - 2ia\sqrt{\frac{\pi}{3}} \delta_{\ell,>1} \sum_{j=\ell-1}^{\ell+1} C_{1j\ell;-110}^m \langle \xi_m^\circ \rangle_{jm} + cc., \quad \forall \ell \geq 0, \end{aligned} \quad (\text{VIII.32e})$$

where

$$\delta_{n,>m} = \begin{cases} 1 & n > m \\ 0 & n \leq m. \end{cases} \quad (\text{VIII.33})$$

To solve the system of mode decomposed algebraic equations, we can start with (VIII.32b) for $\langle \xi_m^\circ \rangle$ for $\ell > 1$ and use it to determine $\langle \xi_\ell^\circ \rangle$ for $\ell > 1$ from . With these we can determine $\langle \xi_n^\circ \rangle$ for $\ell > 1$ with

C. Stationary sector growing-in- u terms and completion scalars α and β

The Held scalars α and β are determined by the $\ell = 0$ and the $\ell = 1, m = 0$ modes of Eqs. (102a) and (102e) or Ref. []

$$\alpha = \sqrt{\frac{1}{4\pi}} \langle a_{00}^\circ \rangle + a\beta, \quad \beta = -i\sqrt{\frac{3}{4\pi}} \langle c_{10}^\circ \rangle. \quad (\text{VIII.34})$$

a. Circular orbit. For a particle on a circular orbit we find that

$$\begin{aligned} \alpha &= \frac{\gamma_p}{16\pi r_p^3} (a\omega_p - 1) (a\omega_p(a^2 + r_p^2) - a^2 + r_p^2), \\ &= \frac{\mu (r_p^2 - a^2 + 2a\sqrt{Mr_p})}{16\pi(r_p^{3/2} + a\sqrt{M})\sqrt{2a\sqrt{M}r_p^{3/2} + r_p^2(r_p - 3M)}} \end{aligned} \quad (\text{VIII.35a})$$

$$\begin{aligned} \beta &= \frac{\gamma_p}{16\pi r_p^3} (a\omega_p - 1) (a^2\omega_p + 2r_p^2\omega_p - a), \\ &= -\frac{\mu (a - 2\sqrt{Mr_p})}{16\pi(r_p^{3/2} + a\sqrt{M})\sqrt{2a\sqrt{M}r_p^{3/2} + r_p^2(r_p - 3M)}}. \end{aligned} \quad (\text{VIII.35b})$$

b. Generic orbits. Recall that for a point particle on a generic orbit a° and c° are given by

$$a^\circ = -\frac{1}{8} \left(\frac{1}{\rho_p} + \frac{1}{\bar{\rho}_p} \right) J_l^\circ \delta_p^2(\Omega). \quad (\text{VIII.36})$$

and

$$\begin{aligned} c^\circ &= -\frac{1}{12} \frac{\bar{\rho}_p}{\rho_p} \tilde{\delta} (J_l^\circ \delta_p^2(\Omega)) + \frac{\bar{\rho}_p(\rho_p + \bar{\rho}_p)\bar{\tau}_p^\circ}{12\rho_p} J_l^\circ \delta_p^2(\Omega) \\ &\quad + \frac{1}{12} \frac{\bar{\rho}_p}{\rho_p} (3\rho_p + \bar{\rho}_p)\tau_p^\circ J_l^\circ \delta_p^2(\Omega) - \frac{1}{6} \left(1 + \frac{\bar{\rho}_p}{\rho_p} \right) J_m^\circ \delta_p^2(\Omega), \end{aligned} \quad (\text{VIII.37})$$

respectively. It is best not to expand the first term unless necessary. Distributionally, $\tilde{\delta} (J_l^\circ \delta_p^2(\Omega))$ means that the angular derivative acts on the full angular distribution.

Explicitly, the averages needed for a generic orbit are

$$\langle a_{00}^\circ \rangle = -\frac{\mu}{8\sqrt{4\pi}} \frac{1}{(2\pi)^2} \int_{\mathbb{T}^2} dq_r dq_z \left[\left(\frac{1}{\rho_p} + \frac{1}{\bar{\rho}_p} \right) \frac{(\dot{x}_p^\mu l_\mu)^2}{\Sigma_p^2 \dot{x}_p^u} \right]_{p(q_r, q_z)} \quad (\text{VIII.38})$$

and

$$\begin{aligned} \langle c_{10}^\circ \rangle = \frac{\mu}{(2\pi)^2} \int_{\mathbb{T}^2} dq_r dq_z & \left[\frac{1}{12} \frac{\bar{\rho}_p}{\rho_p} \left[\tilde{\delta}^\dagger \overline{{}_1Y_{10}} \right]_p \frac{(l_\alpha \dot{x}_p^\alpha)^2}{\Sigma_p^2 \dot{x}_p^u} \right. \\ & + \left(\frac{\bar{\rho}_p(\rho_p + \bar{\rho}_p) \bar{\tau}_p^\circ}{12\rho_p} + \frac{1}{12} \frac{\bar{\rho}_p}{\rho_p} (3\rho_p + \bar{\rho}_p) \tau_p^\circ \right) \overline{{}_1Y_{10}(\Omega_p)} \frac{(l_\alpha \dot{x}_p^\alpha)^2}{\Sigma_p^2 \dot{x}_p^u} \\ & \left. - \frac{1}{6} \left(1 + \frac{\bar{\rho}_p}{\rho_p} \right) \overline{{}_1Y_{10}(\Omega_p)} \frac{(l_\alpha \dot{x}_p^\alpha)(m_\beta \dot{x}_p^\beta)}{\Sigma_p^2 \dot{x}_p^u} \right]_{p(q_r, q_z)}. \end{aligned} \quad (\text{VIII.39})$$

where $\dot{x}_p^\alpha = \Sigma_p u^\alpha = \Sigma_p (dx_p^\alpha/d\tau)$ is Mino time four velocity and $\mathbb{T}^2 \cong \mathbb{S}^1 \times \mathbb{S}^1$ is a torus (since $(q_r, q_z) \sim (q_r + 2\pi, q_z + 2\pi)$).

D. Pushing the point source to a shell density

The action-angle source formulas give a mode-decomposed shell density on the (r, z) domain. The stress tensor mode is

$$T_{\mu\nu}^{(m, \mathbf{k})}(x) = \Theta(z_{\max} - |z|) [\Theta(r - r_p) - \Theta(r - r_a)] \mathcal{J}_{\mu\nu}^{(m, \mathbf{k})}(r, z). \quad (\text{VIII.40})$$

The shell density is obtained by pushing forward the local source to the radial-polar torus.

In terms of the local outgoing-coordinate amplitude $J_{\alpha\beta}^\circ$, the master formula is

$$\boxed{\mathcal{J}_{\alpha\beta}^{(m, \mathbf{k})}(r, z) = \frac{\Upsilon_r \Upsilon_z}{(2\pi)^3 \Upsilon_t} \sum_{\text{branches}} \frac{v_\star^u J_{\alpha\beta, \star}^\circ}{\Sigma_\star \dot{r}_\star \dot{z}_\star} \exp[i\vartheta_{\mathbf{k}, \star} - im\Delta_\vartheta \varphi_\star]}, \quad (\text{VIII.41})$$

where $J_{\alpha\beta}^\circ = \mu \dot{x}_\alpha \dot{x}_\beta / (\dot{z}^u \Sigma^2)$. Here, $x_\star$ means evaluate some quantity x at the inverse-image points of (r, z) for a specific turning-point branch on the orbital torus. The branch sum is over all solutions of

$$r(\psi_r^\star) = r, \quad z(\psi_z^\star) = z. \quad (\text{VIII.42})$$

Generically there are four branches,

$$(\dot{r}, \dot{z}) = (+, +), (+, -), (-, +), (-, -).$$

The factors $1/(\dot{r}_\star \dot{z}_\star)$ are the Jacobian factors from

$$\delta(r - r(\psi_r)) \delta(z - z(\psi_z)).$$

The branch structure is unchanged by the BL-to-outgoing-coordinate transformation, because r and z are the same in both coordinate systems.

The tetrad-projected shell amplitudes are then

$$\mathcal{J}_{ll}^{(m,\mathbf{k})} := l^\alpha l^\beta \mathcal{J}_{\alpha\beta}^{(m,\mathbf{k})}, \quad \mathcal{J}_{lm}^{(m,\mathbf{k})} := l^\alpha m^\beta \mathcal{J}_{\alpha\beta}^{(m,\mathbf{k})}. \quad (\text{VIII.43})$$

E. Stationary sector

The completion only depends on the stationary sector. Therefore we set

$$m = 0, \quad \mathbf{k} = \mathbf{0}.$$

The phase factor drops out, and

$$\boxed{\mathcal{J}_{\alpha\beta}^{(0,0)}(r, z) = \frac{\Upsilon_r \Upsilon_z}{(2\pi)^3 \Upsilon_t} \sum_{\text{branches}} \frac{v_\star^u J_{\alpha\beta,\star}^\circ}{\Sigma_\star \dot{r}_\star \dot{z}_\star}}. \quad (\text{VIII.44})$$

Equivalently, one may work directly with the torus average of the instantaneous point-particle coefficients.

For a Mino-time torus average,

$$\langle F \rangle_\lambda = \frac{1}{(2\pi)^2} \int_0^{2\pi} dq_r \int_0^{2\pi} dq_z F(q_r, q_z). \quad (\text{VIII.45})$$

For a coordinate-time average,

$$\langle F \rangle_t = \frac{1}{(2\pi)^2 \Upsilon_t} \int_0^{2\pi} dq_r \int_0^{2\pi} dq_z \frac{dt}{d\lambda}(q_r, q_z) F(q_r, q_z), \quad (\text{VIII.46})$$

where

$$\Upsilon_t = \left\langle \frac{dt}{d\lambda} \right\rangle_\lambda.$$

The normalization used in the mode source should be chosen consistently with this average. If the Fourier source convention already contains the factor Υ_t^{-1} , then one should not insert an additional time-average weight by hand.

F. Projection of a° onto $\ell = 0$

The monopole of a° is

$$\langle a_{00}^\circ \rangle = \int_{\mathbb{S}^2} d\Omega \overline{Y_{00}} \langle a^\circ \rangle. \quad (\text{VIII.47})$$

Using the angular delta distribution gives the evaluation formula

$$\langle a_{00}^\circ \rangle = -\frac{1}{8} \left\langle \left(\frac{1}{\rho_p} + \frac{1}{\bar{\rho}_p} \right) J_l^\circ \overline{Y_{00}(\Omega_p)} \right\rangle. \quad (\text{VIII.48})$$

Since $Y_{00} = 1/\sqrt{4\pi}$, this becomes

$$\langle a_{00}^\circ \rangle = -\frac{1}{8\sqrt{4\pi}} \left\langle \left(\frac{1}{\rho_p} + \frac{1}{\bar{\rho}_p} \right) J_l^\circ \right\rangle. \quad (\text{VIII.49})$$

For a generic bound orbit, we average over the radial-polar torus. Using action-angle variables (q_r, q_z) ,

$$\langle a_{00}^\circ \rangle = -\frac{1}{8\sqrt{4\pi}} \frac{1}{(2\pi)^2} \int_0^{2\pi} dq_r \int_0^{2\pi} dq_z \left[\left(\frac{1}{\rho_p} + \frac{1}{\bar{\rho}_p} \right) J_l^\circ \right]_{p(q_r, q_z)}. \quad (\text{VIII.50})$$

Substituting the definition of $J_{\alpha\beta}^\circ$,

$$\langle a_{00}^\circ \rangle = -\frac{\mu}{8\sqrt{4\pi}} \frac{1}{(2\pi)^2} \int dq_r dq_z \left[\left(\frac{1}{\rho_p} + \frac{1}{\bar{\rho}_p} \right) \frac{(l^\alpha u_\alpha)^2}{\Sigma_p u^u} \right]. \quad (\text{VIII.51})$$

Or in terms of Mino time,

$$\dot{x}_p^\alpha = \Sigma_p u_\alpha, \quad \dot{x}_p^u = \Sigma_p u^u, \quad (\text{VIII.52})$$

the expression becomes

$$\langle a_{00}^\circ \rangle = -\frac{\mu}{8\sqrt{4\pi}} \frac{1}{(2\pi)^2} \int dq_r dq_z \left[\left(\frac{1}{\rho_p} + \frac{1}{\bar{\rho}_p} \right) \frac{(\dot{x}_p^\mu l_\mu)^2}{\Sigma_p^2 \dot{x}_p^u} \right]. \quad (\text{VIII.53})$$

G. Projection of c° onto $\ell = 1, m = 0$

The dipole coefficient is

$$\langle c_{10}^\circ \rangle = \int d\Omega_1 \overline{Y_{10}} \langle c^\circ \rangle. \quad (\text{VIII.54})$$

For the derivative term, integrate by parts on the sphere:

$$\int d\Omega_1 \overline{Y_{\ell m}} \tilde{\partial} (J_l^\circ \delta_p^2) = - \left[\tilde{\partial}^\dagger_1 \overline{Y_{\ell m}} \right]_p J_l^\circ. \quad (\text{VIII.55})$$

Therefore the general projected coefficient can be written as

$$\begin{aligned} \langle c_{\ell m}^\circ \rangle = & \langle A_p \left[\tilde{\partial}^\dagger_1 \overline{Y_{\ell m}} \right]_p J_l^\circ \\ & + B_{p1} \overline{Y_{\ell m}(\Omega_p)} J_l^\circ \\ & + C_{p1} \overline{Y_{\ell m}(\Omega_p)} J_m^\circ \rangle, \end{aligned} \quad (\text{VIII.56})$$

with

$$A_p = \frac{1}{12} \frac{\bar{\rho}_p}{\rho_p}, \quad (\text{VIII.57})$$

$$B_p = \frac{\bar{\rho}_p(\rho_p + \bar{\rho}_p)\bar{\tau}_p^\circ}{12\rho_p} + \frac{1}{12} \frac{\bar{\rho}_p}{\rho_p} (3\rho_p + \bar{\rho}_p)\tau_p^\circ, \quad (\text{VIII.58})$$

and

$$C_p = -\frac{1}{6} \left(1 + \frac{\bar{\rho}_p}{\rho_p} \right). \quad (\text{VIII.59})$$

For the completion we only need $(\ell, m) = (1, 0)$:

$$\begin{aligned} \langle c_{10}^\circ \rangle = & \langle A_p \left[\tilde{\partial}_1^\dagger \overline{Y_{10}} \right]_p J_l^\circ \\ & + B_{p1} \overline{Y_{10}(\Omega_p)} J_l^\circ \\ & + C_{p1} \overline{Y_{10}(\Omega_p)} J_m^\circ \rangle. \end{aligned} \quad (\text{VIII.60})$$

Averaging over the generic radial-polar torus gives

$$\boxed{\langle c_{10}^\circ \rangle = \frac{1}{(2\pi)^2} \int_0^{2\pi} dq_r \int_0^{2\pi} dq_z \left[\frac{1}{12} \frac{\bar{\rho}_p}{\rho_p} \left[\tilde{\partial}_{s_c}^\dagger \overline{Y_{10}} \right]_p J_l^\circ + B_{p s_c} \overline{Y_{10}(\Omega_p)} J_l^\circ + C_{p s_c} \overline{Y_{10}(\Omega_p)} J_m^\circ \right]_{p(q_r, q_z)}}. \quad (\text{VIII.61})$$

Substituting

$$J_{\alpha\beta}^\circ = \mu \frac{u_\alpha u_\beta}{\Sigma_p u^u}, \quad (\text{VIII.62})$$

we have

$$J_l^\circ = \mu \frac{(l^\alpha u_\alpha)^2}{\Sigma_p u^u}, \quad J_m^\circ = \mu \frac{(l^\alpha u_\alpha)(m^\beta u_\beta)}{\Sigma_p u^u}. \quad (\text{VIII.63})$$

Hence

$$\begin{aligned} \langle c_{10}^\circ \rangle = & \frac{\mu}{(2\pi)^2} \int dq_r dq_z \left[\frac{1}{12} \frac{\bar{\rho}_p}{\rho_p} \left[\tilde{\partial}_{s_c}^\dagger \overline{Y_{10}} \right]_p \frac{(l^\alpha u_\alpha)^2}{\Sigma_p u^u} \right. \\ & + B_{p s_c} \overline{Y_{10}(\Omega_p)} \frac{(l^\alpha u_\alpha)^2}{\Sigma_p u^u} \\ & \left. + C_{p s_c} \overline{Y_{10}(\Omega_p)} \frac{(l^\alpha u_\alpha)(m^\beta u_\beta)}{\Sigma_p u^u} \right]. \end{aligned} \quad (\text{VIII.64})$$

Once the two averaged modes are known, the completion coefficients are

$$\beta = -i\sqrt{\frac{3}{4\pi}} \langle c_{10}^\circ \rangle, \quad \alpha = \sqrt{\frac{1}{4\pi}} \langle a_{00}^\circ \rangle + a\beta. \quad (\text{VIII.65})$$

H. Stationary sector: STF formulation

The stationary gauge-transform equations (VIII.21) contain only algebraic trigonometric factors multiplying angular fields. Rather than employing a spin-weighted spherical-harmonic decomposition, we use the equivalent symmetric trace-free (STF) Cartesian tensor representation of angular dependence.

The STF formulation is advantageous in that it makes manifest that only neighboring multipoles couple, and all angular structure is purely algebraic. Furthermore, annoying phase conventions that plague spherical harmonics aren't an issue with cartesian STF tensors.

Any scalar angular function admits an expansion

$$X(\hat{n}) = \sum_{\ell \geq 0} X_L \hat{n}^{\langle L \rangle}, \quad L = i_1 \cdots i_\ell, \quad (\text{VIII.66})$$

where $\hat{n}^i = x^i/r$ and $\langle L \rangle$ denotes STF projection. Spin-weighted fields are represented using the helicity basis $m_i^\pm$ and STF tensors,

$$X^{(\pm)}(\hat{n}) = \sum_{\ell \geq 1} X_L^{(\pm)} m_i^\pm \hat{n}^{\langle iL-1 \rangle}. \quad (\text{VIII.67})$$

STF representation of trigonometric factors

Using a spherical basis

$$\hat{n}^i = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta) \quad (\text{VIII.68})$$

and unit vector

$$\hat{z}^i = (0, 0, 1), \quad \hat{n}^i \hat{z}_i = \cos \theta, \quad (\text{VIII.69})$$

the elementary angular functions appearing in (VIII.21) correspond to low-rank $s = 0$ STF tensors:

$$\cos \theta = \hat{z}_i \hat{n}^i, \quad (\text{VIII.70})$$

$$\sin \theta = \sqrt{1 - (\hat{z}_i \hat{n}^i)^2}, \quad (\text{VIII.71})$$

$$\sin^2 \theta = \frac{2}{3} - \hat{z}_{\langle i} \hat{z}_{j \rangle} \hat{n}^{\langle ij \rangle}, \quad (\text{VIII.72})$$

$$\sin^2 \theta \cos \theta = -\frac{2}{5} \hat{z}_{\langle i} \hat{z}_j \hat{z}_{k \rangle} \hat{n}^{\langle ijk \rangle}, \quad (\text{VIII.73})$$

$$\sin^3 \theta = \frac{4}{5} \hat{z}_i \hat{n}^i - \frac{2}{5} \hat{z}_{\langle i} \hat{z}_j \hat{z}_{k \rangle} \hat{n}^{\langle ijk \rangle}. \quad (\text{VIII.74})$$

Thus each trigonometric factor corresponds to multiplication by either $\hat{n}^i$ or an STF tensor constructed from $\hat{z}^i$.

For $s \neq 0$ scalars ${}_sX(\hat{n})$ we use the helicity basis

$$\hat{m}_i^\pm = \frac{1}{\sqrt{2}}(\hat{\theta}_i \pm i\hat{\varphi}_i) \quad (\text{VIII.75})$$

and create spin $s = \pm 1$ weighted building blocks via

$$\hat{m}_i^+ \hat{n}^{\langle iL-1 \rangle}, \quad \hat{m}_i^- \hat{n}^{\langle iL-1 \rangle} \quad (\text{VIII.76})$$

and the rule

$${}_sX(\hat{n}) = \sum_{\ell \geq |s|} X_L^{(s)} m_{i_1}^{\sigma_1} \dots m_{i_s}^{\sigma_s} \hat{n}^{\langle i_1 \dots i_s L-s \rangle} \quad (\text{VIII.77})$$

with $\sigma_k = \text{sgn}(s)$.

STF product identity

Products of angular factors with STF multipoles satisfy can be built by repeated application of the reduction formula

$$(\ell) \otimes (1) = (\ell - 1) \oplus (\ell) \oplus (\ell + 1) \quad (\text{VIII.78})$$

which yields

$$\hat{n}^i \hat{n}^{\langle L \rangle} = \text{STF}_{iL}(\hat{n}^i \hat{n}^{\langle L \rangle}) + \frac{\ell}{2\ell + 1} \delta^{i\langle i_\ell} \hat{n}^{L-1 \rangle} \quad (\text{VIII.79})$$

$$= (\hat{n}^{\langle iL \rangle}) + \left(\frac{\ell + 1}{2\ell + 1} \hat{n}^{\langle L-1 \rangle} \right) + \left(\frac{\ell}{2\ell + 1} \delta^{i\langle i_\ell} \hat{n}^{L-1 \rangle} \right), \quad (\text{VIII.80})$$

which shows explicitly that only

$$\ell \rightarrow \ell + 1, \quad \ell, \quad \ell - 1$$

couplings occur.

This is the STF analogue of angular-momentum addition.

Projection rule

Orthogonality of STF tensors gives

$$\int_{\mathbb{S}^2} d\Omega \hat{n}^{\langle L \rangle} \hat{n}^{\langle L' \rangle} = \frac{4\pi}{2\ell + 1} \delta_{\ell\ell'} \delta_{LL'}^L. \quad (\text{VIII.81})$$

Therefore, extracting the ℓ -th multipole reduces to contracting with $\hat{n}^{\langle L \rangle}$ and using algebraic identities—no angular integrals or $3j$ symbols appear.

Nearest-neighbor algorithm (STF)

Given

$$X(\hat{n}) = \sum_{\ell'} X_{L'\hat{n}}^{\langle L' \rangle}, \quad (\text{VIII.82})$$

and a factor proportional to $\hat{n}^i$ or $\hat{z}_{\langle L_1 \rangle} \hat{n}^{\langle L_1 \rangle}$:

1. Multiply using Eq. (VIII.80).

2. Rewrite result as STF tensors of rank $\ell \pm 1, \ell$.

3. Contract with $\hat{n}^{(L)}$ to obtain the coefficient of multipole ℓ .

This procedure generates the same nearest-neighbor structure that in the harmonic approach appears through Clebsch–Gordan coefficients.

Resulting stationary system

I. Stationary sector: STF form of NP components

The stationary Newman–Penrose components therefore admit expansions [7, 8]

$$\langle a^\circ \rangle = \sum_{\ell \geq 0} a_L^\circ \hat{n}^{\langle L \rangle}, \quad (\text{VIII.83a})$$

$$\langle b^\circ \rangle = \sum_{\ell \geq 0} b_L^\circ \hat{n}^{\langle L \rangle}, \quad (\text{VIII.83b})$$

$$\langle c^\circ \rangle = \sum_{\ell \geq 1} c_L^\circ m_i^+ \hat{n}^{\langle iL-1 \rangle}, \quad (\text{VIII.83c})$$

$$\langle d_0^\circ \rangle = \sum_{\ell \geq 2} d_{0,L}^\circ m_i^- m_j^- \hat{n}^{\langle ijL-2 \rangle}, \quad (\text{VIII.83d})$$

$$\langle d_1^\circ \rangle = \sum_{\ell \geq 1} d_{0,L}^\circ m_i^- \hat{n}^{\langle iL-2 \rangle}, \quad (\text{VIII.83e})$$

$$\langle d_2^\circ \rangle = \sum_{\ell \geq 0} d_{0,L}^\circ \hat{n}^{\langle L \rangle}, \quad (\text{VIII.83f})$$

$$\langle d_3^\circ \rangle = \sum_{\ell \geq 1} d_{0,L}^\circ m_i^+ \hat{n}^{\langle iL-1 \rangle}, \quad (\text{VIII.83g})$$

$$\langle \xi_l^\circ \rangle = \sum_{\ell \geq 0} \xi_{l,L}^\circ \hat{n}^{\langle L \rangle}, \quad (\text{VIII.83h})$$

$$\langle \xi_n^\circ \rangle = \sum_{\ell \geq 0} \xi_{n,L}^\circ \hat{n}^{\langle L \rangle}, \quad (\text{VIII.83i})$$

$$\langle \xi_m^\circ \rangle = \sum_{\ell \geq 1} \xi_{m,L}^\circ m_i^+ \hat{n}^{\langle iL-1 \rangle}. \quad (\text{VIII.83j})$$

STF stationary-mode equations

The STF multipoles satisfy the algebraic relations [PZ: Must check: just guessed at these]

$$A_L = (\alpha - a\beta) \delta_{\ell 0} - \frac{1}{2\sqrt{2}} (\Xi_L^{(m)} + \bar{\Xi}_L^{(m)}) + \frac{a\beta}{\sqrt{5}} \delta_{\ell 2} Q_{\langle L \rangle}, \quad \ell \geq 0, \quad (\text{VIII.84a})$$

$$\Xi_L^{(m)} = \sqrt{2} \frac{\lambda_{\ell,-1}}{\lambda_{\ell,-2}} \left(\bar{d}_{2,L} - \frac{2a\beta}{\sqrt{15}} \delta_{\ell 2} Q_{\langle L \rangle} \right), \quad \ell \geq 2, \quad (\text{VIII.84b})$$

$$\Xi_L^{(l)} = 2\lambda_{\ell,2} \left[-\bar{d}_{1,L} + 8ia^2\beta\delta_{\ell 3} Z_{\langle L \rangle} - 4ia\sqrt{\frac{\pi}{3}} \sum_{\ell'=\ell\pm 1} \mathcal{E}_{\ell\ell'} \Xi_{L'}^{(m)} \right], \quad \ell \geq 2, \quad (\text{VIII.84c})$$

$$\Xi_L^{(n)} = b_L^\circ + \frac{1}{2}(1 - \lambda_{\ell,-1}^2)\Xi_L^{(l)} - 2ia \sum_{\ell'=\ell\pm 1} \mathcal{F}_{\ell\ell'} \lambda_{\ell',-1} \xi_{m,L'}^\circ, \quad \ell \geq 0. \quad (\text{VIII.84d})$$

Here $Z_{\langle L \rangle}$ denotes the STF tensor built from products of the unit vector $\hat{z}^i$, and the nearest-neighbor coupling coefficients are purely rational:

$$\mathcal{E}_{\ell,\ell-1} = \frac{\ell+2}{2\ell+1}, \quad \mathcal{E}_{\ell,\ell+1} = \frac{\ell-1}{2\ell+1}, \quad (\text{VIII.85})$$

$$\mathcal{F}_{\ell,\ell-1} = \frac{\ell}{2\ell+1}, \quad \mathcal{F}_{\ell,\ell+1} = \frac{\ell+1}{2\ell+1} \quad (\text{VIII.86})$$

and $Q_{\langle L \rangle} = \hat{z}_i \hat{z}_j - \frac{1}{3}\delta_{ij}$.

J. Oscillatory sector

We now give expression for the gauge vector in the oscillatory sector

$$Z_a^\circ = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{i\omega u} (Z_a^\circ)_\omega \quad (\text{VIII.87})$$

in terms of its ℓm modes

$$\begin{aligned} (Z_m^\circ)_{\omega\ell m} &= \frac{i}{\omega} c_{\omega\ell m}^\circ + (-1)^{m+1} \frac{i\lambda_{l,-2}}{\sqrt{2}\lambda_{l,-1}\omega} (\bar{d}_{3,-\omega}^\circ)_{\ell,-m} \\ &\quad + \frac{i}{\sqrt{2}} \sqrt{\frac{4\pi}{3}} \sum_{j=\ell-1}^{\ell+1} C_{1j\ell;-121}^m (\bar{d}_{3,-\omega}^\circ)_{\ell,-m} \quad \forall \ell \geq 1 \end{aligned} \quad (\text{VIII.88a})$$

$$\begin{aligned} (Z_l^\circ)_{\omega\ell m} &= \frac{i}{\omega} a_{\omega\ell m}^\circ + \frac{i\lambda_{l,-1}}{2\sqrt{2}\omega} ((Z_m^\circ)_{\omega\ell m} + (-1)^m (\bar{Z}_m^\circ)_{-\omega,\ell,-m}) \\ &\quad + \frac{ia}{2} \sqrt{\frac{4\pi}{3}} \sum_j [C_{1j\ell;-110}^m (Z_m^\circ)_{\omega\ell m} + (-1)^m C_{1j\ell;1-10}^m (\bar{Z}_m^\circ)_{-\omega,\ell,-m}], \quad \forall \ell \geq 0, \end{aligned} \quad (\text{VIII.88b})$$

$$\begin{aligned}
(Z_n^\circ)_{\omega\ell m} &= b_{\omega\ell m}^\circ - \frac{1}{2}(\lambda_{l,-1}^2 - 1)(Z_l^\circ)_{\omega\ell m} + \frac{a^2\omega^2}{3} \sum_j \left(2\sqrt{\frac{\pi}{5}} C_{2,j;l;0,0}^m - \sqrt{4\pi} C_{0jl;000}^m \right) (Z_l^\circ)_{\omega jm} \\
&\quad - \frac{a\omega}{\sqrt{2}} \sqrt{\frac{4\pi}{3}} \sum_j C_{1j\ell;000}^m (Z_l^\circ)_{\omega jm} \\
&\quad - ia \sqrt{\frac{2\pi}{3}} \lambda_{\ell,-1} \sum_j C_{1j\ell;000}^m \left((Z_m^\circ)_{\omega\ell m} + (-1)^m (\bar{Z}_m^\circ)_{-\omega,\ell,-m} \right) \\
&\quad + 2ia^2\omega \sqrt{\frac{\pi}{15}} \sum_j \left(C_{2j\ell;-110}^m (Z_m^\circ)_{\omega\ell m} + (-1)^m C_{2j\ell;1-10}^m (\bar{Z}_m^\circ)_{\omega,\ell,-m} \right) \quad \forall \ell \geq 0
\end{aligned} \tag{VIII.88c}$$

K. Oscillatory sector: STF formulation

The oscillatory gauge vector is written as a Fourier integral

$$Z_a^\circ(u, \hat{n}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{i\omega u} Z_{a,\omega}^\circ(\hat{n}). \tag{VIII.89}$$

All angular dependence is expanded in STF tensors,

$$Z_{l,\omega}^\circ(\hat{n}) = \sum_{\ell \geq 0} Z_{\omega,L}^{(l)} \hat{n}^{\langle L \rangle}, \tag{VIII.90}$$

$$Z_{n,\omega}^\circ(\hat{n}) = \sum_{\ell \geq 0} Z_{\omega,L}^{(n)} \hat{n}^{\langle L \rangle}, \tag{VIII.91}$$

$$Z_{m,\omega}^\circ(\hat{n}) = \sum_{\ell \geq 1} Z_{\omega,L}^{(m)} m_i^+ \hat{n}^{\langle iL-1 \rangle}. \tag{VIII.92}$$

STF oscillatory-mode equations

The STF multipoles satisfy [PZ: just guessed at these]

$$Z_{\omega,L}^{(m)} = \frac{i}{\omega} C_{\omega,L} + \frac{i}{\sqrt{2}\omega} \frac{\lambda_{\ell,-2}}{\lambda_{\ell,-1}} \bar{D}_{\omega,L}^{(3)} + i \sqrt{\frac{2\pi}{3}} \sum_{\ell'=\ell\pm 1} \mathcal{F}_{\ell\ell'} \bar{D}_{\omega,L'}^{(3)}, \quad \ell \geq 1, \tag{VIII.93a}$$

$$\begin{aligned}
Z_{\omega,L}^{(l)} &= \frac{i}{\omega} A_{\omega,L} + \frac{i}{2\sqrt{2}\omega} \left(Z_{\omega,L}^{(m)} + \bar{Z}_{-\omega,L}^{(m)} \right) \\
&\quad + \frac{ia}{2} \sum_{\ell'=\ell\pm 1} \mathcal{F}_{\ell\ell'} \left(Z_{\omega,L'}^{(m)} + \bar{Z}_{-\omega,L'}^{(m)} \right), \quad \ell \geq 0,
\end{aligned} \tag{VIII.93b}$$

$$\begin{aligned}
Z_{\omega,L}^{(n)} &= B_{\omega,L} - \frac{1}{2}(\lambda_{\ell,-1}^2 - 1)Z_{\omega,L}^{(l)} - \frac{a\omega}{\sqrt{2}} \sum_{\ell'=\ell\pm 1} \mathcal{F}_{\ell\ell'} Z_{\omega,L'}^{(l)} \\
&\quad - ia \lambda_{\ell,-1} \sum_{\ell'=\ell\pm 1} \mathcal{F}_{\ell\ell'} \left(Z_{\omega,L'}^{(m)} + \bar{Z}_{-\omega,L'}^{(m)} \right) \\
&\quad + 2ia^2\omega \sum_{\ell'=\ell\pm 1} \mathcal{E}_{\ell\ell'} \left(Z_{\omega,L'}^{(m)} + \bar{Z}_{-\omega,L'}^{(m)} \right), \quad \ell \geq 0.
\end{aligned} \tag{VIII.93c}$$

Here the rational STF coupling coefficients are

$$\mathcal{F}_{\ell,\ell-1} = \frac{\ell}{2\ell+1}, \quad \mathcal{F}_{\ell,\ell+1} = \frac{\ell+1}{2\ell+1}, \tag{VIII.94}$$

$$\mathcal{E}_{\ell,\ell-1} = \frac{\ell+2}{2\ell+1}, \quad \mathcal{E}_{\ell,\ell+1} = \frac{\ell-1}{2\ell+1}. \tag{VIII.95}$$

a. Remarks. All angular couplings in the oscillatory sector are nearest-neighbor in ℓ and involve only rational functions of ℓ . The STF formulation removes the need for Clebsch–Gordan or Wigner-3j symbols and makes the structure of the system manifest.

IX. ALGORITHM FOR RECONSTRUCTING ξ_α° IN SCHWARZSCHILD

We work in the Schwarzschild specialization of Appendix F of Ref. [?], where the gauge vector is decomposed into a part growing linearly in u , a stationary (DC) part, and an oscillatory (AC) part. In the notation of Eqs. (F.15) and (96), one writes

$$\xi_\alpha^\circ = \Xi_\alpha^\circ + \langle \xi_\alpha^\circ \rangle + Z_\alpha^\circ, \tag{IX.1}$$

where Ξ_α° is linear in u , $\langle \xi_\alpha^\circ \rangle$ is the stationary/DC part, and Z_α° is the AC part. The growing part takes the form

$$\Xi^\alpha = u (\alpha t^\alpha + \beta z^\alpha + \gamma y^\alpha + \delta x^\alpha), \tag{IX.2}$$

with t^α the time-translation Killing vector and $x^\alpha, y^\alpha, z^\alpha$ the rotational Killing vectors. In spin-weighted harmonic form, the corresponding Held coefficients are given in Eqs. (F.17) and (F.18).

a. Step 1: determine the growing part Ξ_a° . From the $\ell = 0$ mode of Eq. (F.19a) and the $\ell = 1$ mode of Eq. (F.19e), one obtains

$$\alpha = \frac{1}{\sqrt{4\pi}} \langle a_{00}^\circ \rangle, \quad (\text{IX.3})$$

$$\beta = -i\sqrt{\frac{3}{4\pi}} \langle c_{10}^\circ \rangle, \quad (\text{IX.4})$$

$$\gamma = \sqrt{\frac{3}{8\pi}} (\langle c_{11}^\circ \rangle + \langle c_{1,-1}^\circ \rangle), \quad (\text{IX.5})$$

$$\delta = i\sqrt{\frac{3}{8\pi}} (\langle c_{11}^\circ \rangle - \langle c_{1,-1}^\circ \rangle). \quad (\text{IX.6})$$

These are exactly Eq. (F.20). Hence the u -growing low multipoles are fixed directly from a° and c° .

b. Step 2: reconstruct the stationary/DC modes for $\ell \geq 2$. For the stationary sector, Eqs. (F.19b)–(F.19d) give a direct inversion for $\langle \xi_l^\circ \rangle$, $\langle \xi_m^\circ \rangle$, and $\langle \xi_n^\circ \rangle$ when $\ell \geq 2$:

$$\langle \bar{d}_{1,\ell,-m}^\circ \rangle = (-1)^{m+1} \frac{1}{2} \lambda_{\ell,-2} \langle \xi_{l,\ell m}^\circ \rangle, \quad \ell \geq 2, \quad (\text{IX.7})$$

$$\langle \bar{d}_{2,\ell,-m}^\circ \rangle = (-1)^m \frac{\lambda_{\ell,-2}}{\sqrt{2} \lambda_{\ell,-1}} \langle \xi_{m,\ell m}^\circ \rangle, \quad \ell \geq 2, \quad (\text{IX.8})$$

$$\langle b_{\ell m}^\circ \rangle = \frac{1}{2} (\lambda_{\ell,-1}^2 - 1) \langle \xi_{l,\ell m}^\circ \rangle + \langle \xi_{n,\ell m}^\circ \rangle, \quad \ell \geq 0. \quad (\text{IX.9})$$

Therefore, for each $\ell \geq 2$,

$$\langle \xi_{l,\ell m}^\circ \rangle = (-1)^{m+1} \frac{2}{\lambda_{\ell,-2}} \langle \bar{d}_{1,\ell,-m}^\circ \rangle, \quad (\text{IX.10})$$

$$\langle \xi_{m,\ell m}^\circ \rangle = (-1)^m \frac{\sqrt{2} \lambda_{\ell,-1}}{\lambda_{\ell,-2}} \langle \bar{d}_{2,\ell,-m}^\circ \rangle, \quad (\text{IX.11})$$

$$\langle \xi_{n,\ell m}^\circ \rangle = \langle b_{\ell m}^\circ \rangle - \frac{1}{2} (\lambda_{\ell,-1}^2 - 1) \langle \xi_{l,\ell m}^\circ \rangle. \quad (\text{IX.12})$$

This is precisely the inversion procedure summarized in the sentence below Eq. (F.20): first solve for $\langle \xi_l^\circ \rangle$ and $\langle \xi_m^\circ \rangle$ from (F.19d) and (F.19c), then obtain $\langle \xi_n^\circ \rangle$ from (F.19b).

c. Step 3: identify the undetermined stationary low modes. The direct inversion above does *not* determine all stationary low multipoles.

Monopole ($\ell = 0$). From Eq. (F.17a), the Schwarzschild time-translation Killing vector has

$$t_l^\circ = 2t_n^\circ = \sqrt{4\pi} {}_0Y_{00}, \quad t_m^\circ = 0. \quad (\text{IX.13})$$

Thus a stationary addition proportional to t^a shifts only the $\ell = 0$ pieces of ξ_l° and ξ_n° . Eq. (F.19b) at $\ell = 0$ fixes only the combination

$$\langle b_{00}^\circ \rangle = -\frac{1}{2} \langle \xi_{l,00}^\circ \rangle + \langle \xi_{n,00}^\circ \rangle, \quad (\text{IX.14})$$

since $\lambda_{0,-1} = 0$. Hence the stationary monopole contains one residual Killing degree of freedom.

Dipole ($\ell = 1$). From Eqs. (F.17b)–(F.17d), the rotational Killing vectors x^a, y^a, z^a carry only $\ell = 1$ spin-1 content in ξ_m° and no $\ell = 1$ content in ξ_l° or ξ_n° . Since Eqs. (F.19c) and (F.19d) start only at $\ell \geq 2$, the stationary $\ell = 1$ sector is likewise not fixed by the direct inversion formulas; it corresponds to residual rotational Killing freedom.

d. Step 4: fix the residual stationary Killing freedom by convention. To obtain a unique algorithmic reconstruction, one must impose a convention that removes the stationary Killing modes.

A simple choice is:

$$\langle \xi_{l,00}^\circ \rangle_{\text{stat}} = 0, \quad (\text{IX.15})$$

$$\langle \xi_{m,1m}^\circ \rangle_{\text{stat}} = 0 \quad (m = -1, 0, 1). \quad (\text{IX.16})$$

With this choice, the remaining stationary monopole is fixed by

$$\langle \xi_{n,00}^\circ \rangle_{\text{stat}} = \langle b_{00}^\circ \rangle. \quad (\text{IX.17})$$

Any other choice differs only by the addition of a stationary Schwarzschild Killing vector. This is consistent with the structure exhibited in Eqs. (F.15)–(F.18) and with the range of validity of Eqs. (F.19c)–(F.19d).

e. Step 5: reconstruct the oscillatory/AC modes. For nonzero frequency $\omega \neq 0$, the AC modes are determined from Eqs. (F.21a)–(F.21c):

$$Z_{m,\omega\ell m}^\circ = \frac{i}{\omega} c_{\omega\ell m}^\circ + (-1)^{m+1} \frac{i\lambda_{\ell,-2}}{\sqrt{2}\lambda_{\ell,-1}\omega} \bar{d}_{3,-\omega,\ell,-m}^\circ, \quad \ell \geq 1, \quad (\text{IX.18})$$

$$Z_{l,\omega\ell m}^\circ = \frac{i}{\omega} a_{\omega\ell m}^\circ + \frac{i\lambda_{\ell,-1}}{2\sqrt{2}\omega} (Z_{m,\omega\ell m}^\circ + (-1)^m \bar{Z}_{m,-\omega,\ell,-m}^\circ), \quad \ell \geq 0, \quad (\text{IX.19})$$

$$Z_{n,\omega\ell m}^\circ = b_{\omega\ell m}^\circ - \frac{1}{2} (\lambda_{\ell,-1}^2 - 1) Z_{l,\omega\ell m}^\circ, \quad \ell \geq 0. \quad (\text{IX.20})$$

Thus the AC sector is reconstructed mode by mode with no residual low- ℓ ambiguity once $\omega \neq 0$.

f. Final algorithm. The complete Schwarzschild reconstruction of ξ_a° is therefore:

1. Compute the growing coefficients $\alpha, \beta, \gamma, \delta$ from Eq. (F.20), and form $\Xi_a^\circ = u(\alpha t_a + \beta z_a + \gamma y_a + \delta x_a)$.

2. For each stationary mode with $\ell \geq 2$, recover $\langle \xi_\ell^\circ \rangle$ from (F.19d), $\langle \xi_m^\circ \rangle$ from (F.19c), and then $\langle \xi_n^\circ \rangle$ from (F.19b).
3. Fix the residual stationary Killing freedom in the $\ell = 0$ and $\ell = 1$ sectors by an explicit convention, e.g. $\langle \xi_{l,00}^\circ \rangle_{\text{stat}} = 0$ and $\langle \xi_{m,1m}^\circ \rangle_{\text{stat}} = 0$.
4. Reconstruct the AC modes from Eqs. (F.21a)–(F.21c).
5. Sum the three sectors:

$$\xi_a^\circ = \Xi_a^\circ + \langle \xi_a^\circ \rangle + Z_a^\circ. \quad (\text{IX.21})$$

With this understanding, the equations on pp. 61–62 determine ξ_a° uniquely only after quotienting out, or explicitly fixing, the residual Schwarzschild Killing vectors carried by the stationary $\ell = 0, 1$ modes.

Appendix A: Metric and Coordinates

As is standard, we parametrize the Kerr black hole by M for mass and by $J = aM$ for spin angular momentum. We work in outgoing KN coordinates $x^\mu = (u, r, z, \varphi)$, which are related to BL coordinates by

$$u = t - r_* = t - \int \frac{r^2 + a^2}{\Delta} dr, \quad \varphi = \phi - \phi_*(r) = \phi - \int \frac{a}{\Delta} dr, \quad z = \cos \theta. \quad (\text{A.1})$$

The coordinate r is an affine parameter along the outgoing PNC. The coordinate u is a good time coordinate on $\mathcal{H}^-$ and $\mathcal{J}^+$ and labels the geodesics of the outgoing congruence. In these coordinates, the Kerr metric takes the form

$$\begin{aligned} ds^2 = & \left(1 - \frac{2Mr}{\Sigma}\right) du^2 + 2dudr - \frac{\Sigma}{1 - z^2} dz^2 + \frac{4Mar(1 - z^2)}{\Sigma} dud\varphi - 2a(1 - z^2) dr d\varphi \\ & - \frac{1}{\Sigma} ((r^2 + a^2)^2 - a^2 \Delta (1 - z^2)) (1 - z^2) d\varphi^2, \end{aligned} \quad (\text{A.2})$$

where

$$\Delta = r^2 + a^2 - 2Mr \quad (\text{A.3})$$

$$\Sigma = r^2 + a^2 z^2. \quad (\text{A.4})$$

We make heavy use of the Newman-Penrose (NP) formalism. Unless stated otherwise, we work exclusively in the outgoing aligned Kinnersley tetrad $e_{(a)}^\alpha = (\ell^\alpha, n^\alpha, m^\alpha, \bar{m}^\alpha)$

$$l^\alpha = (0, 1, 0, 0), \quad (\text{A.5a})$$

$$n^\alpha = (r^2 + a^2, -\Delta/2, 0, a) / \Sigma, \quad (\text{A.5b})$$

$$m^\alpha = \frac{1}{\sqrt{2}(r +iaz)} \left(ia\sqrt{1-z^2}, 0, -\sqrt{1-z^2}, i/\sqrt{1-z^2} \right). \quad (\text{A.5c})$$

In this tetrad l^α is aligned with the outgoing PNC and n^α is aligned with the ingoing leg and our sign convention is

$$l \cdot n = 1, \quad m \cdot \bar{m} = -1$$

.

The tetrad (A.5a) is “natural basis” in that it takes advantage of the Goldberg-Sachs thm having $\kappa = \kappa' = \sigma = \sigma' = 0$ and $\epsilon = 0$. The remaining GHP coefficients are given by

$$\begin{aligned} \rho &= -1/(r -iaz), & \rho' &= -\Delta\rho/(2\Sigma), \\ \tau &= \frac{ia\sqrt{1-z^2}}{\sqrt{2}\Sigma}, & \tau' &= \frac{ia\rho^2\sqrt{1-z^2}}{\sqrt{2}}, \end{aligned} \quad (\text{A.6})$$

and the Held coefficients (see Appendix C) are

$$\tau^\circ = -\frac{ia\sqrt{1-z^2}}{\sqrt{2}}, \quad (\text{A.7a})$$

$$\rho'^\circ = -1/2, \quad (\text{A.7b})$$

$$\Psi_2^\circ = M \quad (\text{A.7c})$$

$$\Omega^\circ = -2iaz. \quad (\text{A.7d})$$

Furthermore

$$\rho + \bar{\rho} = -2r\rho\bar{\rho}, \quad \rho - \bar{\rho} = -2iaz\rho\bar{\rho} = \Omega^\circ\rho\bar{\rho} \quad (\text{A.8})$$

and

$$r = -\Sigma \operatorname{Re}(\rho) \quad z = \frac{i\Omega^\circ}{2a} = -\frac{\Sigma}{a} \operatorname{Im}(\rho). \quad (\text{A.9})$$

In the outer region $r > r_+$ $\operatorname{Re}(\rho) < 0$ and $\operatorname{Im} \rho$ is bounded in the range $-a/\Sigma < \operatorname{Im}(\rho) < a/\Sigma$. It follows from (C.16) and the definition of arctan that

$$\arg \rho = \tan^{-1}(\operatorname{Im}(\rho)/\operatorname{Re}(\rho)) = \tan^{-1}(az/r) = -\frac{i}{2} \log(\rho/\bar{\rho}). \quad (\text{A.10})$$

Appendix B: Stereographic coordinates on S^2

For calculations at fixed r, u , especially near $\mathcal{I}^+$ where the BL coordinate spheres are asymptotic to the round sphere, it is often useful for to introduce stereographic coordinates $x^A = (\zeta, \bar{\zeta})$ mapping the north pole to ∞ , the equator to the unit circle, and the south pole to 0

$$\zeta = e^{i\varphi} \cot \frac{\theta}{2}. \quad (\text{B.1})$$

Note that

$$\theta = 2 \cot^{-1} |\zeta|, \quad \varphi = \arg \zeta = \frac{1}{2i} \log(\zeta/\bar{\zeta}) \quad (\text{B.2})$$

and

$$\cos \theta = -\frac{1 - \zeta\bar{\zeta}}{1 + \zeta\bar{\zeta}}, \quad \sin \theta = \frac{2\sqrt{\zeta\bar{\zeta}}}{1 + \zeta\bar{\zeta}}, \quad \csc^2 \frac{\theta}{2} = 1 + \zeta\bar{\zeta}. \quad (\text{B.3})$$

The spacelike 2-surfaces of constant u and r near $\mathcal{I}^+$ are nearly spherical and asymptote to standard $\mathbb{S}^2$ surfaces $S_{u,r}$ spanned by the canonical complex null vectors $\hat{m}^\mu = 2^{-1/2}(0, 0, 1, i \csc \theta)$ and $\bar{\hat{m}}^\mu$, which form the projection operators $Z_\alpha^\beta = -\hat{m}_\alpha \bar{\hat{m}}^\beta - \bar{\hat{m}}_\alpha \hat{m}^\beta$ onto $S_{u,v}$ with area element $i\hat{m} \wedge \bar{\hat{m}}$ useful for integration in e.g. the Bondi mass

$$M_B = -i\frac{1}{2} \oint_{\mathbb{S}^2} (\Psi_2 + \sigma\dot{\sigma}) \hat{m} \wedge \bar{\hat{m}} \quad (\text{B.4})$$

In terms of the stereographic representation of the coordinates x^A , the NP tetrad legs spanning unit 2-sphere are

$$\hat{m}^A \partial_A = \frac{1}{\sqrt{2}} (\partial_\theta + i \csc \theta \partial_\varphi) = e^{i\varphi} P \partial_\zeta, \quad (\text{B.5})$$

$$\bar{\hat{m}}^A \partial_A = \frac{1}{\sqrt{2}} (\partial_\theta - i \csc \theta \partial_\varphi) = e^{-i\varphi} P \partial_{\bar{\zeta}}. \quad (\text{B.6})$$

We can bring the m^A into canonical form $m^A \partial_A = P \partial_{\bar{\zeta}}$ by rotating the tetrad by an angle $-\varphi$ (See J. Math. Phys. 7, 863 (1966) for why this is). Then

$$\hat{m}_A dx^A = -P^{-1} d\zeta, \quad \bar{\hat{m}}_A dx^A = -P^{-1} d\bar{\zeta}, \quad (\text{B.7})$$

and the round sphere metric becomes

$$\begin{aligned} d\Omega_2^2 &= -2\hat{m}_{(A} \bar{\hat{m}}_{B)} dx^A dx^B, \\ &= - (d\theta^2 + \sin^2 \theta d\varphi^2) = -2P^{-2}(\zeta, \bar{\zeta}) d\zeta d\bar{\zeta}, \end{aligned} \quad (\text{B.8})$$

$$P(\zeta, \bar{\zeta}) = \frac{1 + \zeta\bar{\zeta}}{\sqrt{2}} \quad (\text{B.9})$$

such that the edth operators on $S_{u,r}$ are

$$\hat{\delta}\eta = P^{1-s} \frac{\partial}{\partial \zeta} (P^s \eta), \quad \bar{\delta} = P^{1+s} \frac{\partial}{\partial \bar{\zeta}} (P^{-s} \eta). \quad (\text{B.10})$$

Note that we choose ζ as our holomorphic coordinate corresponding to $P = \hat{\delta}\zeta$ with “inward” normal.

In Kerr we have that

$$\hat{m}^A = \frac{P}{\Gamma} \partial_\zeta, \quad \bar{\hat{m}}^A = \frac{P}{\bar{\Gamma}} \partial_{\bar{\zeta}} \quad (\text{B.11})$$

where $\Gamma = r + ia \cos(\theta(\zeta, \bar{\zeta}))$.

In terms of the stereographic coordinates the $\hat{\delta}$ operator has a simple Green’s function

$$K(\zeta, \bar{\zeta}, \eta, \bar{\eta}) = \frac{1}{4\pi} \frac{1 + \eta\bar{\eta}}{\bar{\zeta} - \bar{\eta}} \quad (\text{B.12})$$

of spin weight $s = 0$ in $\zeta, \bar{\zeta}$ and spin weight $s = -1$ in $\eta, \bar{\eta}$ such that

$$F(\zeta, \bar{\zeta}) = \int_{\mathbb{S}^2} K(\zeta, \bar{\zeta}, \eta, \bar{\eta}) A(\eta, \bar{\eta}) d\mu_\eta \quad (\text{B.13})$$

is a solution of

$$\hat{\delta}_\zeta F(\zeta, \bar{\zeta}) = A(\zeta, \bar{\zeta}) \quad (\text{B.14})$$

where A is spin $s = 1$ scalar. Here the Möbius invariant volume element is

$$d\mu_\eta = \frac{2d\eta \wedge d\bar{\eta}}{i(1 + \eta\bar{\eta})^2} \quad (\text{B.15})$$

To prove this, we take a test function $g(\zeta, \bar{\zeta})$ with spin weight $s = -1$ and compact support

$$\begin{aligned} -\frac{1}{2\pi i} \int_{\mathbb{S}^2} K(\zeta, \bar{\zeta}, \eta, \bar{\eta}) \hat{\delta} g(\zeta, \bar{\zeta}) d\mu_\zeta &= -\frac{1}{2\pi i} \int_{\mathbb{S}^2} \frac{1 + \eta\bar{\eta}}{\bar{\zeta} - \bar{\eta}} (1 + \zeta\bar{\zeta})^2 \left[\frac{g(\zeta, \bar{\zeta})}{1 + \zeta\bar{\zeta}} \right]_{,\zeta} d\mu_\zeta, \\ &= -\frac{1}{2\pi i} \int_{\mathbb{S}^2} \frac{1 + \eta\bar{\eta}}{\bar{\zeta} - \bar{\eta}} \left[\frac{g(\zeta, \bar{\zeta})}{1 + \zeta\bar{\zeta}} \right]_{,\zeta} d\zeta \wedge d\bar{\zeta}, \\ &= \frac{i}{2\pi} \int_{\mathbb{S}^2} \left[\frac{1 + \eta\bar{\eta}}{\bar{\zeta} - \bar{\eta}} \frac{g(\zeta, \bar{\zeta})}{1 + \zeta\bar{\zeta}} \right]_{,\zeta} d\zeta \wedge d\bar{\zeta}, \\ &= \frac{i}{2\pi} \oint_C \frac{1 + \eta\bar{\eta}}{\bar{\zeta} - \bar{\eta}} \frac{g(\zeta, \bar{\zeta})}{1 + \zeta\bar{\zeta}} d\bar{\zeta}, \\ &= g(\eta, \bar{\eta}), \end{aligned} \quad (\text{B.16})$$

where we have used the residue formula with a countour C encircling $\bar{\zeta} = \bar{\eta}$. This proves that

$$\int_{\mathbb{S}^2} \hat{\delta} K(\zeta, \bar{\zeta}, \eta, \bar{\eta}) g(\zeta, \bar{\zeta}) d\mu_\zeta = g(\eta, \bar{\eta}), \quad (\text{B.17})$$

which demonstrates the K is a Green's function for $\hat{\delta}$.

Note that $\hat{\delta}$ is a non-self-adjoint operator $\hat{\delta}^\dagger = -\hat{\delta}$ and has a non-trivial Kernel so that the solution can only be defined modulo a zero-mode. In particular one can add a homogeneous solution $h = (1 + \zeta\bar{\zeta})^{-s}$ or an $\ell = 0$ harmonic in its kernel.

Another useful kernel M inverts the Laplacian acting on spin-zero functions

$$M = \frac{1}{4\pi} \log |\zeta - \eta|^2. \quad (\text{B.18})$$

Appendix C: GHP and Held Formalisms

We utilize the GHP formalism [9], in which the tetrad legs are assigned a weight $\{p, q\}$ corresponding to how they transform under the group of spins and boosts. When expressing the weight of a GHP tensor f we use the notation $f \doteq \{p, q\}$. The weights of the tetrad legs are

$$\begin{aligned} l &\doteq \{1, 1\}, & n &\doteq \{-1, -1\}, \\ m &\doteq \{1, -1\}, & \bar{m} &\doteq \{-1, 1\}. \end{aligned} \quad (\text{C.1})$$

The GHP formalism includes priming $f' \doteq \{-p, -q\}$ and complex conjugation $\bar{f} \doteq \{q, p\}$ transformations, reducing the written work by half. In type D spacetimes, the GHP vacuum equations may be reduced to

$$\begin{aligned} \mathbb{P}\rho &= \rho^2, & \bar{\partial}\tau &= \tau^2, & \mathbb{P}\tau &= (\tau - \bar{\tau}')\rho, \\ \bar{\partial}\rho &= (\rho - \bar{\rho})\tau, & \mathbb{P}'\rho &= \bar{\partial}'\tau + \rho\bar{\rho}' - \tau\bar{\tau} - \Psi_2 \end{aligned} \quad (\text{C.2})$$

and their primes. The GHP coefficients have weights $\rho \doteq \{1, 1\}$ and $\tau \doteq \{1, -1\}$. The operators appearing in the above equations are the GHP derivative operators

$$\begin{aligned} \mathbb{P} &= l^a \nabla_a - p\epsilon - q\bar{\epsilon}, & \mathbb{P}' &= n^a \nabla_a + p\epsilon' + q\bar{\epsilon}' \\ \bar{\partial} &= m^a \nabla_a - p\beta + q\bar{\beta}', & \bar{\partial}' &= \bar{m}^a \nabla_a + p\beta' - q\bar{\beta} \end{aligned} \quad (\text{C.3})$$

with the obvious weights given by (C.1). The GHP equation are supplemented by the general type-D identities (zero NUT-charge)

$$\frac{\tau}{\bar{\tau}'} = \frac{\tau'}{\bar{\tau}} = -\frac{\rho}{\bar{\rho}} = -\frac{\rho'}{\bar{\rho}'} = -\frac{\Psi_2^{1/3}}{\bar{\Psi}_2^{1/3}}, \quad \mathbb{P}\tau' = 2\rho\tau', \quad \bar{\partial}'\rho = 2\rho\tau' \quad (\text{C.4})$$

which follow from the existence of a Killing tensor, the commutation relations, and the Bianchi identities. The commutation relations also give rise to the identities

$$\mathbf{P}'\rho = \rho\rho' + \tau'(\tau - \bar{\tau}') - \frac{1}{2}\Psi_2 - \frac{\rho}{2\bar{\rho}}\bar{\Psi}_2, \quad (\text{C.5})$$

$$\delta'\tau = \tau\tau' + \rho(\rho' - \bar{\rho}') + \frac{1}{2}\Psi_2 - \frac{\rho}{2\bar{\rho}}\bar{\Psi}_2. \quad (\text{C.6})$$

Note that also $\delta\tau' = \delta'\tau$ and $\mathbf{P}\rho' = \mathbf{P}'\rho$.

In our analysis we utilize the so-called ‘‘Held-formalism’’ [10, 11], a reduction of the GHP formalism that is particularly adapted to case II type-D metrics such as Kerr. The formalism involves a new set of operators

$$\tilde{\mathbf{P}}' = \mathbf{P}' - \bar{\tau}\tilde{\delta} - \tau\tilde{\delta}' + \tau\bar{\tau}\left(\frac{q}{\bar{\rho}} + \frac{p}{\rho}\right) + \frac{1}{2}\left(\frac{q\bar{\Psi}_2}{\bar{\rho}} + \frac{p\Psi_2}{\rho}\right) \doteq (-1, -1), \quad (\text{C.7a})$$

$$\tilde{\delta} = \frac{1}{\bar{\rho}}\delta + \frac{q\tau}{\rho} \doteq (0, -2), \quad (\text{C.7b})$$

$$\tilde{\delta}' = \frac{1}{\rho}\delta' + \frac{p\bar{\tau}}{\bar{\rho}} \doteq (-2, 0), \quad (\text{C.7c})$$

which are good operators at future null infinity and the past horizon. Note that $\bar{\tilde{\delta}} = \tilde{\delta}'$ and $\bar{\tilde{\mathbf{P}}}' = \tilde{\mathbf{P}}'$, however $(\tilde{\delta})' \neq \tilde{\delta}'$. If a Held scalar f° is annihilated by $\mathbf{P}$, then so is its Held derivative. By virtue of the above identities (C.4) we have [10]

$$[\mathbf{P}, \tilde{\delta}]f^\circ = 0, \quad (\text{C.8})$$

for any $f \doteq \{p, q\}$. However, $[\mathbf{P}, \tilde{\mathbf{P}}'] \neq 0$ (see Held [10] for details). In addition to the Held derivatives we have the following Held coefficients

$$\rho' = \rho'^\circ\bar{\rho} - \frac{1}{2}\Psi_2^\circ\rho^2 - (\tilde{\delta}\bar{\tau}^\circ + \frac{1}{2}\Psi_2^\circ)\rho\bar{\rho} - \tau^\circ\bar{\tau}^\circ\rho^2\bar{\rho}, \quad (\text{C.9a})$$

$$\tau' = -\bar{\tau}^\circ\rho^2, \quad (\text{C.9b})$$

$$\tau = \tau^\circ\rho\bar{\rho}, \quad (\text{C.9c})$$

$$\Psi_2 = \Psi_2^\circ\rho^3, \quad (\text{C.9d})$$

with $\tau^\circ \doteq (-1, -3)$, $\Psi_2^\circ \doteq (-3, -3)$, $\rho'^\circ \doteq (-2, -2)$, together with

$$\Omega^\circ = \frac{1}{\bar{\rho}} - \frac{1}{\rho} \quad (\text{C.10})$$

	$\tilde{\mathbf{P}}'$	$\tilde{\delta}$	$\tilde{\delta}'$
ρ	$\rho^2 \rho'^{\circ} - \frac{1}{2} \rho^2 (\rho \Psi^{\circ} + \bar{\rho} \bar{\Psi}_2^{\circ}) - \rho^3 \bar{\rho} \tau^{\circ} \bar{\tau}^{\circ}$	$\tau^{\circ} \rho^2$	$-\bar{\tau}^{\circ} \rho^2$
Ω°	0	$2\tau^{\circ}$	$-2\bar{\tau}^{\circ}$
ρ'°	0	0	0
τ°	0	0	$\frac{1}{2}(\rho'^{\circ} + \bar{\rho}'^{\circ})\Omega^{\circ} + \frac{1}{2}(\Psi^{\circ} - \bar{\Psi}^{\circ})$
Ψ°	0	0	0

TABLE I. Action of Held's operators on GHP background quantities in type D spacetimes of Kinnersley's Case II, in a tetrad aligned with the principal null directions [10, 11].

where $\Omega^{\circ} \doteq (-1, -1)$. Note that

$$\tilde{\delta}\rho = 0 \quad (C.11)$$

$$\tilde{\delta}\bar{\Omega}^{\circ} = -\tau^{\circ} \quad (C.12)$$

$$\tilde{\delta}\bar{\rho} = \bar{\rho}^2 \tilde{\delta}\bar{\Omega}^{\circ} = -\bar{\rho}^2 \tau^{\circ} \quad (C.13)$$

$$\tilde{\delta}\Omega^{\circ} = 2\tau^{\circ}. \quad (C.14)$$

In the Kinnersley tetrad the Held scalars reduce to

$$\tau^{\circ} = -\frac{ia \sin \theta}{\sqrt{2}}, \quad \rho'^{\circ} = -1/2, \quad \Psi^{\circ} = M, \quad \Omega^{\circ} = -2ia \cos \theta. \quad (C.15a)$$

Furthermore, in a Kinnersley frame

$$\rho + \bar{\rho} = -2r\rho\bar{\rho}, \quad \rho - \bar{\rho} = -2iaz\rho\bar{\rho} = \Omega^{\circ}\rho\bar{\rho} \quad (C.16)$$

and

$$r = -\Sigma \operatorname{Re}(\rho) \quad z = \frac{i\Omega^{\circ}}{2a} = -\frac{\Sigma}{a} \operatorname{Im}(\rho). \quad (C.17)$$

In the outer region $r > r_+$ $\operatorname{Re}(\rho) < 0$ and $\operatorname{Im} \rho$ is bounded in the range $-a/\Sigma < \operatorname{Im}(\rho) < a/\Sigma$.

When acting on Held scalars, the Held operators in outgoing Kerr-Newman coordinates and Kinnersley's frame give

$$\begin{aligned} \tilde{\delta} &= -\frac{1}{\sqrt{2}} \left(\partial_{\theta} + i \csc \theta \partial_{\varphi} + ia \sin \theta \partial_u - \frac{1}{2}(p - q) \cot \theta \right), \\ \tilde{\delta}' &= -\frac{1}{\sqrt{2}} \left(\partial_{\theta} - i \csc \theta \partial_{\varphi} - ia \sin \theta \partial_u + \frac{1}{2}(p - q) \cot \theta \right), \\ \tilde{\mathbf{P}}' &= \partial_u, \end{aligned} \quad (C.18)$$

On ℓ, m, ω -modes, the Held operators above related to the Chandrasekhar spin-raising ${}_s\mathcal{L}_\omega^\dagger$ and lowering ${}_s\mathcal{L}_\omega$ by

$$\tilde{\delta} = {}_s\mathcal{L}_\omega^\dagger, \quad \tilde{\delta}' = {}_s\mathcal{L}_\omega \quad (\text{C.19})$$

1. Teukolsky Equation and distribution character of source T_0

a. Teukolsky equation as a charged scalar. The Teukolsky operators have a particularly nice form in terms of Θ_a and a null vector

$$B^\alpha = -\rho n^\alpha + \tau \bar{m}^\alpha \doteq \{0, 0\} \quad (\text{C.20})$$

given by

$${}_sO\psi_s = \left[g^{\alpha\beta}(\Theta_\alpha + 2sB_\alpha)(\Theta_\beta + 2sB_\beta) - 4s^2\Psi^2 \right] \psi_s \quad (\text{C.21})$$

which has the structure of a Klein-Gordon equation for a massive charged in an external potential.

In particular, on any scalar $\eta_s \doteq \{2s, 0\}$ we can define a “gauge covariant” derivative

$$\begin{aligned} D_\alpha \eta_s &\equiv (\Theta_\alpha + 2sB_\alpha)\eta_s \\ &= (\nabla_\alpha + 2s(B_\alpha - \beta_\alpha))\eta_s \\ &= (\nabla_\alpha + iqA_\alpha)\eta_s \end{aligned} \quad (\text{C.22})$$

with $q \equiv -is$ and complex potential $A_\alpha \equiv 2[(-\epsilon - \rho)n_\alpha + \epsilon' l_\alpha - \beta' m_\alpha + (\beta + \tau)\bar{m}_\alpha]$.

b. Teukolsky $\{4, 0\}$ equation. In particular the Teukolsky operators in the equation for ψ_0 in Held form read

$$\begin{aligned} \frac{1}{2}\mathcal{O} &= \mathbb{P}\tilde{\mathbb{P}}' + \rho\bar{\rho}(\bar{\tau}^\circ\mathbb{P}\tilde{\delta} + \tau^\circ\mathbb{P}\tilde{\delta}') - (4\rho + \bar{\rho})\tilde{\mathbb{P}}' - \rho\bar{\rho}\tilde{\delta}\tilde{\delta}' \\ &\quad - \frac{1}{2}(\Psi^\circ\rho(3\rho + \bar{\rho}) + 2\bar{\rho}\rho'^\circ + \rho\bar{\rho}(-\bar{\Psi}^\circ + 6\rho\tau^\circ\bar{\tau}^\circ + \Omega^\circ(\rho'^\circ + \bar{\rho}^\circ)))\mathbb{P} + \frac{1}{2}\rho\bar{\rho}(3\Psi^\circ\rho + 8\rho'^\circ), \end{aligned} \quad (\text{C.23})$$

and

$$\begin{aligned} T_0 &= -\mathbb{P}\mathbb{P}T_{mm} + \bar{\rho}\mathbb{P}\tilde{\delta}T_{lm} + \bar{\rho}\tilde{\delta}\mathbb{P}T_{lm} - \bar{\rho}^2\tilde{\delta}\tilde{\delta}T_{ll} + 2\bar{\rho}(-2\rho + \bar{\rho})\tau^\circ\mathbb{P}T_{lm} - 4\rho\mathbb{P}T_{mm} \\ &\quad + 4\bar{\rho}^2(\bar{\rho} - \rho)\tau^\circ\tilde{\delta}T_{ll} - 2\bar{\rho}(2\rho + \bar{\rho})\tilde{\delta}T_{lm} + \bar{\rho}[2(2\rho + \bar{\rho})T_{mm} + 4\bar{\rho}^2\tau^\circ(T_{lm} + \tau^\circ(\rho - \bar{\rho})T_{ll})]. \end{aligned} \quad (\text{C.24})$$

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⁷ In the last line we have used that in a Kinnersley frame

$$\bar{\tau}^\circ = -\tau^\circ, \quad \bar{\Omega}^\circ = -\Omega^\circ, \quad \bar{\rho}'^\circ = \rho'^\circ, \quad \bar{\Psi}^\circ = \Psi^\circ \quad (\text{C.25})$$

c. *Teukolsky $\{4, 0\}$ equation in retarded outgoing chart.* In retarded coordinates we can write the Teukolsky operator (C.23) in retarded-Held-Kinnersley form⁸ as

$$\hat{A}\partial_r^2\psi_0 + 2\partial_u\partial_r\psi_0 + \hat{B}\partial_r\psi_0 + \hat{C}\psi_0 = T_0, \quad (\text{C.26})$$

where the operators appearing in (C.26) are given by

$$\hat{A} = -\Delta\rho\bar{\rho}, \quad (\text{C.27a})$$

$$\begin{aligned} \hat{B} = & -(\Psi^\circ\rho(3\rho + \bar{\rho}) + 2\bar{\rho}\rho'^\circ + \rho\bar{\rho}(-\bar{\Psi}^\circ + 6\rho\tau^\circ\bar{\tau}^\circ + \Omega^\circ(\rho'^\circ + \bar{\rho}'^\circ))) \\ & + 2\rho\bar{\rho}(\bar{\tau}^\circ\tilde{\delta} + \tau^\circ\tilde{\delta}'), \end{aligned} \quad (\text{C.27b})$$

$$\hat{C} = \partial_r(\Delta\rho\bar{\rho}) - (4\rho + \bar{\rho})\partial_u + \frac{1}{2}\rho\bar{\rho}(3\Psi^\circ\rho + 8\rho'^\circ) - \rho\bar{\rho}\tilde{\delta}\tilde{\delta}'. \quad (\text{C.27c})$$

Note that $\hat{C}$ has no r -derivatives.

d. *Point source and distributional identities.* Using the distributional identities for derivatives of the $\delta_p(r)$ multiplied by a test function $F(\rho(r))$ and noting that $\partial_r F(\rho) = \rho^2 F'(\rho)$

$$F(\rho)\delta_p(r) = F(\rho_p)\delta_p(r) \quad (\text{C.28})$$

$$F(\rho)\delta'_p(r) = F(\rho_p)\delta'_p(r) - \rho^2 F'(\rho)\delta_p(r) \quad (\text{C.29})$$

$$F(\rho)\delta''_p(r) = F(\rho_p)\delta''_p(r) - 2\rho^2 F'(\rho)\delta'_p(r) - 2\rho^3 F'(\rho)\delta_p(r) - \rho^2 F''(\rho)\delta_p(r) \quad (\text{C.30})$$

the source T_0 for a point particle can be written as

$$T_0^{\text{pp}} = T_0^{(2)}\delta''_p + T_0^{(1)}\delta'_p + T_p^{(0)}\delta_p, \quad (\text{C.31})$$

where

$$T_0^{(2)} = -T_{mm}^\circ, \quad (\text{C.32a})$$

$$T_0^{(1)} = 2\bar{\rho}_p(\bar{\rho}_p - 2\rho_p)\tau^\circ T_{lm}^\circ + 4\rho_p T_{mm}^\circ + 2\bar{\rho}_p T_{lm}^\circ, \quad (\text{C.32b})$$

$$\begin{aligned} T^{(0)} = & 4\bar{\rho}_p(2\bar{\rho}_p^2 - \bar{\rho}_p\rho_p + \rho_p^2)\tau^\circ T_{lm}^\circ - 4\rho_p^2 T_{mm}^\circ - 2\bar{\rho}_p^2\tilde{\delta}T_{lm}^\circ \\ & + 4\bar{\rho}^2(\bar{\rho}_p\rho_p)\tau^\circ\tilde{\delta}T_{ll}^\circ - 2\bar{\rho}(2\rho_p + \bar{\rho}_p)\tilde{\delta}T_{lm}^\circ \\ & + \bar{\rho}_p[2(2\rho_p + \bar{\rho}_p)T_{mm}^\circ + 4\bar{\rho}_p^2\tau^\circ(T_{lm}^\circ + \tau^\circ(\rho_p - \bar{\rho}_p)T_{ll}^\circ)] - \bar{\rho}_p^2\tilde{\delta}^2 T_{lm}^\circ. \end{aligned} \quad (\text{C.32c})$$

which is the reason for the appearance of the incorrect GHP weights. We have also used that $[\mathbf{P}, \tilde{\delta}] = 0$.

⁸ By this we mean in tetrad where $\mathbf{P} = \partial_r$, $\tilde{\mathbf{P}}' = \partial_u - \Delta\rho\bar{\rho}\partial_r$, which accounts for the Held non-covariant

terms containing Δ due to fact that n^μ has a component along r , $n^\mu\partial_\mu r = -\Delta/(2\Sigma)$

e. ψ_0 distributional content. Substituting the ansatz

$$\psi_0(u, r, z, \varphi) = \psi_0^\circ(u, z, \varphi) \delta_p(r) + \psi^+(u, z, \varphi, r) \Theta_p^+(r) + \psi^-(u, z, \varphi, r) \Theta_p^-(r) \quad (\text{C.33})$$

and equating coefficients gives

$$\delta_p'' : \quad T_{mm}^\circ = -\hat{A}B^\circ, \quad (\text{C.34a})$$

$$\delta_p' : \quad 2\hat{A}B^\circ + \hat{A}(\psi_p^+ - \psi_p^-) + \hat{B}B^\circ - 2\partial_u B^\circ = T_0^{(1)} \quad (\text{C.34b})$$

$$\begin{aligned} \delta_p : \quad & \hat{A}(\partial_r \psi_p^+ - \partial_r \psi_p^-) + (\partial_r^2 \hat{A})B^\circ + \hat{B}\psi_p^+ - \hat{B}\psi_p^- + \hat{C}B^\circ \\ & - 2\partial_u(\psi_p^+ - \psi_p^-) = T_0^{(0)} - \partial_r T_0^{(1)} \end{aligned} \quad (\text{C.34c})$$

where we've used hats to emphasize that $\hat{A}$, $\hat{B}$, and $\hat{C}$ are differential operators. Assuming C^1 differentiability of the radial functions across the location of the particle gives the equations

$$T_{mm}^\circ = \Delta_p \rho_p \bar{\rho}_p \psi_0^\circ. \quad (\text{C.35})$$

2. Fourth-order radial inversion and the string polynomial for point particle source

Suppose the Weyl scalar has the distributional decomposition

$$\psi_0(u, r, z, \varphi) = B(u, z, \varphi) \delta_p(r) + \psi^+(u, r, z, \varphi) \Theta_p^+(r) + \psi^-(u, r, z, \varphi) \Theta_p^-(r), \quad (\text{C.36})$$

where

$$\Theta_p^+(r) := \Theta(r - r_p), \quad \Theta_p^-(r) := \Theta(r_p - r), \quad \delta_p(r) := \delta(r - r_p). \quad (\text{C.37})$$

We wish to solve

$$\partial_r^4 \Phi = -\frac{1}{4} \bar{\psi}_0. \quad (\text{C.38})$$

a. Ansatz. To allow explicitly for a one-sided string piece, we write

$$\Phi = \Phi^+(u, r, z, \varphi) \Theta_p^+(r) + \Phi^-(u, r, z, \varphi) \Theta_p^-(r) + \Theta_p^+(r) P_3(r - r_p), \quad (\text{C.39})$$

where

$$P_3(x) = b_0 + b_1 x + b_2 x^2 + b_3 x^3, \quad x := r - r_p. \quad (\text{C.40})$$

b. Distributional identities. For a piecewise smooth field

$$F = F^+ \Theta_p^+ + F^- \Theta_p^-, \quad (\text{C.41})$$

one has

$$\partial_r^4 F = (\partial_r^4 F^+) \Theta_p^+ + (\partial_r^4 F^-) \Theta_p^- + [F''']_p \delta_p + [F'']_p \delta'_p + [F']_p \delta''_p + [F]_p \delta'''_p, \quad (\text{C.42})$$

where the jump of any quantity f is defined by

$$[f]_p := f^+(u, r_p, z, \varphi) - f^-(u, r_p, z, \varphi). \quad (\text{C.43})$$

Moreover,

$$\partial_r^4 (\Theta_p^+ P_3(x)) = b_0 \delta_p''' + b_1 \delta_p'' + 2b_2 \delta_p' + 6b_3 \delta_p. \quad (\text{C.44})$$

Substituting (C.39) into (C.38), and using (C.36), (C.42), and (C.44), gives

$$\begin{aligned} & (\partial_r^4 \Phi^+) \Theta_p^+ + (\partial_r^4 \Phi^-) \Theta_p^- + ([\partial_r^3 \Phi]_p + 6b_3) \delta_p \\ & + ([\partial_r^2 \Phi]_p + 2b_2) \delta'_p + ([\partial_r \Phi]_p + b_1) \delta''_p + ([\Phi]_p + b_0) \delta'''_p \\ & = -\frac{1}{4} \bar{B} \delta_p - \frac{1}{4} \overline{\psi^+} \Theta_p^+ - \frac{1}{4} \overline{\psi^-} \Theta_p^-. \end{aligned} \quad (\text{C.45})$$

Equating coefficients of the independent distributions yields

$$\delta_p'''(r) : \quad [\Phi]_p + b_0 = 0, \quad (\text{C.46a})$$

$$\delta_p''(r) : \quad [\partial_r \Phi]_p + b_1 = 0, \quad (\text{C.46b})$$

$$\delta_p'(r) : \quad [\partial_r^2 \Phi]_p + 2b_2 = 0, \quad (\text{C.46c})$$

$$\delta_p(r) : \quad [\partial_r^3 \Phi]_p + 6b_3 = -\frac{1}{4} \bar{B}, \quad (\text{C.46d})$$

$$\Theta_p^+(r) : \quad \partial_r^4 \Phi^+ = -\frac{1}{4} \overline{\psi^+}, \quad (\text{C.46e})$$

$$\Theta_p^-(r) : \quad \partial_r^4 \Phi^- = -\frac{1}{4} \overline{\psi^-}. \quad (\text{C.46f})$$

c. Interpretation. Equations (C.46a)–(C.46d) show that the coefficients of the one-sided string polynomial are determined by the jumps of the piecewise-ular part:

$$b_0 = -[\Phi]_p, \quad b_1 = -[\partial_r \Phi]_p, \quad b_2 = -\frac{1}{2} [\partial_r^2 \Phi]_p, \quad (\text{C.47})$$

and

$$b_3 = -\frac{1}{24}\bar{B} - \frac{1}{6}[\partial_r^3 \Phi]_p. \quad (\text{C.48})$$

Thus the lower-order coefficients b_0, b_1, b_2 are pure matching data for the homogeneous/string sector, while only the cubic coefficient b_3 contains a source-forced contribution coming from the delta-function coefficient B in ψ_0 .

d. General piecewise solution. Integrating (C.38) four times gives

$$\begin{aligned} \Phi^+(u, z, \varphi, r) = & a_+(u, z, \varphi) + b_+(u, z, \varphi)x + \frac{1}{2}c_+(u, z, \varphi)x^2 + \frac{1}{6}d_+(u, z, \varphi)x^3 \\ & - \frac{1}{24} \int_{r_p}^r (r-s)^3 \overline{\psi^+(u, z, \varphi, s)} ds, \end{aligned} \quad (\text{C.49})$$

$$\begin{aligned} \Phi^-(u, z, \varphi, r) = & a_-(u, z, \varphi) + b_-(u, z, \varphi)x + \frac{1}{2}c_-(u, z, \varphi)x^2 + \frac{1}{6}d_-(u, z, \varphi)x^3 \\ & - \frac{1}{24} \int_{r_p}^r (r-s)^3 \overline{\psi^-(u, z, \varphi, s)} ds. \end{aligned} \quad (\text{C.50})$$

The jump conditions (??)–(??) imply

$$a_+ = a_- =: \mathcal{B}_0, \quad b_+ = b_- =: \mathcal{B}_1, \quad c_+ = c_- =: \mathcal{B}_2, \quad d_+ - d_- =: \mathcal{B}_3 = -\frac{1}{4}\bar{B}. \quad (\text{C.51})$$

To parametrize the cubic coefficients in same was (V.9) we write

$$d_+ = d - \frac{1}{8}\bar{B}, \quad d_- = d + \frac{1}{8}\bar{B}, \quad (\text{C.52})$$

where $d(u, z, \varphi)$ is arbitrary and corresponds to adding a homogeneous cubic solution in r .

Substituting this into (C.49)–(C.50), one obtains

$$\begin{aligned} \Phi(u, r, z, \varphi) = & a + bx + \frac{1}{2}cx^2 + \frac{1}{6}dx^3 - \frac{1}{48}\bar{B}|x|^3 \\ & - \frac{1}{24}\Theta_p^+(r) \int_{r_p}^r (r-s)^3 \overline{\psi^+(u, z, \varphi, s)} ds \\ & - \frac{1}{24}\Theta_p^-(r) \int_{r_p}^r (r-s)^3 \overline{\psi^-(u, z, \varphi, s)} ds. \end{aligned} \quad (\text{C.53})$$

The singular part is therefore

$$\Phi_{\text{sing}} = -\frac{1}{48}\bar{B}|r - r_p|^3, \quad (\text{C.54})$$

since

$$\partial_r^4 |r - r_p|^3 = 12\delta_p(r), \quad (\text{C.55})$$

and hence

$$\partial_r^4 \Phi_{\text{sing}} = -\frac{1}{48}\bar{B}\partial_r^4 |r - r_p|^3 = -\frac{1}{4}\bar{B}\delta_p(r), \quad (\text{C.56})$$

exactly reproducing the delta-function contribution in (??).

e. Summary. The inversion of a Weyl scalar of the form (??) produces a Hertz potential of the form (C.53): it is C^2 at the particle, but not C^3 . Equivalently,

$$\psi_0 \sim \delta_p + \Theta_p^\pm \quad \implies \quad \Phi \sim |r - r_p|^3 + \Theta_p^\pm. \quad (\text{C.57})$$

In particular, the coefficient of the cubic cusp is fixed entirely by the coefficient of the $\delta_p(r)$ term in ψ_0 .

3. Spin weighted spherical harmonics

First, we introduce the differential operators $\hat{\delta}$ and $\hat{\delta}'$ operators related to Chandra's operators by $\hat{\delta} \equiv \mathcal{L}_0^\dagger$ and $\hat{\delta}' = \bar{\hat{\delta}} = {}_s\mathcal{L}_0$. These take the form

$$\begin{aligned} \hat{\delta} &= -2^{-1/2} \left(-\sqrt{1-z^2} \partial_z + \frac{i}{\sqrt{1-z^2}} \partial_\varphi - \frac{1}{2}(p-q) \frac{z}{\sqrt{1-z^2}} \right), \\ \hat{\delta}' &= -2^{-1/2} \left(-\sqrt{1-z^2} \partial_z - \frac{i}{\sqrt{1-z^2}} \partial_\varphi + \frac{1}{2}(p-q) \frac{z}{\sqrt{1-z^2}} \right). \end{aligned} \quad (\text{C.58})$$

or in spherical coordinates and using $s = (p - q)/2$,

$$\begin{aligned} \hat{\delta} &= -2^{-1/2} (\partial_\theta + i \csc \theta \partial_\varphi - s \cot \theta), \\ \hat{\delta}' &= -2^{-1/2} (\partial_\theta - i \csc \theta \partial_\varphi + s \cot \theta). \end{aligned} \quad (\text{C.59})$$

The operators $\hat{\delta}$ and $\hat{\delta}'$ act as spin raising and lowering operators on spin-weighted spherical harmonics

$${}_s Y_{\ell m} = (\sqrt{2})^s \lambda_{\ell, s} \hat{\delta}^s Y_{\ell m}, \quad 0 \leq s \leq \ell, \quad (\text{C.60a})$$

$${}_s Y_{\ell m} = (-\sqrt{2})^{-s} \lambda_{\ell, -s} (\hat{\delta}')^{-s} Y_{\ell m}, \quad -\ell \leq s \leq 0. \quad (\text{C.60b})$$

where

$$\lambda_{\ell, s} = \sqrt{\frac{(\ell - s)!}{(\ell + s)!}}. \quad (\text{C.61})$$

It follows that

$$\hat{\delta}_s Y_{\ell, m} = 2^{-1/2} \sqrt{(\ell - s)(\ell + s + 1)}_{s+1} Y_{\ell, m} \quad (\text{C.62})$$

$$\hat{\delta}'_s Y_{\ell, m} = -2^{-1/2} \sqrt{(\ell + s)(\ell - s + 1)}_{s-1} Y_{\ell, m}, \quad (\text{C.63})$$

Repeated action gives

$$\hat{\mathfrak{d}}^k {}_s Y_{\ell m} = \left[\prod_{j=0}^{k-1} \sqrt{\frac{(\ell - s - j)(\ell + s + j + 1)}{2}} \right] {}_{s+k} Y_{\ell m}, \quad (\text{C.64a})$$

$$(\hat{\mathfrak{d}}')^k {}_s Y_{\ell m} = (-1)^k \left[\prod_{j=0}^{k-1} \sqrt{\frac{(\ell + s - j)(\ell - s + j + 1)}{2}} \right] {}_{s-k} Y_{\ell m}. \quad (\text{C.64b})$$

paragraphSingle-step ladder identities.

$$\hat{\mathfrak{d}} {}_s Y_{\ell m} = + \frac{\lambda_{\ell,s}}{\sqrt{2}\lambda_{\ell,s+1}} {}_{s+1} Y_{\ell m}, \quad (\text{C.65})$$

$$\bar{\mathfrak{d}} {}_s Y_{\ell m} = - \frac{\lambda_{\ell,s-1}}{\sqrt{2}\lambda_{\ell,s}} {}_{s-1} Y_{\ell m}. \quad (\text{C.66})$$

a. *Repeated raising.* For $k \geq 0$,

$$\hat{\mathfrak{d}}^k {}_s Y_{\ell m} = 2^{-k/2} \left(\prod_{j=0}^{k-1} \lambda_{\ell,s+j+1} \right) {}_{s+k} Y_{\ell m} = \frac{1}{(\sqrt{2})^k} \frac{\lambda_{\ell,s}}{\lambda_{\ell,s+k}} {}_{s+k} Y_{\ell m} \quad (\text{C.67})$$

b. *Repeated lowering.* For $k \geq 0$,

$$(\hat{\mathfrak{d}}')^k {}_s Y_{\ell m} = (-1)^k 2^{-k/2} \left(\prod_{j=0}^{k-1} \lambda_{\ell, -(s-j)} \right) {}_{s-k} Y_{\ell m} = \frac{1}{(-\sqrt{2})^k} \frac{\lambda_{\ell,s-k}}{\lambda_{\ell,s}} {}_{s-k} Y_{\ell m} \quad (\text{C.68})$$

Note also the adjoint relations under integration by parts (they complex conjugate also under the sesquilinear $L_2(\mathbb{S}^2)$ inner product, however).

$$\hat{\mathfrak{d}}^\dagger = -\bar{\mathfrak{d}}', \quad \bar{\mathfrak{d}}'^\dagger = -\hat{\mathfrak{d}} \quad (\text{C.69})$$

As $\tilde{\mathfrak{d}} = \hat{\mathfrak{d}} - 2^{-1/2}ia \sin \theta \partial_u$, it follows that $\tilde{\mathfrak{d}}^\dagger = -\bar{\mathfrak{d}}$. Note that the usual GHP $\mathfrak{d}$ operators do not have such nice conjugation properties and instead one must add τ factors under adjoint operation.

a. *Decomposition of Spin-Weighted Spherical Harmonics Product*

The product of two spin-weighted spherical harmonics⁹ ${}_s Y_{\ell_1 m_1}(\theta, \phi)$ and ${}_s Y_{\ell_2 m_2}(\theta, \phi)$ can be expanded as a sum over spin-weighted harmonics with total spin $s = s_1 + s_2$ and total

⁹ Our conventions chosen to be consistent with Mathematica, (Wikipedia?) and the spin-2 paper []

azimuthal number $m = m_1 + m_2$:

$$\begin{aligned}
{}_s Y_{j_1 m_1}(\theta, \phi) {}_s Y_{j_2 m_2}(\theta, \phi) &= \sum_{j=|j_1-j_2|}^{|j_1+j_2|} c_{j_1 j_2 J; s_1 s_2 s}^{m_1, m_2, M} Y_{j m}(\theta, \phi) \\
&= \sum_{J=|j_1-j_2|}^{|j_1+j_2|} \underbrace{(-1)^{s+M} \sqrt{\frac{(2j_1+1)(2j_2+1)(2J+1)}{4\pi}} \begin{pmatrix} j_1 & j_2 & J \\ m_1 & m_2 & -M \end{pmatrix} \begin{pmatrix} j_1 & j_2 & J \\ -s_1 & -s_2 & s \end{pmatrix}}_{\text{Clebsch-Gordan coefficient for spin-weighted harmonics}} {}_s Y_{J m}(\theta, \phi),
\end{aligned} \tag{C.70}$$

where the sum runs over all allowed values of j satisfying the triangular inequality $|j_1 - j_2| \leq J \leq j_1 + j_2$, and $m = m_1 + m_2$, $s = s_1 + s_2$. The Wigner 3- j symbols related to the Clebsch-Gordan coefficients $\langle j_1 j_2 m_1 m_2 | j_1 j_2 j_3 m_3 \rangle$ by

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j_3+1}} \langle j_1 j_2 m_1 m_2 | j_1 j_2 j_3 (-m_3) \rangle \tag{C.71}$$

which are zero unless $m_1 + m_2 + m_3 = 0$ and $|j_1 + j_2| \leq j_3 \leq j_1 + j_2$. The quantity C introduced above relates to c by

$$C_{j_1 j_2 j; s_1 s_2 s}^m = C_{j_1 j_2 j; s_1 s_2 s}^{0, m, m} \tag{C.72}$$

Note also the identity

$$C_{l_1 l_2 j; -s_1, -s_2, -s}^m = (-1)^{l_1+l_2+j} C_{l_1 l_2 j; s_1 s_2 s}^m \tag{C.73}$$

which derives from the reflection property

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ s_1 & s_2 & s_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ -s_1 & -s_2 & -s_3 \end{pmatrix} \tag{C.74}$$

These relations, along with the conjugation property ${}_s \bar{Y}_{j, m} = (-1)^{s+j} {}_{-s} Y_{j, -m}$ can be used to derive a spin weighted form of the ‘‘Slater integral’’

$$\begin{aligned}
\int_{\mathbb{S}^2} {}_s \bar{Y}_{j_1, m_1} {}_s Y_{j_2, m_2} {}_s Y_{j_3, m_3} &= (-1)^{j_1+j_2+j_3} \sqrt{\frac{(2j_1+1)(2j_2+1)(2j_3+1)}{4\pi}} \\
&\quad \times \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3 \\ s_1 & s_2 & s_3 \end{pmatrix}
\end{aligned} \tag{C.75}$$

$$= (-1)^{j_1+j_2+j_3-s_3-M} C_{j_1 j_2 j; -s_1, -s_2, s_3}^{m_1, m_2, -M} \tag{C.76}$$

$$= (-1)^{-s_3-M} C_{j_1 j_2 j; s_1, s_2, -s_3}^{m_1, m_2, -M} \tag{C.77}$$

The product of spin-weighted harmonics with the Held scalars Ω° and τ° in outgoing KN coordinates and Kinnersley tetrad are given by

$$\tau^\circ {}_s Y_{\ell,m} = -ia \sqrt{\frac{4\pi}{3}} {}_{-1}Y_{1,0} {}_s Y_{\ell,m} \quad (\text{C.78a})$$

$$\bar{\tau}^\circ {}_s Y_{\ell,m} = -ia \sqrt{\frac{4\pi}{3}} {}_{-1}Y_{1,0} {}_s Y_{\ell,m} \quad (\text{C.78b})$$

$$\Omega^\circ {}_s Y_{\ell,m} = -2ia \sqrt{\frac{4\pi}{3}} {}_0Y_{1,0} {}_s Y_{\ell,m} \quad (\text{C.78c})$$

With these results we can derive the following results for the difference between the angular Held operators $\tilde{\partial}, \tilde{\partial}'$ and spin-weight raising/lowering operators $\hat{\partial}, \hat{\partial}'$:

$$\begin{aligned} (\tilde{\partial}' - \hat{\partial}') {}_s Y_{\ell,m} e^{-i\omega u} &= -a\omega \sqrt{\frac{4\pi}{3}} {}_{-1}Y_{1,0} {}_s Y_{\ell,m} e^{-i\omega u} \\ &= a\omega \sum_{j=\ell-1}^{\ell+1} (-1)^m \sqrt{(2\ell+1)(2j+1)} \begin{pmatrix} \ell & 1 & j \\ m & 0 & -m \end{pmatrix} \times \\ &\quad \begin{pmatrix} \ell & 1 & j \\ s & -1 & -s+1 \end{pmatrix} {}_{s-1}Y_{j,m} e^{-i\omega u} \end{aligned} \quad (\text{C.79})$$

$$\begin{aligned} (\tilde{\partial} - \hat{\partial}) {}_s Y_{\ell,m} e^{-i\omega u} &= -a\omega \sqrt{\frac{4\pi}{3}} {}_{-1}Y_{1,0} {}_s Y_{\ell,m} e^{-i\omega u} \\ &= a\omega \sum_{j=\ell-1}^{\ell+1} (-1)^{s+m} \sqrt{(2\ell+1)(2j+1)} \begin{pmatrix} \ell & 1 & j \\ m & 0 & -m \end{pmatrix} \times \\ &\quad \begin{pmatrix} \ell & 1 & j \\ -s & -1 & s+1 \end{pmatrix} {}_{s+1}Y_{j,m} e^{-i\omega u} \end{aligned} \quad (\text{C.80})$$

Appendix D: Spin-Weighted Spherical Harmonics vs. GHP Scalars

There is a subtle but important distinction between *spin weight* as used in the theory of spin-weighted spherical harmonics ${}_s Y_{\ell,m}$, and *spin weight* in the Newman–Penrose/GHP formalism. Although these notions share the same name, they arise from **different transformation groups**.

1. Spin weight in the GHP sense

In the GHP formalism one introduces a complex null tetrad $(l^a, n^a, m^a, \bar{m}^a)$. A *tetrad spin rotation* is a local $U(1)$ gauge transformation

$$m^a \mapsto e^{i\chi} m^a, \quad \bar{m}^a \mapsto e^{-i\chi} \bar{m}^a.$$

A field η of GHP type (p, q) transforms as

$$\eta \mapsto e^{i(p-q)\chi} \eta.$$

The quantity $s = \frac{1}{2}(p - q)$ is the *GHP spin weight*. The GHP derivative operators $\eth, \eth', \mathbb{P}, \mathbb{P}'$ and the associated Held operators $\tilde{\eth}, \tilde{\eth}', \tilde{\mathbb{P}}'$ are defined precisely so that they change s and $b = (p + q)/2$ in a controlled and covariant way. A GHP scalar of spin weight s is therefore a field whose transformation under *tetrad spin rotations* is specified by this rule.

2. Spin weight in the representation–theory sense

The spin-weighted spherical harmonics ${}_s Y_{\ell m}(\theta, \phi)$ are defined on the *unit round two-sphere*. They are obtained by contracting symmetric trace-free tensor harmonics $Y_{\ell m}^{A_1 \dots A_s}$ with the *coordinate dyad* of the round sphere,

$$m_A^\pm = \frac{1}{\sqrt{2}}(\hat{\theta}_A \pm i\hat{\phi}_A), \quad \hat{\theta}^A, \hat{\phi}^A \text{ the orthonormal basis of } (\mathbb{S}^2, q_{AB}).$$

Thus

$${}_s Y_{\ell m} \propto Y_{\ell m}^{A_1 \dots A_s} m_{A_1} \dots m_{A_s}.$$

Under a *geometric rotation* of the sphere (i.e. an $SO(3)$ rotation of the coordinates), m_A transforms as a spin-1 representation and the result transforms with the familiar phase $e^{is\psi}$ associated with the rotation angle ψ .

This is the usual sense in which ${}_s Y_{\ell m}$ have “spin weight s ”. It has nothing to do with tetrad rotations.

3. Why ${}_s Y_{\ell m}$ are *not* GHP scalars on $\mathcal{M}_{\text{Kerr}}(g_{a,M})$

A tetrad spin rotation acts on the *spacetime* vector m^a . The spin-weighted harmonics are built instead from the dyad $m_A^\pm$ constructed purely from the round-sphere metric q_{AB} .

These two structures coincide *only* if the NP tetrad is chosen to be globally aligned with the round-sphere coordinate basis.

In curved spacetimes such as Kerr, the NP tetrad satisfies

$$m^\alpha \partial_\alpha y^A = -\bar{\rho} \hat{m}^A$$

where e_θ^a, e_ϕ^a are obtained by projecting the spacetime tetrad onto the angular directions. These vectors do *not* coincide with $\hat{\theta}_A, \hat{\phi}_A$ of the unit round sphere, due to frame dragging, normalization by Σ , the ρ -spin coefficient, etc, which transforms under a boost. So although, the direction is the same (modulo a phase), the GHP transformation behavior is different, because ρ .

4. Why this causes “spin mixing” when projecting

Objects like

$$\tilde{\delta}\delta_p^2(\Omega)$$

are genuine GHP spin-weighted distributions, because the delta function here is defined using the tetrad-adapted angular coordinates.

However, when one projects such an object onto ${}_s Y_{\ell m}$,

$$\int_{\mathbb{S}^2} (\tilde{\delta}\delta_p^2(\Omega)) {}_s \bar{Y}_{\ell m} d\Omega,$$

the integral mixes:

- GHP-spin-weighted objects (the left-hand factor),
- with $SO(3)$ representation-theory spin-weighted harmonics (the right-hand factor),
- not to mention also the coordinate measure $d\Omega$, which is not GHP covariant.

Because the two notions of spin are different, this projection does *not* preserve GHP spin weight, and one can obtain contributions involving ${}_s Y$, ${}_{s+1} Y$, and ${}_{s-1} Y$ harmonics.

Appendix E: Cartesian tensor decompositions of spin-weighted scalars

In many applications it is convenient to replace the standard spin-weighted spherical harmonic (SWSH) expansion with a *Cartesian symmetric trace-free (STF)* decomposition.

This avoids explicit sums over m and provides a natural geometric representation of angular structures.

1. Orthonormal triad and helicity basis

Introduce an orthonormal triad on the unit sphere:

$$(\hat{n}^i, \hat{e}_\theta^i, \hat{e}_\phi^i), \quad \hat{n}^i \hat{n}_i = 1, \quad \hat{e}_\theta^i \hat{e}_{\theta i} = \hat{e}_\phi^i \hat{e}_{\phi i} = 1, \quad \hat{n}^i \hat{e}_{\theta i} = \hat{n}^i \hat{e}_{\phi i} = \hat{e}_\theta^i \hat{e}_{\phi i} = 0. \quad (\text{E.1})$$

Define the *helicity basis vectors*:

$$m_i^\pm \equiv \frac{1}{\sqrt{2}} (\hat{e}_{\theta i} \pm i \hat{e}_{\phi i}), \quad m_i^+ m^{-i} = 1, \quad m_i^\pm m^{\pm i} = 0, \quad m_i^\pm \hat{n}^i = 0. \quad (\text{E.2})$$

The m_i^- vector lowers spin by one, while m_i^+ raises spin by one.

2. STF expansion of spin-weighted scalars

A spin- s scalar field $X_s(\hat{n})$ admits an expansion in terms of STF tensors:

$$X_s(\hat{n}) = \sum_{\ell \geq |s|} X_{i_1 \dots i_\ell} m_{i_1}^{-s} \dots m_{i_{|s|}}^{-s} \hat{n}^{\langle i_{|s|+1} \dots i_\ell \rangle}, \quad (\text{E.3})$$

where $\langle \dots \rangle$ denotes the symmetric trace-free projection. For a scalar ($s = 0$) we have the familiar expansion

$$f(\hat{n}) = \sum_{\ell \geq 0} F_L \hat{n}^{\langle L \rangle}, \quad F_L \equiv F_{i_1 \dots i_\ell}. \quad (\text{E.4})$$

Cartesian decomposition of products

Let $X_{s_1}(\hat{n})$ and $Y_{s_2}(\hat{n})$ be two spin-weighted fields with STF expansions

$$X_{s_1} = X_{L_1} m_{i_1}^{-s_1} \dots m_{i_{|s_1|}}^{-s_1} \hat{n}^{\langle L_1 \rangle}, \quad Y_{s_2} = Y_{L_2} m_{j_1}^{-s_2} \dots m_{j_{|s_2|}}^{-s_2} \hat{n}^{\langle L_2 \rangle}. \quad (\text{E.5})$$

The product $X_{s_1} Y_{s_2}$ can be expanded without Clebsch–Gordan coefficients by simply performing a symmetric trace-free decomposition of the combined tensor:

$$X_{s_1} Y_{s_2} = \sum_L (X \otimes Y)_{i_1 \dots i_\ell}^{\langle \text{STF} \rangle} m_{i_1}^{-s} \dots m_{i_{|s|}}^{-s} \hat{n}^{\langle i_{|s|+1} \dots i_\ell \rangle}, \quad s = s_1 + s_2. \quad (\text{E.6})$$

Algorithmically, this corresponds to:

1. Form the outer product of STF tensors $X_{L_1} \otimes Y_{L_2}$.
2. Symmetrize and remove traces to obtain a new STF tensor of rank $\ell = |L_1| + |L_2|, |L_1| + |L_2| - 2, \dots$.
3. Contract with the appropriate number of m_i^- to encode the total spin $s = s_1 + s_2$.

This procedure automatically accounts for nearest-neighbor couplings of angular modes, in direct analogy with Clebsch–Gordan rules, but in a fully Cartesian, coordinate-free form.

Example: spin-1 field times scalar

For a scalar field $f(\hat{n}) = F_L \hat{n}^{\langle L \rangle}$ and a spin-1 field $X_i = X_L m_i^- \hat{n}^{\langle L \rangle}$, the product is

$$X_i f = \sum_{\ell \geq 1} (X_{iL_1} F_{L_2})^{\langle \text{STF} \rangle} m_i^- \hat{n}^{\langle L_1 L_2 \rangle}, \quad (\text{E.7})$$

which reproduces the same mode couplings as the usual spin-weighted Clebsch–Gordan decomposition.

Appendix F: Lie derivative in Held form for gauge transformation

$$(\mathcal{L}_\xi g)_{ll} = 2\mathbb{P}\xi_l, \quad (\text{F.1a})$$

$$(\mathcal{L}_\xi g)_{ln} = \mathbb{P}\xi_n + \mathbb{P}'\xi_l + (\tau + \bar{\tau}')\xi_{\bar{m}} + (\bar{\tau} + \tau')\xi_m, \quad (\text{F.1b})$$

$$\begin{aligned} &= \tilde{\mathbb{P}}'\xi_l + \rho\bar{\rho} \left(\bar{\tau}^\circ\tilde{\mathfrak{d}} + \tau^\circ\tilde{\mathfrak{d}}' - (\rho + \bar{\rho})\tau^\circ\bar{\tau}^\circ - \frac{1}{2} \left(\frac{\Psi^\circ\rho}{\bar{\rho}} + \frac{\bar{\Psi}^\circ\bar{\rho}}{\rho} \right) \right) \xi_l \\ &\quad + \mathbb{P}\xi_n + \rho\bar{\rho}\tau^\circ\xi_{\bar{m}} + \rho\bar{\rho}\bar{\tau}^\circ\xi_m - \bar{\rho}^2\tau^\circ\xi_{\bar{m}} - \rho^2\bar{\tau}^\circ\xi_m \end{aligned}$$

$$(\mathcal{L}_\xi g)_{lm} = (\mathbb{P} + \bar{\rho})\xi_m + (\mathfrak{d} + \bar{\tau}')\xi_l, \quad (\text{F.1c})$$

$$= (\mathbb{P} + \bar{\rho})\xi_m + \bar{\rho}(\tilde{\mathbb{P}}' - 2\bar{\rho}\tau^\circ)\xi_l$$

$$(\mathcal{L}_\xi g)_{m\bar{m}} = \mathfrak{d}\xi_{\bar{m}} + \mathfrak{d}'\xi_m + (\rho + \bar{\rho})\xi_n + (\rho' + \bar{\rho}')\xi_l, \quad (\text{F.1d})$$

$$\begin{aligned} &= -\frac{1}{2} \left(\Psi^\circ\rho^2 + \bar{\Psi}^\circ\bar{\rho}^2 + 2(\rho + \bar{\rho})\rho\bar{\rho}\tau^\circ\bar{\tau}^\circ - 2(\rho\bar{\rho}^\circ + \bar{\rho}\rho^\circ) \right) \xi_l \\ &\quad + (\rho + \bar{\rho})\xi_n + \rho(\tilde{\mathfrak{d}}' + \rho\tau^\circ)\xi_m + \bar{\rho}(\tilde{\mathfrak{d}} - \bar{\rho}\bar{\tau}^\circ)\xi_{\bar{m}}, \end{aligned}$$

$$(\mathcal{L}_\xi g)_{nn} = 2\mathbb{P}'\xi_n, \quad (\text{F.1e})$$

$$= 2\tilde{\mathbb{P}}'\xi_n + 2\rho\bar{\rho}(\bar{\tau}^\circ\tilde{\mathfrak{d}} + \tau^\circ\tilde{\mathfrak{d}}')\xi_n + (\Psi^\circ\rho^2 + \bar{\Psi}^\circ\bar{\rho}^2 + 2\rho\bar{\rho}(\rho + \bar{\rho})\tau^\circ\bar{\tau}^\circ)\xi_n.$$

$$(\mathcal{L}_\xi g)_{nm} = (\mathbb{P}' + \rho')\xi_m + (\mathfrak{d} + \tau)\xi_n, \quad (\text{F.1f})$$

$$(\mathcal{L}_\xi g)_{mm} = 2\mathfrak{d}\xi_m, \quad (\text{F.1g})$$

$$= 2\bar{\rho}(\tilde{\mathfrak{d}} + \tau^\circ)\xi_m.$$

Appendix G: Leibniz rule for $\hat{\mathfrak{d}}$ on distributions

When working with spin-weighted distributions such as $\delta(z - z_0)$ or $\delta(\varphi - \varphi_0)$, the usual pointwise Leibniz rule does not apply directly. Instead, one must define the action of $\hat{\mathfrak{d}}$ in the ****distributional sense****, via its adjoint.

Distributional definition

For any smooth test function f and a distribution g ,

$$\int_{\mathbb{S}^2} f(\hat{\mathfrak{d}}g) d\Omega := \int_{\mathbb{S}^2} (\hat{\mathfrak{d}}^\dagger f) g d\Omega, \quad (\text{G.1})$$

where $\hat{\mathfrak{d}}^\dagger = -\bar{\hat{\mathfrak{d}}}$ is the adjoint of $\hat{\mathfrak{d}}$ on the sphere with respect to the standard measure $d\Omega$.

This is analogous to the standard definition of a derivative of a delta function:

$$\int f(\theta) \delta'(\theta - \theta_0) d\theta = -f'(\theta_0).$$

Leibniz-type identity for distributions

Let f be smooth and g a distribution. Then

$$\int_{\mathbb{S}^2} \phi f \hat{\mathfrak{D}} g d\Omega = \int_{\mathbb{S}^2} (\hat{\mathfrak{D}}^\dagger(\phi f)) g d\Omega \quad (\text{G.2})$$

for any smooth test function ϕ .

Equivalently, one can think of $f \hat{\mathfrak{D}} g$ as acting on test functions via

$$f \hat{\mathfrak{D}} g \longrightarrow \phi \mapsto \int_{\mathbb{S}^2} (\hat{\mathfrak{D}}^\dagger(f\phi)) g d\Omega. \quad (\text{G.3})$$

a. Conventions (complex L^2 pairing). We use the standard complex inner product on S^2 ,

$$\langle f | g \rangle := \int_{S^2} d\Omega \overline{f(\Omega)} g(\Omega), \quad d\Omega = \sin \theta d\theta d\varphi. \quad (\text{G.4})$$

For the point-supported delta distribution $\delta \equiv \delta^2(\Omega - \Omega_p)$ we then have

$$\langle f | \delta \rangle = \overline{f(\Omega_p)}. \quad (\text{G.5})$$

Let $\tilde{\mathfrak{D}}$ be the Held-modified edth operator, and define its adjoint $\tilde{\mathfrak{D}}^\dagger$ by the *definition* of adjoint:

$$\langle f | \tilde{\mathfrak{D}} g \rangle = \langle \tilde{\mathfrak{D}}^\dagger f | g \rangle. \quad (\text{G.6})$$

We further assume the standard GHP/Held relation

$$\tilde{\mathfrak{D}}^\dagger = -\tilde{\mathfrak{D}}'. \quad (\text{G.7})$$

b. Derivative of a distribution. Define $\tilde{\mathfrak{D}}\delta$ as the distributional derivative via the sesquilinear adjoint:

$$\langle \psi | \tilde{\mathfrak{D}}\delta \rangle_{L^2} := -\langle \tilde{\mathfrak{D}}'\psi | \delta \rangle_{L^2}, \quad \forall \psi \in C^\infty(S^2). \quad (\text{G.8})$$

Using (G.5) and (G.7) gives

$$\langle \psi | \tilde{\mathfrak{D}}\delta \rangle = \overline{(\tilde{\mathfrak{D}}^\dagger \psi)(\Omega_p)} = \overline{(-\tilde{\mathfrak{D}}'\psi)(\Omega_p)} = -\overline{(\tilde{\mathfrak{D}}'\psi)(\Omega_p)}. \quad (\text{G.9})$$

Thus, with the complex L^2 pairing, $\tilde{\mathfrak{D}}\delta$ evaluates $\tilde{\mathfrak{D}}'\psi$ at Ω_p up to the overall complex conjugation coming from (G.5).

c. Product with a smooth function. Let $f(\Omega)$ be smooth. Then, for any test function $\psi \in C^\infty(S^2)$,

$$\begin{aligned}\langle \psi | f \tilde{\delta} \rangle &= \langle \psi \bar{f} | \tilde{\delta} \delta \rangle = \langle \tilde{\delta}^\dagger(\psi \bar{f}) | \delta \rangle \\ &= \overline{(\tilde{\delta}^\dagger(\psi \bar{f}))(\Omega_p)} = -\overline{(\tilde{\delta}'(\psi \bar{f}))(\Omega_p)} \\ &= -\overline{(\tilde{\delta}'\psi) \bar{f} + \psi \tilde{\delta}'(\bar{f})}(\Omega_p).\end{aligned}\tag{G.10}$$

Equivalently, expanding the conjugation,

$$\langle \psi | f \tilde{\delta} \rangle = -f(\Omega_p) \overline{(\tilde{\delta}'\psi)(\Omega_p)} - \overline{\psi(\Omega_p)} \overline{(\tilde{\delta}'\bar{f})(\Omega_p)}.\tag{G.11}$$

d. Distributional product rule (intrinsic form). As a distributional identity (i.e. independent of test-function pairing conventions), one may write the standard product rule

$$f(\Omega) \tilde{\delta} \delta^2(\Omega - \Omega_p) = f|_{\Omega_p} \tilde{\delta} \delta^2(\Omega - \Omega_p) - (\tilde{\delta}' f)|_{\Omega_p} \delta^2(\Omega - \Omega_p),\tag{G.12}$$

provided $\tilde{\delta}^\dagger = -\tilde{\delta}'$ is understood with the same conventions used to define the distributional derivative. When working explicitly with the complex L^2 pairing, the evaluation against a test function is given by (G.11).

e. Application. Taking $f = (\bar{\rho}/\rho) J_l$ in (G.12) gives

$$\frac{\bar{\rho}}{\rho} J_l \tilde{\delta} \delta^2(\Omega - \Omega_p) = \left(\frac{\bar{\rho}}{\rho} J_l \right) \Big|_{\Omega_p} \tilde{\delta} \delta^2(\Omega - \Omega_p) - \left(\tilde{\delta}' \left[\frac{\bar{\rho}}{\rho} J_l \right] \right) \Big|_{\Omega_p} \delta^2(\Omega - \Omega_p).\tag{G.13}$$

In particular, if J_l is treated as angle-independent at fixed (u, r) , then

$$\left(\tilde{\delta}' \left[\frac{\bar{\rho}}{\rho} J_l \right] \right) \Big|_{\Omega_p} = J_l \Big|_{\Omega_p} \left(\tilde{\delta}' \left[\frac{\bar{\rho}}{\rho} \right] \right) \Big|_{\Omega_p}.\tag{G.14}$$

Appendix H: Projection of a spin- s source onto spin-weighted harmonics

We consider a source of the form

$$F^\circ(\Omega) = f_l^\circ \hat{\delta} \delta(\Omega - \Omega_p), \quad \text{spin} = s.\tag{H.1}$$

where f° is constant. We want its ℓ, m mode coefficient in the standard L^2 inner product:

$$F_{\ell m}^\circ = \int_{\mathbb{S}^2} {}_s\bar{Y}_{\ell m}(\Omega) F^\circ(\Omega)\tag{H.2}$$

In the standard L^2 inner product

$$\langle a|b\rangle_{L^2} = \int_{\mathbb{S}^2} \bar{a} b \quad (\text{H.3})$$

the adjoint of $\hat{\delta}$ is

$$\hat{\delta}^\dagger = -\bar{\delta}, \quad (\text{H.4})$$

so we can “integrate by parts”

$$\int_{\mathbb{S}^2} {}_s\bar{Y}_{\ell m} f^\circ \hat{\delta} \delta(\Omega - \Omega_p) d\Omega = -f^\circ \int_{\mathbb{S}^2} \delta(\Omega - \Omega_p) \overline{\hat{\delta} {}_sY_{\ell m}} \quad (\text{H.5})$$

Evaluating at the “particle” location, the delta function picks out $\Omega = \Omega_p$:

$$F_{\ell m}^\circ = -f^\circ \overline{(\hat{\delta} {}_sY_{\ell m})(\Omega_p)}. \quad (\text{H.6})$$

Using the spin-lowering relation

$$\bar{\delta} {}_sY_{\ell m} = -\frac{\lambda_{\ell,s}}{\sqrt{2}\lambda_{\ell,s-1}} {}_{s-1}Y_{\ell m}, \quad (\text{H.7})$$

we find

$$F_{\ell m}^\circ = \frac{f^\circ \lambda_{\ell,s}}{\sqrt{2}\lambda_{\ell,s-1}} {}_{s-1}\bar{Y}_{\ell m}(\Omega_p). \quad (\text{H.8})$$

We often encounter integrals of the form

$$\langle {}_sY_{\ell,m}|f(\Omega) \hat{\delta}[g(\Omega)\delta_p(\Omega)]\rangle_{L^2} \quad (\text{H.9})$$

Using $\langle a|bc\rangle_{L^2} = \langle a\bar{b}|c\rangle_{L^2}$ and $\langle a|\mathcal{O}b\rangle_{L^2} = \langle \mathcal{O}^\dagger a|b\rangle_{L^2}$ we get

$$\begin{aligned} \langle {}_sY_{\ell,m}(\Omega)|f(\Omega) \hat{\delta}[g(\Omega)\delta_p(\Omega)]\rangle_{L^2} &= \langle {}_sY_{\ell,m}(\Omega)\bar{f}(\Omega)|\hat{\delta}\delta_p(\Omega)\rangle_{L^2}, \\ &= \langle \hat{\delta}^\dagger {}_sY_{\ell,m}(\Omega)\bar{f}(\Omega)|g(\Omega)\delta_p(\Omega)\rangle_{L^2}, \\ &= -\int_{\mathbb{S}^2} \overline{\hat{\delta}({}_sY_{\ell,m}\bar{f})(\Omega)} g(\Omega)\delta_p(\Omega), \\ &= -g(\Omega_p) \left(-\frac{\lambda_{\ell,s}}{\sqrt{2}\lambda_{\ell,s-1}} {}_{s-1}\bar{Y}_{\ell,m}^p f(\Omega_p) + {}_s\bar{Y}_{\ell,m}^p(\hat{\delta}f)(\Omega_p) \right) \end{aligned} \quad (\text{H.10})$$

Appendix I: Derivative mode projections for f° scalar (pp on circular orbit): Integration by parts and chain rule for $\tilde{\partial}$ on Held scalars

There are three terms in the mode-decomposition of f° involving derivative which must be performed carefully due to their distributional character. These are

$$A := \left[-\frac{1}{2} \frac{2\rho^2 - \bar{\rho}^2}{\rho^3(\rho + \bar{\rho})} \right]_{r=r_p} \tilde{\partial}' T_m^\circ, \quad (\text{I.1a})$$

$$B := \left[-\frac{1}{4} \frac{\rho^2}{\bar{\rho}^3(\rho + \bar{\rho})} \right]_{r=r_p} \tilde{\partial}' \tilde{\partial} T_m^\circ, \quad (\text{I.1b})$$

$$C := \left[\frac{1}{4} \frac{\rho^6 + 2\rho^5\bar{\rho} - 3\rho^4\bar{\rho}^2 + \rho^2\bar{\rho}^4 - 2\rho\bar{\rho}^5 - \bar{\rho}^6}{\rho^3\bar{\rho}^3(\rho + \bar{\rho})} \bar{\tau}^\circ \right]_{r=r_p} \tilde{\partial} T_l^\circ. \quad (\text{I.1c})$$

Projection of the A -term

We consider

$$I_{\ell m}^{(A)} = \int_{S^2} \bar{Y}_{\ell m} \left[-\frac{1}{2} \frac{2\rho^2 - \bar{\rho}^2}{\rho^3(\rho + \bar{\rho})} \tilde{\partial}' T_m^\circ \right] d\Omega. \quad (\text{I.2})$$

where

$$T_m^\circ = J_m^\circ \delta_p(\Omega) \quad (\text{I.3})$$

where J° is taken at the point (u_p, z_p, φ_p) . Define

$$G(\rho, \bar{\rho}) := -\frac{1}{2} \frac{2\rho^2 - \bar{\rho}^2}{\rho^3(\rho + \bar{\rho})}. \quad (\text{I.4})$$

Using the adjoint property

$$\langle \phi | \tilde{\partial}' \psi \rangle_{L^2} = \langle \tilde{\partial}'^\dagger \phi | \psi \rangle_{L^2}, \quad \tilde{\partial}'^\dagger = -\tilde{\partial}, \quad (\text{I.5})$$

we move $\tilde{\partial}'$ off T_m° ,

$$I_{\ell m}^{(A)} = - \int_{S^2} T_m^\circ \tilde{\partial} [\bar{Y}_{\ell m} G] d\Omega. \quad (\text{I.6})$$

Expanding,

$$\tilde{\partial}(\bar{Y}_{\ell m} G) = (\tilde{\partial} \bar{Y}_{\ell m}) G + \bar{Y}_{\ell m} \tilde{\partial} G. \quad (\text{I.7})$$

Using the Held identities

$$\tilde{\partial} \rho = \rho^2 \tau^\circ, \quad \tilde{\partial} \bar{\rho} = -\bar{\rho}^2 \tau^\circ, \quad (\text{I.8})$$

a short chain-rule computation gives

$$\begin{aligned}\tilde{\partial}G &= \tau^\circ (\rho^2 \partial_\rho G - \bar{\rho}^2 \partial_{\bar{\rho}} G), \\ &= \frac{(2\rho + \bar{\rho})(2\rho^2 - 2\rho\bar{\rho} - \bar{\rho}^2)}{2\rho^2(\rho + \bar{\rho})} \tau^\circ.\end{aligned}\tag{I.9}$$

On the worldline

$$\rho_p = \bar{\rho}_p = -\frac{1}{r_p}, \quad \tau_p^\circ = -\frac{ia}{\sqrt{2}},\tag{I.10}$$

and therefore

$$G_p = -\frac{r_p^2}{4}, \quad (\tilde{\partial}G)_p = -\frac{3iar_p}{4\sqrt{2}r_p}.\tag{I.11}$$

Hence

$$I_{\ell m}^{(A)} = J_m^\circ \left[(\tilde{\partial}Y_{\ell m}) \frac{r_p^2}{4} + Y_{\ell m} \frac{3iar_p}{4\sqrt{2}} \right]_{\Omega=\Omega_p}.\tag{I.12}$$

Lastly GHP covariance to work at the level of fixed modes in KN coordinates where

$$\tilde{\partial} \mapsto {}_s\mathcal{L}_\omega^\dagger, \quad \tilde{\partial}' \mapsto {}_s\mathcal{L}_\omega.\tag{I.13}$$

Projection of the B -term

We now consider

$$I_{\ell m}^{(B)} = \int_{S^2} \bar{Y}_{\ell m} \frac{1}{4} \frac{\rho^2}{\bar{\rho}^3(\rho + \bar{\rho})} \tilde{\partial}' \tilde{\partial} T_l^\circ d\Omega.\tag{I.14}$$

Define

$$H(\rho, \bar{\rho}) := \frac{1}{4} \frac{\rho^2}{\bar{\rho}^3(\rho + \bar{\rho})}.\tag{I.15}$$

Move $\tilde{\partial}'$ by parts,

$$I_{\ell m}^{(B)} = - \int_{S^2} \tilde{\partial} [\bar{Y}_{\ell m} H] \tilde{\partial} T_l^\circ d\Omega.\tag{I.16}$$

Integrate by parts again,

$$I_{\ell m}^{(B)} = \int_{S^2} T_l^\circ \tilde{\partial}^\dagger \tilde{\partial} [Y_{\ell m} H] d\Omega.\tag{I.17}$$

Expanding,

$$\tilde{\partial}^\dagger \tilde{\partial}(\bar{Y}_{\ell m} H) = (\tilde{\partial}^\dagger \tilde{\partial} \bar{Y}_{\ell m}) H + (\tilde{\partial}^\dagger H)(\tilde{\partial} \bar{Y}_{\ell m}) + (\tilde{\partial}^\dagger \bar{Y}_{\ell m})(\tilde{\partial} H) + \bar{Y}_{\ell m} \tilde{\partial}^\dagger \tilde{\partial} H. \quad (\text{I.18})$$

Using

$$\tilde{\partial}^\dagger \rho = \rho^2 \bar{\tau}^\circ, \quad \tilde{\partial}^\dagger \bar{\rho} = -\bar{\rho}^2 \bar{\tau}^\circ, \quad (\text{I.19})$$

and the observation that

$$\tilde{\partial} H = -\frac{\tau^\circ}{\bar{\tau}^\circ} \tilde{\partial}' H = \frac{\tau^\circ}{\bar{\tau}^\circ} \tilde{\partial}^\dagger H \quad (\text{I.20})$$

one finds that

$$\tilde{\partial} H = \frac{\rho^2(\rho + 4\bar{\rho})\tau^\circ}{4\bar{\rho}^2(\rho + \bar{\rho})} = \tau^\circ(\rho + 4\bar{\rho})H \quad (\text{I.21})$$

and

$$\tilde{\partial}^\dagger \tilde{\partial} H = \tau^\circ \bar{\tau}^\circ (\rho^2 + 4\rho\bar{\rho} + 6\bar{\rho}^2)H. \quad (\text{I.22})$$

For an equatorial circular orbit one gets

$$H_p = \frac{r_p^2}{8}, \quad (\tilde{\partial}^\dagger H)_p = -\frac{5iar_p}{8\sqrt{2}}, \quad (\tilde{\partial} H)_p = \frac{5iar_p}{8\sqrt{2}}, \quad (\tilde{\partial}^\dagger \tilde{\partial} H)_p = -\frac{11a^2}{8}. \quad (\text{I.23})$$

Therefore

$$I_{\ell m}^{(B)} = J_l^\circ \left[\frac{r_p^2}{8} (\tilde{\partial}^\dagger \tilde{\partial} \bar{Y}_{\ell m}) - \frac{5iar_p}{8\sqrt{2}} (\tilde{\partial} \bar{Y}_{\ell m}) + \frac{5iar_p}{8\sqrt{2}} (\tilde{\partial}^\dagger \bar{Y}_{\ell m}) - \frac{11a^2}{8} \bar{Y}_{\ell m} \right]_{\Omega=\Omega_p}. \quad (\text{I.24})$$

C term

We consider the mode projection integral so that

$$I_{\ell m}^{(C)} = \int_{S^2} \bar{Y}_{\ell m} F(\rho, \bar{\rho}) \tilde{\partial} T_l^\circ d\Omega. \quad (\text{I.25})$$

with

$$F(\rho, \bar{\rho}) := \frac{1}{4} \frac{\rho^6 + 2\rho^5\bar{\rho} - 3\rho^4\bar{\rho}^2 + \rho^2\bar{\rho}^4 - 2\rho\bar{\rho}^5 - \bar{\rho}^6}{\rho^3\bar{\rho}^3(\rho + \bar{\rho})} \bar{\tau}^\circ, \quad (\text{I.26})$$

so that

$$I_{\ell m}^{(C)} = \int_{S^2} \bar{Y}_{\ell m}(\theta, \varphi) \frac{1}{4} \frac{\rho^6 + 2\rho^5\bar{\rho} - 3\rho^4\bar{\rho}^2 + \rho^2\bar{\rho}^4 - 2\rho\bar{\rho}^5 - \bar{\rho}^6}{\rho^3\bar{\rho}^3(\rho + \bar{\rho})} \bar{\tau}^\circ \tilde{\partial} T_l^\circ d\Omega. \quad (\text{I.27})$$

Using the adjoint property of $\tilde{\partial}$ on the sphere, we move $\tilde{\partial}$ off T_l° :

$$\begin{aligned} I_{\ell m}^{(C)} &= \int_{S^2} T_l^\circ \tilde{\partial}^\dagger [\bar{Y}_{\ell m} F(\rho, \bar{\rho})] d\Omega \\ &= - \int_{S^2} T_l^\circ \tilde{\partial}' [\bar{Y}_{\ell m} F(\rho, \bar{\rho})] d\Omega. \end{aligned} \quad (\text{I.28})$$

Expanding the derivative,

$$\tilde{\partial}' [\bar{Y}_{\ell m} F] = (\tilde{\partial}' \bar{Y}_{\ell m}) F + \bar{Y}_{\ell m} \tilde{\partial}' F. \quad (\text{I.29})$$

Chain rule on $F(\rho, \bar{\rho})$

Using the Ricci/Held identities (??) again and $\tilde{\partial}^\dagger = -\tilde{\partial}'$, we find

$$\tilde{\partial}^\dagger F = \rho^2 \bar{\tau}^\circ \frac{\partial F}{\partial \rho} - \bar{\rho}^2 \bar{\tau}^\circ \frac{\partial F}{\partial \bar{\rho}}. \quad (\text{I.30})$$

Explicitly, letting

$$P(\rho, \bar{\rho}) = \rho^6 + 2\rho^5 \bar{\rho} - 3\rho^4 \bar{\rho}^2 + \rho^2 \bar{\rho}^4 - 2\rho \bar{\rho}^5 - \bar{\rho}^6, \quad (\text{I.31})$$

$$Q(\rho, \bar{\rho}) = 4\rho^3 \bar{\rho}^3 (\rho + \bar{\rho}), \quad (\text{I.32})$$

so that $F = P/Q \bar{\tau}^\circ$, the partial derivatives are

$$\frac{\partial F}{\partial \rho} = \bar{\tau}^\circ \frac{Q(\partial_\rho P) - P(\partial_\rho Q)}{Q^2}, \quad (\text{I.33})$$

$$\frac{\partial F}{\partial \bar{\rho}} = \bar{\tau}^\circ \frac{Q(\partial_{\bar{\rho}} P) - P(\partial_{\bar{\rho}} Q)}{Q^2}, \quad (\text{I.34})$$

with

$$\partial_\rho P = 6\rho^5 + 10\rho^4 \bar{\rho} - 12\rho^3 \bar{\rho}^2 + 2\rho \bar{\rho}^4 - 2\bar{\rho}^5, \quad (\text{I.35})$$

$$\partial_{\bar{\rho}} P = 2\rho^5 - 6\rho^4 \bar{\rho} + 4\rho^2 \bar{\rho}^3 - 10\rho \bar{\rho}^4 - 6\bar{\rho}^5, \quad (\text{I.36})$$

$$\partial_\rho Q = 4\bar{\rho}^3 (4\rho^3 + 3\rho^2 \bar{\rho}), \quad (\text{I.37})$$

$$\partial_{\bar{\rho}} Q = 4\rho^3 (4\bar{\rho}^3 + 3\rho \bar{\rho}^2). \quad (\text{I.38})$$

The fully explicit chain-rule expression is then

$$\tilde{\partial}^\dagger F = \rho^2 \bar{\tau}^\circ \frac{\bar{\tau}^\circ [Q(\partial_\rho P) - P(\partial_\rho Q)]}{Q^2} - \bar{\rho}^2 \bar{\tau}^\circ \frac{\bar{\tau}^\circ [Q(\partial_{\bar{\rho}} P) - P(\partial_{\bar{\rho}} Q)]}{Q^2}. \quad (\text{I.39})$$

This is the expression to substitute into the mode projection integral:

$$I_{\ell m}^{(C)} = - \int_{\mathbb{S}^2} J_l^\circ \delta_p(\Omega) \left[(\tilde{\partial}^\dagger \bar{Y}_{\ell m}) F + \bar{Y}_{\ell m} \tilde{\partial}^\dagger F \right] \quad (\text{I.40})$$

where $T_l^\circ = J_l^\circ \delta_p(\Omega)$ is delta-supported at the particle and all functions can then be evaluated at the particle's position. On the worldline

$$F_p = \frac{ir_p a}{4\sqrt{2}}, \quad \tilde{\partial}^\dagger F_p = -\frac{5a^2}{8} \quad (\text{I.41})$$

Appendix J: Held integration for x_{ab}

Here we utilize Held's geometric integration formalism [10, 11] to derive integral expressions for corrector field x_{ab} in type-D spacetimes. We then relate these results to the general solution for x_{ab} to derive expressions for the coefficients $a^\circ, b^\circ, c^\circ, e^\circ$ and f° in terms of explicit radial integrals of the NP components of the stress energy. We take the source T_{ab} to have spatially compact support away from the past horizon in the radial interval $r_{\min} < r < r_{\max}$ but otherwise leave it arbitrary. We integrate the source from the past horizon $r_+ < r_{\min}$, where it vanishes, outwards along the principal null direction l^a to $r = \infty$. We utilize the general solution for the string piece x_{ab}^S in the region outside the source and ignore the contribution inside the source x_{ab}^M , given already in integral form elsewhere.

In practice we exchange integration variables from r to ρ using the relations $dr = \rho^{-2} d\rho$, $\bar{\rho} = \rho/(1 + \rho\Omega^\circ)$ and $\rho_{\max/\min} = \rho(r_{\min/\max})$. To derive the coefficients $a^\circ, b^\circ, \dots, f^\circ$, we put the both the general solution and our integral forms of the solution to follow in a canonical Laurent polynomial form. Then the coefficients of ρ can be easily equated in both forms, leading to algebraic relations for constants $a^\circ, b^\circ, \dots, f^\circ$. To put things in canonical form we simply multiply by an appropriate factor which eliminates all positive powers of $\bar{\rho}$.

We start with the seed equation governing the trace component $x_{m\bar{m}}$, Eq. (I.1). Integrating (I.1) with our assumptions $\text{supp}(T_{ab})$ gives

$$x_{m\bar{m}} = \frac{\bar{\rho}}{\rho} \int_{\rho_{\max}}^{\rho} d\rho_2 (1 + \rho_2 \Omega^\circ) \int_{\rho_{\max}}^{\rho_2} d\rho_1 \rho_1^{-4} T_{ll}. \quad (\text{J.1})$$

In the source-free region $r > r_{\max}$ the solution (J.1) can be rewritten as

$$x_{m\bar{m}}^S = \frac{C_1}{2}(\rho + \bar{\rho}) + C_3 \frac{\bar{\rho}}{\rho}, \quad (\text{J.2})$$

where

$$\begin{aligned} C_1 &= \int_{\rho_{\max}}^{\rho_{\min}} d\rho_1 \frac{T_{ll}}{\rho_1^4}, \\ C_3 &= \int_{\rho_{\max}}^{\rho_{\min}} d\rho_2 (1 + \rho_2 \Omega^\circ) \int_{\rho_{\max}}^{\rho_2} d\rho_1 \frac{T_{ll}}{\rho_1^4} - C_1 \rho_{\min} (1 + \tfrac{1}{2} \rho_{\min} \Omega^\circ) \end{aligned} \quad (\text{J.3a})$$

To relate C_1 and C_3 in (J.2) to the coefficients a° and b° in the general solution

$$x_{m\bar{m}}^S = a^\circ \left(\frac{\rho}{\bar{\rho}} + \frac{\bar{\rho}}{\rho} \right) + b^\circ (\rho + \bar{\rho}), \quad (\text{J.4})$$

we multiply both sides of Eqs. (J.2) and (J.4) by the homogeneous solution $\rho/\bar{\rho} = 1 + \rho\Omega^\circ$ and replace all remaining instances of $\bar{\rho}$ with ρ and Ω° to put the solution in canonical form

$$\frac{\rho}{\bar{\rho}} x_{m\bar{m}}^S = C_3 + \frac{C_1}{2} \rho (2 + \rho \Omega^\circ) \quad (\text{J.5})$$

$$= a^\circ (2 + 2\rho\Omega^\circ + \rho^2\Omega^\circ) + b^\circ \rho (2 + \rho\Omega^\circ) \quad (\text{J.6})$$

to find that

$$a^\circ = \tfrac{1}{2} C_3, \quad b^\circ = \tfrac{1}{2} (C_1 - \Omega^\circ C_3). \quad (\text{J.7})$$

Moving now to x_{nm} , recall that

$$x_{nm} = 2\rho(\bar{\rho} + \rho) \int_{\rho_{\max}}^{\rho} \frac{d\rho_2}{\rho_2^4} \left(\frac{1 + \rho_2 \Omega^\circ}{2 + \rho_2 \Omega^\circ} \right)^2 \int_{\rho_{\max}}^{\rho_2} \frac{d\rho_1}{\rho_1^2} \frac{2 + \rho_1 \Omega^\circ}{1 + \rho_1 \Omega^\circ} (T_{lm} + \mathcal{N} x_{m\bar{m}}), \quad (\text{J.8})$$

where $\mathcal{N}$ is the differential operator appearing on the RHS of (III.2). In the vacuum region $r > r_{\max}$ we have

$$x_{nm}^S = 2\rho(\rho + \bar{\rho}) \left(D_3 - \frac{D_1}{6\rho^3} (1 + \rho\Omega^\circ)(2 + \rho\Omega^\circ) + I_1(\rho) \right) \quad (\text{J.9})$$

where the coefficients D_1 and D_3 are given by the integrals

$$D_1 = \int_{\rho_{\max}}^{\rho_{\min}} d\rho \frac{2 + \rho\Omega}{\rho^2(1 + \rho\Omega)} (T_{lm} + \mathcal{N} x_{m\bar{m}}), \quad (\text{J.10})$$

$$\begin{aligned} D_3 &= \int_{\rho_{\max}}^{\rho_{\min}} d\rho_2 \left(\frac{1 + \rho_2 \Omega^\circ}{\rho_2^2(2 + \rho_2 \Omega^\circ)} \right)^2 \int_{\rho_{\max}}^{\rho_2} \frac{d\rho_1}{\rho_1^2} \frac{2 + \rho_1 \Omega^\circ}{1 + \rho_1 \Omega^\circ} (T_{lm} + \mathcal{N} x_{m\bar{m}}) \\ &\quad - \frac{1 + 2\rho_{\min}\Omega^\circ}{\rho_{\min}^3(2 + \rho_{\min}\Omega^\circ)} D_1 \end{aligned} \quad (\text{J.11})$$

in the interior of the source and

$$I_1(\rho) = \int_{\rho_{\min}}^{\rho} \frac{d\rho_2}{\rho_2^4} \left(\frac{1 + \rho_2 \Omega^\circ}{2 + \rho_2 \Omega^\circ} \right)^2 \int_{\rho_{\min}}^{\rho_2} \frac{d\rho_1}{\rho_1^2} \frac{2 + \rho_1 \Omega^\circ}{1 + \rho_1 \Omega^\circ} (\mathcal{N} x_{m\bar{m}}) \quad (\text{J.12})$$

results from integration of $x_{m\bar{m}}$ in the vacuum region $r > r_{\max}$. The integral I_1 is needed to determine c° and e° . It has the general structure $I_1 = \frac{(\rho - \rho_{\min})^2 \bar{\rho}^2}{(\rho + \bar{\rho}) \rho^4} p_3(\rho)$, where $p_3(\rho)$ is a third-order polynomial in ρ . The $\bar{\rho}^2$ contribution from the numerator of I_1 is the highest power of $\bar{\rho}$ in x_{nm} , and must be factored out to put the solution in canonical Laurent form. To compute the coefficients c° and e° we also put the general solution (III.3) in Laurent form. After putting both the solutions in canonical form, equating coefficients of ρ^{-3} and ρ^2 gives

$$c^\circ = -\frac{1}{6} \left\{ \frac{2\rho_{\min}^2}{1 + \rho_{\min}\Omega^\circ} (\tilde{\partial}b^\circ + \Omega^\circ \tilde{\partial}a^\circ) + 8D_1 + \frac{\tau^\circ \rho_{\min}^2 (2 + \rho_{\min}\Omega^\circ)^2}{(1 + \rho_{\min}\Omega^\circ)^4} [\rho_{\min}^2 \Omega^\circ b^\circ + (2 + \rho_{\min}\Omega^\circ (4 + \rho_{\min}\Omega^\circ)) a^\circ] \right\}, \quad (\text{J.13})$$

$$e^\circ = 2D_3 - \frac{\tau^\circ (2 + \rho_{\min}\Omega^\circ)}{6\rho_{\min}(1 + \rho_{\min}\Omega^\circ)^4} (4 + 10\rho_{\min}\Omega^\circ + 9\rho_{\min}^2 \Omega^{\circ 2} + 4\rho_{\min}^3 \Omega^{\circ 3}) a^\circ - \frac{\tau^\circ (6 + 16\rho_{\min}\Omega^\circ + 13\rho_{\min}^2 \Omega^{\circ 2} + 4\rho_{\min}^3 \Omega^{\circ 3})}{6(1 + \rho_{\min}\Omega^\circ)^4} b^\circ - \frac{\tilde{\partial}b^\circ + \Omega^\circ \tilde{\partial}a^\circ}{3\rho_{\min}(1 + \rho_{\min}\Omega^\circ)}. \quad (\text{J.14})$$

Lastly, for x_{nn} , we take on task of computing f° from the final integral

$$x_{nn} = 2(\rho + \bar{\rho}) \int_{\rho_{\max}}^{\rho} \frac{d\rho_1}{\rho_1^4} \left(\frac{1 + \rho_1 \Omega^\circ}{2 + \rho_1 \Omega^\circ} \right)^2 (T_{ln} + \text{Re}\mathcal{U}x_{m\bar{m}} + \text{Re}(\mathcal{V}x_{nm})), \quad (\text{J.15})$$

where $\mathcal{U}$ and $\mathcal{V}$ are the differential operators given in (III.5) and (III.6), respectively. We first rewrite the solution as

$$x_{nn} = 2(\rho + \bar{\rho})(E_1 + I_2(\rho)), \quad (\text{J.16})$$

where

$$E_1 = \int_{\rho_{\max}}^{\rho_{\min}} \frac{d\rho_1}{\rho_1^4} \left(\frac{1 + \rho_1 \Omega^\circ}{2 + \rho_1 \Omega^\circ} \right)^2 (T_{ln} + \text{Re}\mathcal{U}x_{m\bar{m}} + \text{Re}(\mathcal{V}x_{nm})), \quad (\text{J.17})$$

is a constant involving the stress energy and

$$I_2(\rho) = \int_{\rho_{\min}}^{\rho} \frac{d\rho_1}{\rho_1^4} \left(\frac{1 + \rho_1 \Omega^\circ}{2 + \rho_1 \Omega^\circ} \right)^2 (\text{Re}\mathcal{U}x_{m\bar{m}} + \text{Re}(\mathcal{V}x_{nm})) \quad (\text{J.18})$$

is determined by $x_{m\bar{m}}$ and x_{nm} in the source free region $r > r_{\max}$. The form of I_2 is simply $I_2 = \frac{(\rho - \rho_{\min})^2 \bar{\rho}^2}{\rho^3 (\rho + \bar{\rho})} p_5(\rho)$ where $p_5(\rho)$ is a fifth-order polynomial in ρ . We therefore solve for f° by multiplying the general solution and the particular solution by $1/\bar{\rho}^2$ to put them in canonical

form and equate coefficients. The result for f° is

$$\begin{aligned}
f^\circ = -\text{Re} \left\{ \frac{1}{2\rho_{\min}^2(1+\rho_{\min}\Omega^\circ)(2+\rho_{\min}\Omega^\circ)} \right. & \left(-2(\tilde{\delta}'c^\circ + \tilde{\delta}\bar{c}^\circ - \tilde{\mathbf{P}}'a^\circ) \right. \\
& + \left[4\tilde{\delta}\tilde{\delta}'a^\circ - 8\rho'^\circ a^\circ + 4\tilde{\mathbf{P}}'b^\circ + 8\bar{\tau}^\circ c^\circ + 8\tau^\circ \bar{c}^\circ + 2\Omega^\circ(6\tilde{\mathbf{P}}'a^\circ - 5\tilde{\delta}'c^\circ) \right] \rho_{\min} \\
& + \left[2a^\circ(\bar{\Psi}^\circ + \Psi^\circ) - 8E_1 + 8\Omega^\circ \left(\tilde{\delta}\tilde{\delta}a^\circ + \tilde{\mathbf{P}}'b^\circ - 2\rho'^\circ a^\circ + 3\tau^\circ \bar{c}^\circ + \bar{\tau}^\circ c^\circ \right) \right. \\
& \quad \left. + 2\Omega^{\circ 2}(6\tilde{\delta}\bar{c}^\circ - 4\tilde{\delta}'c^\circ + 7\tilde{\mathbf{P}}'a^\circ) \right] \rho_{\min}^2 \\
& + \left[2(\Psi^\circ + \bar{\Psi}^\circ)b^\circ + 2\tilde{\delta}\bar{e}^\circ + 2\tilde{\delta}'e^\circ + 4\tau^\circ \tilde{\delta}'b^\circ + 4\bar{\tau}^\circ \tilde{\delta}b^\circ + 24\tau^\circ \bar{\tau}^\circ a^\circ \right. \\
& + 3\Omega^\circ(a^\circ(3\bar{\Psi}^\circ - \Psi^\circ) - 4\Omega^\circ E_1) \\
& + 4\Omega^{\circ 2}(\tilde{\mathbf{P}}'b^\circ + 2\tilde{\delta}\tilde{\delta}'a^\circ - 2\rho'^\circ a^\circ + 6\tau^\circ \Omega^{\circ 2} \bar{c}^\circ) + \Omega^{\circ 3}(16\tilde{\delta}\bar{c}^\circ + 9\tilde{\mathbf{P}}'a^\circ) \left. \right] \rho_{\min}^3 \\
& + \left[8\tau^\circ \bar{\tau}^\circ b^\circ + 4\bar{\tau}^\circ e^\circ + 4\tau^\circ \bar{e}^\circ + 4\Omega^\circ \left(6\tau^\circ \bar{\tau}^\circ a^\circ + (\Psi^\circ + \bar{\tau}^\circ \tilde{\delta} + \tau^\circ \tilde{\delta}')b^\circ + \tilde{\delta}'e^\circ \right) \right. \\
& + \Omega^{\circ 2}((9\Psi^\circ - 3\bar{\Psi}^\circ)a^\circ - E_1) + 4\Omega^{\circ 3}(\tilde{\delta}\tilde{\delta}'a^\circ + 2\tau^\circ \bar{c}^\circ) + 3\Omega^{\circ 4}(\tilde{\mathbf{P}}'a^\circ + 2\tilde{\delta}\bar{c}^\circ) \left. \right] \rho_{\min}^4 \\
& \left. + 2\Omega^\circ \left[2\bar{\tau}^\circ(e^\circ + \tau^\circ b^\circ) + \Omega^\circ \left(2\tau^\circ \bar{\tau}^\circ a^\circ + \Psi^\circ b^\circ + \tilde{\delta}'e^\circ \right) + \Omega^{\circ 2}\Psi^\circ a^\circ \right] \rho_{\min}^5 \right) \left. \right\}
\end{aligned} \tag{J.19}$$

In the Schwarzschild limit $\Omega^\circ = \tau^\circ = 0$, the coefficients simplify to

$$\begin{aligned}
a^\circ &= \frac{1}{2} \int_{\rho_{\max}}^{\rho_{\min}} d\rho_2 \int_{\rho_{\max}}^{\rho_2} d\rho_1 \frac{T_{ll}}{\rho^4} - \frac{1}{2} \rho_{\min} \int_{\rho_{\max}}^{\rho_{\min}} d\rho_1 \frac{T_{ll}}{\rho_1^4}, \\
b^\circ &= \frac{1}{2} \int_{\rho_{\max}}^{\rho_{\min}} d\rho_1 \frac{T_{ll}}{\rho_1^4},
\end{aligned} \tag{J.20}$$

$$c^\circ = -\frac{1}{3} \left(\rho_{\min}^2 \tilde{\delta}b^\circ + 8 \int_{\rho_{\max}}^{\rho_{\min}} \frac{d\rho}{\rho^2} (T_{lm} + \mathcal{N}x_{m\bar{m}}) \right), \tag{J.21}$$

$$e^\circ = \int_{\rho_{\max}}^{\rho_{\min}} \frac{d\rho_2}{\rho_2^4} \int_{\rho_{\max}}^{\rho_2} \frac{d\rho_1}{\rho_1^2} (T_{lm} + \mathcal{N}x_{m\bar{m}}) + \frac{3c^\circ}{4\rho_{\min}^3} - \frac{\tilde{\delta}b^\circ}{12\rho_{\min}} \tag{J.22}$$

$$\begin{aligned}
f^\circ &= \frac{1}{4} \int_{\rho_{\max}}^{\rho_{\min}} \frac{d\rho_1}{\rho_1^4} (T_{ln} + \text{Re}\mathcal{U}x_{m\bar{m}} + \text{Re}(\mathcal{V}x_{nm})) + \frac{1}{4\rho_{\min}^2} \left(\tilde{\delta}'c^\circ + \tilde{\delta}\bar{c}^\circ + \rho_{\min}^3(\tilde{\delta}'e^\circ + \tilde{\delta}\bar{e}^\circ) \right) \\
&+ \frac{\rho'^\circ}{\rho_{\min}} a^\circ - \frac{1}{4}(\Psi^\circ + \bar{\Psi}^\circ)(\rho_{\min}b^\circ + a^\circ) - \frac{1}{2\rho_{\min}^2} \tilde{\mathbf{P}}'a^\circ - \frac{1}{2\rho_{\min}} \tilde{\mathbf{P}}'b^\circ - \frac{1}{2\rho_{\min}} \tilde{\delta}\tilde{\delta}'a^\circ.
\end{aligned} \tag{J.23}$$

Appendix K: Transport boundary data for a box window

1. Left WT boundary data

a. Box window function

It is necessary in the effective source approach to taper the puncture to zero outside of the worldtube. For the purpose of comparing to past work in Schwarzschild, we here derive the continuity conditions at the worldtube boundaries imposed by windowing with “Box” function of width 2λ centered on the particle

$$h_{\mu\nu}^{\mathcal{P}}(x; \lambda) = h_{\mu\nu}^{\mathcal{P}}(x) \Theta_{\lambda}^M(x) \quad (\text{K.1})$$

where in outgoing KN coordinates

$$\Theta_{\lambda}^M = \Theta(r - |r_p(u) - \lambda|) \quad (\text{K.2})$$

Through the EFE, the action of the operator

$$\mathcal{E}_{\mu\nu}[h^{\mathcal{P}}](x) = -T_{\mu\nu}^{\mathcal{R}}(x), \quad x \notin \gamma_p \quad (\text{K.3})$$

gives rise to certain continuity conditions at the boundaries $r_{\max} = r_p + \lambda$ and $r_{\min} = r_p - \lambda$.

The second-order nature of the Einstein tensor gives rise to the distributional form

$$T_{\mu\nu}^{\mathcal{R}} = T^M \Theta^M + T_{\mu\nu}^{(\delta, -)} \delta_-(r) + T_{\mu\nu}^{(\delta', -)} \delta'_-(r) + T_{\mu\nu}^{(\delta, +)} \delta_+(r) + T_{\mu\nu}^{(\delta', +)} \delta'_+(r) \quad (\text{K.4})$$

where $\delta_-(r) = \Theta'(r - r_{\min})$ and $\delta_+(r) = \Theta'(r - r_{\max})$.

Let us now make the simplifying assumption of zero eccentricity

$$\partial_u \Theta^M = -\delta_-(r) \dot{r}_p + \delta_+(r) \dot{r}_p = 0 \quad (\text{K.5})$$

and introduce the notation

$$x_{\mu\nu}^- = \lim_{r \rightarrow (r_{\min})^+} x_{\mu\nu}, \quad x_{\mu\nu}^+ = \lim_{r \rightarrow (r_{\max})^-} x_{\mu\nu} \quad (\text{K.6})$$

Equating distributions gives (in Outgoing KN chart) $x^\mu = (u, r, \theta, \varphi)$ and Kinnersley tetrad¹⁰

$$\begin{aligned} x_{m\bar{m}}^- &= T_{ll}^{(\delta', -)}, \\ (\partial_r x_{m\bar{m}})^- &= T_{ll}^{(\delta, -)} + (\rho + \bar{\rho}) T_{ll}^{(\delta', -)}, \quad r = r_{\min} \end{aligned} \quad (\text{K.7})$$

¹⁰ The extraction of jump conditions using $\partial_r \Theta(r - r_p)$ breaks the manifest boost-covariant form of operators such as $\mathbf{P}'$, since it singles out a preferred coordinate normal to the matching surface. However, this does not break spin covariance: under a spin rotation $m \rightarrow e^{i\chi} m$, $\bar{m} \rightarrow e^{-i\chi} \bar{m}$, the vectors l^a and n^a are unchanged, and hence the u - and r -pieces of $\mathbf{P}'$ remain spin invariant. Thus the loss of covariance is only in the boost sector, not in the spin sector.

$$\begin{aligned}
x_{nm}^- &= 2T_{lm}^{(\delta', -)}, \\
(\partial_r x_{nm})^- &= 4\rho T_{lm}^{(\delta', -)} + 2T_{lm}^{(\delta, -)} - (\bar{\rho} \tilde{\partial} T_{ll}^{(\delta', -)}) + 3\bar{\rho}(\rho + \bar{\rho})\tau^\circ T_{lm}^{(\delta', -)}, \quad r = r_{\min} \quad (\text{K.8})
\end{aligned}$$

and

$$\begin{aligned}
\frac{1}{2}(\rho + \bar{\rho})x_{nn}^- &= T_{ln}^{(\delta, -)} - \frac{\Delta}{2\Sigma} T_{ll}^{(\delta, -)} + \frac{r^2 + a^2}{\Sigma} (\partial_u T_{ll}^{(\delta', -)}) - \frac{r\Delta}{\Sigma^2} T_{ll}^{(\delta', -)} \\
&\quad - \left(\rho \tilde{\partial}' T_{lm}^{(\delta', -)} + \bar{\rho} \tilde{\partial} T_{lm}^{(\delta', -)} \right) + 2\rho\bar{\rho} \left(\tau^\circ T_{lm}^{(\delta', -)} + \bar{\tau}^\circ T_{lm}^{(\delta', -)} \right), \quad r = r_{\min} \quad (\text{K.9})
\end{aligned}$$

using that

$$\partial_r (\Delta/(2\Sigma)) - \Delta r/\Sigma^2 + (r\Delta/\Sigma - (r - M)/\Sigma) = -r\Delta/\Sigma^2 \quad (\text{K.10})$$

Using that $\rho'/\rho = -\Delta/(2\Sigma)$ and that $r = -(\rho + \bar{\rho})/(2\rho\bar{\rho})$ the simplified expression may also be written as

$$\begin{aligned}
\frac{1}{2}(\rho + \bar{\rho})x_{nn}^- &= T_{ln}^{(\delta, -)} - \frac{\Delta}{2\Sigma} T_{ll}^{(\delta, -)} + \frac{r^2 + a^2}{\Sigma} \partial_u T_{ll}^{(\delta', -)} - \frac{(\rho + \bar{\rho})\rho'}{\rho} T_{ll}^{(\delta', -)} \\
&\quad - \left(\rho \tilde{\partial}' T_{lm}^{(\delta', -)} + \bar{\rho} \tilde{\partial} T_{lm}^{(\delta', -)} \right) + 2\rho\bar{\rho} \left(\tau^\circ T_{lm}^{(\delta', -)} + \bar{\tau}^\circ T_{lm}^{(\delta', -)} \right). \quad (\text{K.11})
\end{aligned}$$

$$\begin{aligned}
T_{ln}^{(\delta', -)} &= \frac{\Delta}{2\Sigma} T_{ll}^{(\delta', -)} \\
&= -\frac{\rho'}{\rho} T_{ll}^{(\delta', -)} \quad (\text{K.12})
\end{aligned}$$

The last equation Eq.(K.12) provides a useful numerical check.

2. Right WT boundary data

a. Box window function

We now repeat the same extraction of boundary data, but with the initial data imposed at the right worldtube boundary

$$r_{\max} = r_p + \lambda. \quad (\text{K.13})$$

The relevant interior limit is now

$$x_{\mu\nu}^+ = \lim_{r \rightarrow (r_{\max})^-} x_{\mu\nu}, \quad (\partial_r x_{\mu\nu})^+ = \lim_{r \rightarrow (r_{\max})^-} \partial_r x_{\mu\nu}. \quad (\text{K.14})$$

The only additional subtlety is the orientation of the box window at the right boundary. Writing the box window as

$$\Theta_\lambda^M = \Theta(r - r_{\min})\Theta(r_{\max} - r) = \Theta(r - r_{\min}) - \Theta(r - r_{\max}), \quad (\text{K.15})$$

one has

$$\partial_r \Theta_\lambda^M = \delta_-(r) - \delta_+(r), \quad \partial_r^2 \Theta_\lambda^M = \delta'_-(r) - \delta'_+(r), \quad (\text{K.16})$$

where

$$\delta_-(r) = \Theta'(r - r_{\min}), \quad \delta_+(r) = \Theta'(r - r_{\max}). \quad (\text{K.17})$$

Thus, keeping the convention

$$T_{\mu\nu}^{\mathcal{R}} = T^M \Theta^M + T_{\mu\nu}^{(\delta,-)} \delta_-(r) + T_{\mu\nu}^{(\delta',-)} \delta'_-(r) + T_{\mu\nu}^{(\delta,+)} \delta_+(r) + T_{\mu\nu}^{(\delta',+)} \delta'_+(r), \quad (\text{K.18})$$

the right-boundary data are obtained from the left-boundary formulae by the replacement

$$x_{\mu\nu}^- \rightarrow x_{\mu\nu}^+, \quad T_{\mu\nu}^{(\delta,-)} \rightarrow -T_{\mu\nu}^{(\delta,+)}, \quad T_{\mu\nu}^{(\delta',-)} \rightarrow -T_{\mu\nu}^{(\delta',+)}, \quad r = r_{\min} \rightarrow r = r_{\max}. \quad (\text{K.19})$$

The minus sign is a consequence of the fact that the box window jumps from one to zero at the right boundary.

For the $m\bar{m}$ corrector scalar, the right-boundary conditions are therefore

$$x_{m\bar{m}}^+ = -T_{ll}^{(\delta',+)}, \quad (\partial_r x_{m\bar{m}})^+ = -T_{ll}^{(\delta,+)} - (\rho + \bar{\rho}) T_{ll}^{(\delta',+)}, \quad r = r_{\max}. \quad (\text{K.20})$$

For the nm corrector scalar one obtains

$$x_{nm}^+ = -2T_{lm}^{(\delta',+)}, \quad (\partial_r x_{nm})^+ = -4\rho T_{lm}^{(\delta',+)} - 2T_{lm}^{(\delta,+)} + \bar{\rho} \tilde{\partial} T_{ll}^{(\delta',+)} - 3\bar{\rho}(\rho + \bar{\rho}) \tau^\circ T_{lm}^{(\delta',+)}, \quad r = r_{\max}. \quad (\text{K.21})$$

Finally, for the nn corrector scalar we find

$$\begin{aligned} \frac{1}{2}(\rho + \bar{\rho})x_{nn}^+ &= -T_{ln}^{(\delta,+)} + \frac{\Delta}{2\Sigma} T_{ll}^{(\delta,+)} - \frac{r^2 + a^2}{\Sigma} (\partial_u T_{ll}^{(\delta',+)}) + \frac{r\Delta}{\Sigma^2} T_{ll}^{(\delta',+)} \\ &\quad + \left(\rho \tilde{\partial}' T_{lm}^{(\delta',+)} + \bar{\rho} \tilde{\partial} T_{l\bar{m}}^{(\delta',+)} \right) - 2\rho\bar{\rho} \left(\tau^\circ T_{l\bar{m}}^{(\delta',+)} + \bar{\tau}^\circ T_{lm}^{(\delta',+)} \right), \quad r = r_{\max}. \end{aligned} \quad (\text{K.22})$$

Using again

$$\frac{\rho'}{\rho} = -\frac{\Delta}{2\Sigma}, \quad r = -\frac{\rho + \bar{\rho}}{2\rho\bar{\rho}}, \quad (\text{K.23})$$

this may equivalently be written as

$$\begin{aligned} \frac{1}{2}(\rho + \bar{\rho})x_{nn}^+ &= -T_{ln}^{(\delta,+)} + \frac{\Delta}{2\Sigma} T_{ll}^{(\delta,+)} - \frac{r^2 + a^2}{\Sigma} \partial_u T_{ll}^{(\delta',+)} + \frac{(\rho + \bar{\rho})\rho'}{\rho} T_{ll}^{(\delta',+)} \\ &\quad + \left(\rho \tilde{\partial}' T_{lm}^{(\delta',+)} + \bar{\rho} \tilde{\partial} T_{l\bar{m}}^{(\delta',+)} \right) - 2\rho\bar{\rho} \left(\tau^\circ T_{l\bar{m}}^{(\delta',+)} + \bar{\tau}^\circ T_{lm}^{(\delta',+)} \right), \quad r = r_{\max}. \end{aligned} \quad (\text{K.24})$$

The analogue of the useful numerical constraint is

$$\begin{aligned} T_{ln}^{(\delta',+)} &= \frac{\Delta}{2\Sigma} T_{ll}^{(\delta',+)} \\ &= -\frac{\rho'}{\rho} T_{ll}^{(\delta',+)} . \end{aligned} \quad (\text{K.25})$$

This relation has the same form as the left-boundary constraint because both sides involve the same right-boundary δ' coefficients.

If the ODE system is integrated inward from $r = r_{\max}$, the data supplied to the solver are the interior values

$$(x_{\mu\nu}, \partial_r x_{\mu\nu}) \big|_{r=r_{\max}^-} = (x_{\mu\nu}^+, (\partial_r x_{\mu\nu})^+) . \quad (\text{K.26})$$

If instead the inward integration is formulated using the inward radial coordinate

$$R = r_{\max} - r, \quad (\text{K.27})$$

then the derivative data must be converted according to

$$\partial_R x_{\mu\nu} = -\partial_r x_{\mu\nu}. \quad (\text{K.28})$$

3. Corrector scalars in the vacuum region as determined by right boundary data

a. $x_{m\bar{m}}$

$$a^\circ = \frac{r_{\max}^2 - a^2 z^2}{2\Sigma(r_{\max})} x_{m\bar{m}}^+ + \frac{r_{\max}}{2} (\partial_r x_{m\bar{m}})^+ + a_{\text{src}}^\circ \quad (\text{K.29})$$

$$b^\circ = -\frac{2a^2 z^2}{\Sigma(r_{\max})} r_{\max} x_{m\bar{m}}^+ + b_{\text{src}}^\circ \quad (\text{K.30})$$

where

$$a_{\text{src}}^\circ = -\frac{r_{\max}}{2} T_{ll}^{(\delta,+)} - \frac{1}{2} T_{ll}^{(\delta',+)} \quad (\text{K.31})$$

$$b_{\text{src}}^\circ = \frac{r_{\max}^2 - a^2 z^2}{2} T_{ll}^{(\delta,+)} - r_{\max} T_{ll}^{(\delta',+)} \quad (\text{K.32})$$

So that

$$x_{m\bar{m}}^> = \left(\frac{r_{\max}}{2} T_{ll}^{(\delta,+)} - \frac{1}{2} T_{ll}^{(\delta',+)} \right) \left(\frac{\rho}{\bar{\rho}} + \frac{\bar{\rho}}{\rho} \right) + \left(\frac{r_{\max}^2 - a^2 z^2}{2} T_{ll}^{(\delta,+)} - r_{\max} T_{ll}^{(\delta',+)} \right) (\rho + \bar{\rho}). \quad (\text{K.33})$$

b. x_{nm}

We now determine the remaining homogeneous coefficients c° and d° for the outer solution x_{nm}^+ . In the notation of Eq. (63b), the coefficient denoted here by d° is the same object that is called e° in the paper.

Define

$$\rho_{r_{\max}} := \rho(r_{\max}), \quad \bar{\rho}_{r_{\max}} := \bar{\rho}(r_{\max}). \quad (\text{K.34})$$

From Eq. (63b), the homogeneous outer solution has the form

$$x_{nm}^+(r) = X(r) + \frac{2\rho - \bar{\rho}}{\rho^2} c^\circ + \rho(\rho + \bar{\rho}) d^\circ, \quad (\text{K.35})$$

where the part already fixed by a° and b° is

$$X(r) := \bar{\rho} \Omega^\circ \tilde{\partial} a^\circ + \frac{(\rho + \bar{\rho})^2}{\rho} \tau^\circ a^\circ + \rho \tilde{\partial} b^\circ + (\rho + \bar{\rho})^2 \tau^\circ b^\circ. \quad (\text{K.36})$$

It is therefore convenient to define the reduced field

$$\hat{x}_{nm}(r) := x_{nm}(r) - X(r),$$

so that

$$\hat{x}_{nm}(r) = u_c(r) c^\circ + u_d(r) d^\circ,$$

with basis functions

$$u_c(r) := \frac{2\rho - \bar{\rho}}{\rho^2}, \quad u_d(r) := \rho(\rho + \bar{\rho}).$$

Evaluating at $r = r_{\max}^+$, define

$$\hat{x}_{nm}^+ := \hat{x}_{nm}(r_{\max}^+), \quad \partial_r \hat{x}_{nm}^+ := \partial_r \hat{x}_{nm}(r_{\max}^+). \quad (\text{K.37})$$

Then

$$\hat{x}_{nm}^+ = u_c(r_{\max}) c^\circ + u_d(r_{\max}) d^\circ, \quad (\text{K.38})$$

$$\partial_r \hat{x}_{nm}^+ = u'_c(r_{\max}) c^\circ + u'_d(r_{\max}) d^\circ. \quad (\text{K.39})$$

Using

$$\partial_r \rho = \rho^2, \quad \partial_r \bar{\rho} = \bar{\rho}^2,$$

one finds

$$u_c(r_{\max}) = \frac{2\rho_{r_{\max}} - \bar{\rho}_{r_{\max}}}{\rho_{r_{\max}}^2}, \quad u_d(r_{\max}) = \rho_{r_{\max}}(\rho_{r_{\max}} + \bar{\rho}_{r_{\max}}),$$

and

$$u'_c(r_{\max}) = -2 + 2\frac{\bar{\rho}_{r_{\max}}}{\rho_{r_{\max}}} - \frac{\bar{\rho}_{r_{\max}}^2}{\rho_{r_{\max}}^2}, \quad u'_d(r_{\max}) = 2\rho_{r_{\max}}^3 + \rho_{r_{\max}}^2 \bar{\rho}_{r_{\max}} + \rho_{r_{\max}} \bar{\rho}_{r_{\max}}^2.$$

The Wronskian is

$$W := u_c(r_{\max})u'_d(r_{\max}) - u'_c(r_{\max})u_d(r_{\max}) = 6\rho_{r_{\max}}^2. \quad (\text{K.40})$$

Hence the coefficients are given by

$$c^\circ = \frac{\hat{x}_{nm}^+ u'_d(r_{\max}) - (\partial_r \hat{x}_{nm}^+) u_d(r_{\max})}{6\rho_{r_{\max}}^2}, \quad (\text{K.41})$$

$$d^\circ = \frac{-\hat{x}_{nm}^+ u'_c(r_{\max}) + (\partial_r \hat{x}_{nm}^+) u_c(r_{\max})}{6\rho_{r_{\max}}^2}. \quad (\text{K.42})$$

After simplification this becomes

$$c^\circ = \frac{(2\rho_{r_{\max}}^2 + \rho_{r_{\max}} \bar{\rho}_{r_{\max}} + \bar{\rho}_{r_{\max}}^2) \hat{x}_{nm}^+ - (\rho_{r_{\max}} + \bar{\rho}_{r_{\max}}) \partial_r \hat{x}_{nm}^+}{6\rho_{r_{\max}}},$$

and

$$d^\circ = \frac{(2\rho_{r_{\max}}^2 - 2\rho_{r_{\max}} \bar{\rho}_{r_{\max}} + \bar{\rho}_{r_{\max}}^2) \hat{x}_{nm}^+ + (2\rho_{r_{\max}} - \bar{\rho}_{r_{\max}}) \partial_r \hat{x}_{nm}^+}{6\rho_{r_{\max}}^4}.$$

Now suppose the total reduced source has endpoint distributional structure

$$\check{T}_{nm} = \check{T}_{nm}^{(\delta)} \delta(r - r_{\max}) + \check{T}_{nm}^{(\delta')} \delta'(r - r_{\max}) + \dots,$$

where

$$\check{T}_{nm} = T_{lm} + \mathcal{N}[x_{m\bar{m}}]$$

The relevant operator is

$$\mathcal{N}[x_{m\bar{m}}] := -\frac{1}{2} \left\{ \bar{\rho}(\mathbf{P} + \rho - \bar{\rho}) \check{\partial} - \bar{\rho}(3\bar{\rho} + \rho) \tau^\circ \mathbf{P} + 4\rho \bar{\rho}^2 \tau^\circ \right\} x_{m\bar{m}}.$$

Since this operator is only first order in the radial derivative $\mathbf{P}$, acting on a field $x_{m\bar{m}}$ with a jump can generate a $\delta(r - r_{\max})$ contribution, but not a $\delta'(r - r_{\max})$ contribution.

Write

$$x_{m\bar{m}} = x_{m\bar{m}}^< \Theta(r_{\max} - r) + x_{m\bar{m}}^> \Theta(r - r_{\max}),$$

and define the jump

$$[x_{m\bar{m}}] := \lim_{\epsilon \rightarrow 0} \left(x_{m\bar{m}}^>(r_{\max} + \epsilon) - x_{m\bar{m}}^<(r_{\max} - \epsilon) \right).$$

Then

$$\mathbb{P}x_{m\bar{m}} = (\mathbb{P}x_{m\bar{m}}^<) \Theta(r_{\max} - r) + (\mathbb{P}x_{m\bar{m}}^>) \Theta(r - r_{\max}) + [x_{m\bar{m}}] \delta(r - r_{\max}),$$

and similarly

$$\mathbb{P}(\tilde{\partial}x_{m\bar{m}}) = \tilde{\partial}[x_{m\bar{m}}] \delta(r - r_{\max}) + \dots,$$

since $\tilde{\partial}$ does not differentiate the Heaviside factor and since the two operators commute.

Therefore the distributional part of the $x_{m\bar{m}}$ -operator is

$$-\frac{1}{2}\bar{\rho}_{r_{\max}} \tilde{\partial}[x_{m\bar{m}}] \delta(r - r_{\max}) + \frac{1}{2}\bar{\rho}_{r_{\max}} (3\bar{\rho}_{r_{\max}} + \rho_{r_{\max}}) \tau^\circ [x_{m\bar{m}}] \delta(r - r_{\max}).$$

Hence, if the total reduced source in the x_{nm} equation is written as

$$\check{T}_{nm} = \check{T}_{lm}^{(\delta)} \delta(r - r_{\max}) + \check{T}_{lm}^{(\delta')} \delta'(r - r_{\max}) + \dots,$$

then the endpoint coefficients are

$$\check{T}_{lm}^{(\delta')} = T_{lm}^{(\delta',+)},$$

and

$$\check{T}_{lm}^{(\delta)} = T_{lm}^{(\delta,+)} - \frac{1}{2}\bar{\rho}_{r_{\max}} \tilde{\partial}[x_{m\bar{m}}] + \frac{1}{2}\bar{\rho}_{r_{\max}} (3\bar{\rho}_{r_{\max}} + \rho_{r_{\max}}) \tau^\circ [x_{m\bar{m}}].$$

Using the jump condition from the $x_{m\bar{m}}$ equation,

$$[x_{m\bar{m}}] = x_{m\bar{m}}^+ = T_{ll}^{(\delta',+)},$$

we obtain the explicit formulas

$$\boxed{\check{T}_{lm}^{(\delta')} = T_{lm}^{(\delta',+)}}$$

and

$$\boxed{\check{T}_{lm}^{(\delta)} = T_{lm}^{(\delta,+)} - \frac{1}{2}\bar{\rho}_{r_{\max}} \tilde{\partial}T_{ll}^{(\delta',+)} + \frac{1}{2}\bar{\rho}_{r_{\max}} (3\bar{\rho}_{r_{\max}} + \rho_{r_{\max}}) \tau^\circ T_{ll}^{(\delta',+)}.}$$

Thus the formulas for c° and d° obtained previously remain valid, provided one substitutes these corrected source coefficients.

In particular, the source-only contributions become

$$c_{\text{src}}^\circ = -\frac{\rho_{r_{\max}} + \bar{\rho}_{r_{\max}}}{6\rho_{r_{\max}}} \check{T}_{lm}^{(\delta,+)} + \frac{\bar{\rho}_{r_{\max}} (\bar{\rho}_{r_{\max}} - \rho_{r_{\max}})}{6\rho_{r_{\max}}} T_{lm}^{(\delta',+)},$$

$$d_{\text{src}}^\circ = \frac{2\rho_{r_{\max}} - \bar{\rho}_{r_{\max}}}{6\rho_{r_{\max}}^4} \check{T}_{lm}^{(\delta,+)} + \frac{6\rho_{r_{\max}}^2 - 4\rho_{r_{\max}}\bar{\rho}_{r_{\max}} + \bar{\rho}_{r_{\max}}^2}{6\rho_{r_{\max}}^4} T_{lm}^{(\delta',+)},$$

with

$$\check{T}_{lm}^{(\delta)} = T_{lm}^{(\delta,+)} - \frac{1}{2}\bar{\rho}_{r_{\max}} \tilde{\partial} T_{ll}^{(\delta',+)} + \frac{1}{2}\bar{\rho}_{r_{\max}} (3\bar{\rho}_{r_{\max}} + \rho_{r_{\max}}) \tau^\circ T_{ll}^{(\delta',+)}.$$

Equivalently, writing these out fully,

$$\begin{aligned} c_{\text{src}}^\circ = & -\frac{\rho_{r_{\max}} + \bar{\rho}_{r_{\max}}}{6\rho_{r_{\max}}} \left[T_{lm}^{(\delta)} - \frac{1}{2}\bar{\rho}_{r_{\max}} \tilde{\partial} T_{ll}^{(\delta',+)} + \frac{1}{2}\bar{\rho}_{r_{\max}} (3\bar{\rho}_{r_{\max}} + \rho_{r_{\max}}) \tau^\circ T_{ll}^{(\delta',+)} \right] \\ & + \frac{\bar{\rho}_{r_{\max}} (\bar{\rho}_{r_{\max}} - \rho_{r_{\max}})}{6\rho_{r_{\max}}} T_{lm}^{(\delta',+)} \end{aligned} \quad (\text{K.43})$$

and

$$\begin{aligned} d_{\text{src}}^\circ = & \frac{2\rho_{r_{\max}} - \bar{\rho}_{r_{\max}}}{6\rho_{r_{\max}}^4} \left[T_{lm}^{(\delta,+)} - \frac{1}{2}\bar{\rho}_{r_{\max}} \tilde{\partial} T_{ll}^{(\delta',+)} + \frac{1}{2}\bar{\rho}_{r_{\max}} (3\bar{\rho}_{r_{\max}} + \rho_{r_{\max}}) \tau^\circ T_{ll}^{(\delta',+)} \right] \\ & + \frac{6\rho_{r_{\max}}^2 - 4\rho_{r_{\max}}\bar{\rho}_{r_{\max}} + \bar{\rho}_{r_{\max}}^2}{6\rho_{r_{\max}}^4} T_{lm}^{(\delta',+)} . \end{aligned} \quad (\text{K.44})$$

c. x_{nn}

We now extract the $r = r_{\max}$ data for x_{nn} from the final equation in the hierarchy,

$$\frac{1}{2}(\rho + \bar{\rho})^2 \mathbb{P} \left(\frac{x_{nn}}{\rho + \bar{\rho}} \right) = T_{ln} + \text{Re}[\mathcal{U}x_{m\bar{m}} + \mathcal{V}x_{nm}],$$

where

$$\begin{aligned} \mathcal{U} = & \rho\bar{\rho} \left[-\tilde{\partial}'\tilde{\partial} - \bar{\tau}^\circ(\rho + \bar{\rho})\tilde{\partial} + \bar{\tau}^\circ\tilde{\partial}' \mathbb{P} + \tau^\circ(\bar{\rho} - \rho)\tilde{\partial}' + \tau^\circ\tilde{\partial}' \mathbb{P} \right] + \tilde{\mathbb{P}}' \mathbb{P} \\ & + \frac{1}{2} \left[\Psi^\circ(\rho^2 - \bar{\rho}^2) - 2\rho\bar{\rho}(\bar{\rho} - \rho)\tau^\circ\bar{\tau}^\circ - 4\bar{\rho}\rho'^\circ(1 + \rho\Omega^\circ) \right] \mathbb{P} - 2\rho\tilde{\mathbb{P}}' + \dots, \end{aligned} \quad (\text{K.45})$$

and

$$\mathcal{V} = -\rho \left\{ -3\rho\tilde{\partial}' + [\tilde{\partial}' - \bar{\tau}^\circ(2\bar{\rho} + \rho)] \mathbb{P} + 2\rho^2\bar{\tau}^\circ \right\}.$$

The omitted terms in $\mathcal{U}$ are purely multiplicative and therefore do not contribute distributionally at $r = r_{\max}$.

It is convenient to define the total effective source

$$\check{T}_{ln} := T_{ln} + \text{Re}[\mathcal{U}x_{m\bar{m}} + \mathcal{V}x_{nm}]. \quad (\text{K.46})$$

Near $r = r_{\max}$, write

$$\check{T}_{ln} = \check{T}_{ln}^{(\delta')} \delta'(r - r_{\max}) + \check{T}_{ln}^{(\delta)} \delta(r - r_{\max}) + \dots \quad (\text{K.47})$$

Since the left-hand side is only first order in $\mathbb{P}$, a piecewise x_{nn} can generate a $\delta(r - r_{\max})$ term but not a $\delta'(r - r_{\max})$ term. Therefore consistency requires

$$\boxed{\check{T}_{ln}^{(\delta')} = 0.}$$

Now write

$$x_{nn} = x_{nn}^< \Theta(r_{\max} - r) + x_{nn}^> \Theta(r - r_{\max}).$$

Then

$$\mathbb{P}\left(\frac{x_{nn}}{\rho + \bar{\rho}}\right) = \left[\frac{x_{nn}}{\rho + \bar{\rho}}\right] \delta(r - r_{\max}) + \dots,$$

and so matching the δ -coefficient gives

$$\boxed{\left[\frac{x_{nn}}{\rho + \bar{\rho}}\right] = \frac{2}{(\rho_{r_{\max}} + \bar{\rho}_{r_{\max}})^2} \check{T}_{ln}^{(\delta)}.$$

Equivalently,

$$\boxed{[x_{nn}] = \frac{2}{\rho_{r_{\max}} + \bar{\rho}_{r_{\max}}} \check{T}_{ln}^{(\delta)}.$$

Thus the outer boundary value is

$$\boxed{x_{nn}^>(r_{\max}) = x_{nn}^<(r_{\max}) + \frac{2}{\rho_{r_{\max}} + \bar{\rho}_{r_{\max}}} \check{T}_{ln}^{(\delta)}.$$

To evaluate $\check{T}_{ln}^{(\delta')}$ and $\check{T}_{ln}^{(\delta)}$, note first that

$$[x_{m\bar{m}}] = T_{ll}^{(\delta')}, \quad [\mathbb{P}x_{m\bar{m}}] = T_{ll}^{(\delta)} - \frac{2r_{\max}}{\Sigma_{r_{\max}}} T_{ll}^{(\delta')}, \quad \Sigma_{r_{\max}} := r_{\max}^2 + a^2 z^2,$$

and from the previous stage

$$[x_{nm}] = \check{T}_{lm}^{(\delta')} = T_{lm}^{(\delta')}.$$

The operator $\mathcal{V}$ contains only one explicit $\mathbb{P}$, so it contributes only to the δ -coefficient:

$$\boxed{(\mathcal{V}x_{nm})^{(\delta)} = -\rho_{r_{\max}} \left(\check{\partial}' - \bar{\tau}^\circ (2\bar{\rho}_{r_{\max}} + \rho_{r_{\max}}) \right) T_{lm}^{(\delta')}.$$

For $\mathcal{U}$, the only possible source of a δ' -term is $\tilde{\mathbb{P}}' \mathbb{P}x_{m\bar{m}}$. Write the radial part of $\tilde{\mathbb{P}}'$ as

$$\tilde{\mathbb{P}}' = A(r, z) \mathbb{P} + B,$$

where B contains no $\mathbb{P}$. Then

$$(\tilde{\mathbb{P}}' \mathbb{P}x_{m\bar{m}})^{(\delta')} = A_{r_{\max}} [x_{m\bar{m}}] = A_{r_{\max}} T_{ll}^{(\delta')}.$$

Hence

$$\boxed{\mathcal{S}_{ln}^{(\delta')} = T_{ln}^{(\delta')} + \text{Re}[A_{r_{\max}} T_{ll}^{(\delta')}].}$$

Since $\check{T}_{ln}^{(\delta')} = 0$, this gives the endpoint consistency condition

$$\boxed{T_{ln}^{(\delta')} + \text{Re}[A_{r_{\max}} T_{ll}^{(\delta')}] = 0.}$$

For the δ -coefficient, one finds

$$(\tilde{\mathbf{P}}' x_{m\bar{m}})^{(\delta)} = A_{r_{\max}}[x_{m\bar{m}}] = A_{r_{\max}} T_{ll}^{(\delta')},$$

$$(\tilde{\mathbf{P}}' \mathbf{P} x_{m\bar{m}})^{(\delta)} = A_{r_{\max}}[\mathbf{P} x_{m\bar{m}}] + (\mathbf{P} A)_{r_{\max}}[x_{m\bar{m}}] + B_{r_{\max}}[x_{m\bar{m}}],$$

and therefore

$$\boxed{\check{T}_{ln}^{(\delta)} = T_{ln}^{(\delta)} + \text{Re} \left\{ \rho_{r_{\max}} \bar{\rho}_{r_{\max}} \left(\bar{\tau}^{\circ} \tilde{\theta}' + \tau^{\circ} \tilde{\theta}' \right) T_{ll}^{(\delta')} \right. \\ + \left[\frac{1}{2} \left(\Psi^{\circ}(\rho_{r_{\max}}^2 - \bar{\rho}_{r_{\max}}^2) - 2\rho_{r_{\max}} \bar{\rho}_{r_{\max}} (\bar{\rho}_{r_{\max}} - \rho_{r_{\max}}) \tau^{\circ} \bar{\tau}^{\circ} - 4\bar{\rho}_{r_{\max}} \rho'^{\circ} (1 + \rho_{r_{\max}} \Omega^{\circ}) \right) - 2\rho_{r_{\max}} \right. \\ \left. \left. + (\mathbf{P} A)_{r_{\max}} + B_{r_{\max}} \right] T_{ll}^{(\delta')} + A_{r_{\max}} \left(T_{ll}^{(\delta)} - \frac{2r_{\max}}{\Sigma_{r_{\max}}} T_{ll}^{(\delta')} \right) \right. \\ \left. \left. - \rho_{r_{\max}} \left(\tilde{\theta}' - \bar{\tau}^{\circ} (2\bar{\rho}_{r_{\max}} + \rho_{r_{\max}}) \right) T_{lm}^{(\delta')} \right\}.$$

This is the quantity that determines the outer boundary value of x_{nn} through

$$\boxed{x_{nn}^{>}(r_{\max}) = x_{nn}^{<}(r_{\max}) + \frac{2}{\rho_{r_{\max}} + \bar{\rho}_{r_{\max}}} \check{T}_{ln}^{(\delta)}.$$

Finally, write the homogeneous outer solution (Eq. (63c)) as

$$x_{nn}^{+}(r) = 2 \text{Re}[Y(r)] + 2(\rho + \bar{\rho}) f^{\circ},$$

where $Y(r)$ denotes all the terms in Eq. (63c) already fixed by $a^{\circ}, b^{\circ}, c^{\circ}, d^{\circ}$, i.e. everything except $(\rho + \bar{\rho}) f^{\circ}$. Evaluating at $r = r_{\max}$ gives

$$x_{nn}^{+}(r_{\max}) = 2 \text{Re}[Y(r_{\max})] + 2(\rho_{r_{\max}} + \bar{\rho}_{r_{\max}}) f^{\circ},$$

since $\rho_{r_{\max}} + \bar{\rho}_{r_{\max}}$ is real. Thus

$$\boxed{f^{\circ} = \frac{x_{nn}^{+}(r_{\max}) - 2 \text{Re}[Y(r_{\max})]}{2(\rho_{r_{\max}} + \bar{\rho}_{r_{\max}})}.$$

If the interior field vanishes, $x_{nn}^-(r_{\max}) = 0$, then

$$x_{nn}^+(r_{\max}) = \frac{2}{\rho_{r_{\max}} + \bar{\rho}_{r_{\max}}} \check{T}_{ln}^{(\delta)},$$

and hence

$$f_{\text{src}}^\circ = \frac{\check{T}_{ln}^{(\delta)}}{(\rho_{r_{\max}} + \bar{\rho}_{r_{\max}})^2} - \frac{\text{Re}[Y(r_{\max})]}{\rho_{r_{\max}} + \bar{\rho}_{r_{\max}}}.$$

Define

$$Y(r_{\max}) := 2 \text{Re}[\mathfrak{a}_{r_{\max}} a^\circ + \mathfrak{b}_{r_{\max}} b^\circ + \mathfrak{c}_{r_{\max}} c^\circ + \mathfrak{d}_{r_{\max}} d^\circ],$$

where $\mathfrak{a}_{r_{\max}}, \mathfrak{b}_{r_{\max}}, \mathfrak{c}_{r_{\max}}, \mathfrak{d}_{r_{\max}}$ are the operator coefficients of $a^\circ, b^\circ, c^\circ, d^\circ$, evaluated at $r = r_{\max}$.

Then the source-only part is

$$Y_{\text{src}}(r_{\max}) = 2 \text{Re}[\mathfrak{a}_{r_{\max}} a_{\text{src}}^\circ + \mathfrak{b}_{r_{\max}} b_{\text{src}}^\circ + \mathfrak{c}_{r_{\max}} c_{\text{src}}^\circ + \mathfrak{d}_{r_{\max}} d_{\text{src}}^\circ].$$

Substituting the previously derived coefficients gives

$$\begin{aligned} Y_{\text{src}}(r_{\max}) = 2 \text{Re} \left\{ \left(\frac{r_{\max}}{2} \mathfrak{a}_{r_{\max}} + \frac{r_{\max}^2 - a^2 z^2}{2} \mathfrak{b}_{r_{\max}} \right) T_{ll}^{(\delta)} + \left(-\frac{1}{2} \mathfrak{a}_{r_{\max}} - r_{\max} \mathfrak{b}_{r_{\max}} \right) T_{ll}^{(\delta')} \right. \\ \left. + \left(-\frac{\rho_{r_{\max}} + \bar{\rho}_{r_{\max}}}{6\rho_{r_{\max}}} \mathfrak{c}_{r_{\max}} + \frac{2\rho_{r_{\max}} - \bar{\rho}_{r_{\max}}}{6\rho_{r_{\max}}^4} \mathfrak{d}_{r_{\max}} \right) \check{T}_{lm}^{(\delta)} \right. \\ \left. + \left(\frac{\bar{\rho}_{r_{\max}}(\bar{\rho}_{r_{\max}} - \rho_{r_{\max}})}{6\rho_{r_{\max}}} \mathfrak{c}_{r_{\max}} + \frac{6\rho_{r_{\max}}^2 - 4\rho_{r_{\max}}\bar{\rho}_{r_{\max}} + \bar{\rho}_{r_{\max}}^2}{6\rho_{r_{\max}}^4} \mathfrak{d}_{r_{\max}} \right) T_{lm}^{(\delta')} \right\}, \end{aligned}$$

where

$$\check{T}_{lm}^{(\delta)} = T_{lm}^{(\delta)} - \frac{1}{2} \bar{\rho}_{r_{\max}} \tilde{\delta} T_{ll}^{(\delta')} + \frac{1}{2} \bar{\rho}_{r_{\max}} (3\bar{\rho}_{r_{\max}} + \rho_{r_{\max}}) \tau^\circ T_{ll}^{(\delta')}.$$

Equivalently, defining

$$\begin{aligned} \mathfrak{A}_{ll}^{(\delta)} &:= \frac{r_{\max}}{2} \mathfrak{a}_{r_{\max}} + \frac{r_{\max}^2 - a^2 z^2}{2} \mathfrak{b}_{r_{\max}}, \\ \mathfrak{A}_{lm}^{(\delta)} &:= -\frac{\rho_{r_{\max}} + \bar{\rho}_{r_{\max}}}{6\rho_{r_{\max}}} \mathfrak{c}_{r_{\max}} + \frac{2\rho_{r_{\max}} - \bar{\rho}_{r_{\max}}}{6\rho_{r_{\max}}^4} \mathfrak{d}_{r_{\max}}, \\ \mathfrak{A}_{lm}^{(\delta')} &:= \frac{\bar{\rho}_{r_{\max}}(\bar{\rho}_{r_{\max}} - \rho_{r_{\max}})}{6\rho_{r_{\max}}} \mathfrak{c}_{r_{\max}} + \frac{6\rho_{r_{\max}}^2 - 4\rho_{r_{\max}}\bar{\rho}_{r_{\max}} + \bar{\rho}_{r_{\max}}^2}{6\rho_{r_{\max}}^4} \mathfrak{d}_{r_{\max}}, \end{aligned}$$

one obtains

$$Y_{\text{src}}(r_{\max}) = 2 \text{Re} \left\{ \mathfrak{A}_{ll}^{(\delta)} T_{ll}^{(\delta)} + \mathfrak{A}_{lm}^{(\delta)} T_{lm}^{(\delta)} + \mathfrak{A}_{lm}^{(\delta')} T_{lm}^{(\delta')} + \mathfrak{A}_{ll}^{(\delta')} T_{ll}^{(\delta')} \right\},$$

with

$$\mathfrak{A}_{ll}^{(\delta')} = -\frac{1}{2} \mathfrak{a}_{r_{\max}} - r_{\max} \mathfrak{b}_{r_{\max}} + \mathfrak{A}_{lm}^{(\delta)} \left[-\frac{1}{2} \bar{\rho}_{r_{\max}} \tilde{\delta} + \frac{1}{2} \bar{\rho}_{r_{\max}} (3\bar{\rho}_{r_{\max}} + \rho_{r_{\max}}) \tau^\circ \right].$$

4. Corrector scalars in the inner vacuum region as determined by left WT boundary data

We now perform the complementary construction in which the transport hierarchy is initialized at the outer worldtube boundary and integrated inward. We write

$$R_+ := r_{\max}, \quad R_- := r_{\min}, \quad (\text{K.48})$$

and use the shorthand

$$\rho_{\pm} := \rho(R_{\pm}), \quad \bar{\rho}_{\pm} := \bar{\rho}(R_{\pm}), \quad \Sigma_{\pm} := R_{\pm}^2 + a^2 z^2. \quad (\text{K.49})$$

The goal is to determine the homogeneous coefficients of the solution in the inner vacuum region $r < R_-$, assuming that the corrector scalars vanish in the outer vacuum region,

$$x_{\mu\nu}^>(r) = 0, \quad r > R_+. \quad (\text{K.50})$$

We use the jump conventions

$$[x]_+ := x^>(R_+^+) - x^{\text{WT}}(R_+^-), \quad [x]_- := x^{\text{WT}}(R_-^+) - x^<(R_-^-). \quad (\text{K.51})$$

Thus, imposing $x^> = 0$ at the outer boundary gives

$$x^{\text{WT}}(R_+^-) = -[x]_+. \quad (\text{K.52})$$

After inward transport through the worldtube, the inner vacuum data are obtained from

$$x^<(R_-^-) = x^{\text{WT}}(R_-^+) - [x]_-. \quad (\text{K.53})$$

a. Outer-boundary initial data at $r = R_+$

For the first equation in the hierarchy, the right-boundary jump conditions are

$$[x_{m\bar{m}}]_+ = T_u^{(\delta',+)}, \quad (\text{K.54})$$

and

$$[\partial_r x_{m\bar{m}}]_+ = T_u^{(\delta,+) + (\rho_+ + \bar{\rho}_+)T_u^{(\delta',+)}. \quad (\text{K.55})$$

Therefore, since $x_{m\bar{m}}^> = 0$, the worldtube data at R_+ are

$$\boxed{x_{m\bar{m}}^{\text{WT}}(R_+^-) = -T_u^{(\delta',+)},} \quad (\text{K.56})$$

and

$$\boxed{\partial_r x_{m\bar{m}}^{\text{WT}}(R_+^-) = -T_l^{(\delta,+)} - (\rho_+ + \bar{\rho}_+)T_l^{(\delta',+)} .} \quad (\text{K.57})$$

For the x_{nm} equation, define

$$J_{nm}^{(+)} := [x_{nm}]_+, \quad K_{nm}^{(+)} := [\partial_r x_{nm}]_+. \quad (\text{K.58})$$

The jump in the field is

$$J_{nm}^{(+)} = 2T_{lm}^{(\delta',+)} , \quad (\text{K.59})$$

while the derivative jump is

$$\begin{aligned} K_{nm}^{(+)} &= 4\rho_+ T_{lm}^{(\delta',+)} + 2T_{lm}^{(\delta,+)} - \bar{\rho}_+ \tilde{\partial} T_l^{(\delta',+)} \\ &\quad + 3\bar{\rho}_+(\rho_+ + \bar{\rho}_+)\tau^\circ T_{lm}^{(\delta',+)} . \end{aligned} \quad (\text{K.60})$$

Thus the corresponding inward-shooting data are

$$\boxed{x_{nm}^{\text{WT}}(R_+^-) = -2T_{lm}^{(\delta',+)} ,} \quad (\text{K.61})$$

and

$$\boxed{\partial_r x_{nm}^{\text{WT}}(R_+^-) = -K_{nm}^{(+)} .} \quad (\text{K.62})$$

Finally, for x_{nn} , define the effective endpoint source

$$\check{T}_{ln} := T_{ln} + \text{Re}[\mathcal{U}x_{m\bar{m}} + \mathcal{V}x_{nm}] , \quad (\text{K.63})$$

and write its endpoint distributional part as

$$\check{T}_{ln} = \check{T}_{ln}^{(\delta,+)}\delta(r - R_+) + \check{T}_{ln}^{(\delta',+)}\delta'(r - R_+) + \cdots . \quad (\text{K.64})$$

Consistency of the first-order x_{nn} equation requires

$$\check{T}_{ln}^{(\delta',+)} = 0, \quad (\text{K.65})$$

and the jump condition gives

$$[x_{nn}]_+ = \frac{2}{\rho_+ + \bar{\rho}_+} \check{T}_{ln}^{(\delta,+)} . \quad (\text{K.66})$$

Hence, assuming $x_{nn}^> = 0$, the initial data at R_+ are

$$\boxed{x_{nn}^{\text{WT}}(R_+^-) = -\frac{2}{\rho_+ + \bar{\rho}_+} \check{T}_{ln}^{(\delta,+)} .} \quad (\text{K.67})$$

b. Transported worldtube data at $r = R_-$

Integrating the hierarchy inward from R_+ to R_- , define the transported worldtube values

$$X_{m\bar{m}}^- := x_{m\bar{m}}^{\text{WT}}(R_-^+), \quad P_{m\bar{m}}^- := \partial_r x_{m\bar{m}}^{\text{WT}}(R_-^+), \quad (\text{K.68})$$

$$X_{nm}^- := x_{nm}^{\text{WT}}(R_-^+), \quad P_{nm}^- := \partial_r x_{nm}^{\text{WT}}(R_-^+), \quad (\text{K.69})$$

and

$$X_{nn}^- := x_{nn}^{\text{WT}}(R_-^+). \quad (\text{K.70})$$

These quantities contain the finite worldtube transport from the outer boundary to the inner boundary.

c. $x_{m\bar{m}}$ in the inner vacuum region

At the inner boundary, the left jump conditions are

$$[x_{m\bar{m}}]_- = T_l^{(\delta', -)}, \quad (\text{K.71})$$

and

$$[\partial_r x_{m\bar{m}}]_- = T_l^{(\delta, -)} + (\rho_- + \bar{\rho}_-) T_l^{(\delta', -)}. \quad (\text{K.72})$$

Therefore the inner vacuum data at R_- are

$$\boxed{x_{m\bar{m}}^<(R_-) = X_{m\bar{m}}^- - T_l^{(\delta', -)}}, \quad (\text{K.73})$$

and

$$\boxed{\partial_r x_{m\bar{m}}^<(R_-) = P_{m\bar{m}}^- - T_l^{(\delta, -)} - (\rho_- + \bar{\rho}_-) T_l^{(\delta', -)}}. \quad (\text{K.74})$$

For brevity, define

$$x_- := x_{m\bar{m}}^<(R_-), \quad p_- := \partial_r x_{m\bar{m}}^<(R_-). \quad (\text{K.75})$$

The inner homogeneous solution is written as

$$x_{m\bar{m}}^<(r) = a_-^\circ \left(\frac{\rho}{\bar{\rho}} + \frac{\bar{\rho}}{\rho} \right) + b_-^\circ (\rho + \bar{\rho}). \quad (\text{K.76})$$

Solving for the two homogeneous coefficients gives

$$\boxed{a_-^\circ = \frac{R_-^2 - a^2 z^2}{2\Sigma_-} x_- + \frac{R_-}{2} p_-}, \quad (\text{K.77})$$

and

$$\boxed{b_-^\circ = -\frac{2a^2 z^2 R_-}{\Sigma_-} x_- + \frac{R_-^2 - a^2 z^2}{2} p_-}. \quad (\text{K.78})$$

Equivalently, the source-separated form is obtained by substituting

$$x_- = X_{m\bar{m}}^- - T_{ll}^{(\delta', -)}, \quad (\text{K.79})$$

and

$$p_- = P_{m\bar{m}}^- - T_{ll}^{(\delta, -)} - (\rho_- + \bar{\rho}_-) T_{ll}^{(\delta', -)}. \quad (\text{K.80})$$

d. x_{nm} in the inner vacuum region

The inner homogeneous solution for x_{nm} has the form

$$x_{nm}^<(r) = X_-(r) + \frac{2\rho - \bar{\rho}}{\rho^2} c_-^\circ + \rho(\rho + \bar{\rho}) d_-^\circ, \quad (\text{K.81})$$

where the piece already determined by a_-° and b_-° is

$$\begin{aligned} X_-(r) := & \bar{\rho} \Omega^\circ \tilde{\partial} a_-^\circ + \frac{(\rho + \bar{\rho})^2}{\rho} \tau^\circ a_-^\circ \\ & + \rho \tilde{\partial} b_-^\circ + (\rho + \bar{\rho})^2 \tau^\circ b_-^\circ. \end{aligned} \quad (\text{K.82})$$

At R_- , define the left-boundary jump data

$$J_{nm}^{(-)} := [x_{nm}]_-, \quad K_{nm}^{(-)} := [\partial_r x_{nm}]_-. \quad (\text{K.83})$$

Explicitly,

$$J_{nm}^{(-)} = 2T_{lm}^{(\delta', -)}, \quad (\text{K.84})$$

and

$$\begin{aligned} K_{nm}^{(-)} = & 4\rho_- T_{lm}^{(\delta', -)} + 2T_{lm}^{(\delta, -)} - \bar{\rho}_- \tilde{\partial} T_{ll}^{(\delta', -)} \\ & + 3\bar{\rho}_- (\rho_- + \bar{\rho}_-) \tau^\circ T_{lm}^{(\delta', -)}. \end{aligned} \quad (\text{K.85})$$

Thus the inner one-sided data are

$$x_{nm}^<(R_-) = X_{nm}^- - J_{nm}^{(-)}, \quad (\text{K.86})$$

and

$$\partial_r x_{nm}^<(R_-) = P_{nm}^- - K_{nm}^{(-)}. \quad (\text{K.87})$$

Define the reduced data

$$\widehat{x}_{nm}^- := x_{nm}^<(R_-) - X_-(R_-), \quad (\text{K.88})$$

and

$$\partial_r \widehat{x}_{nm}^- := \partial_r x_{nm}^<(R_-) - \partial_r X_-(R_-). \quad (\text{K.89})$$

Using the basis functions

$$u_c(r) := \frac{2\rho - \bar{\rho}}{\rho^2}, \quad u_d(r) := \rho(\rho + \bar{\rho}), \quad (\text{K.90})$$

whose Wronskian is

$$W = u_c u_d' - u_c' u_d = 6\rho^2, \quad (\text{K.91})$$

one obtains

$$c_-^\circ = \frac{(2\rho_-^2 + \rho_- \bar{\rho}_- + \bar{\rho}_-^2) \widehat{x}_{nm}^- - (\rho_- + \bar{\rho}_-) \partial_r \widehat{x}_{nm}^-}{6\rho_-}. \quad (\text{K.92})$$

Similarly,

$$d_-^\circ = \frac{(2\rho_-^2 - 2\rho_- \bar{\rho}_- + \bar{\rho}_-^2) \widehat{x}_{nm}^- + (2\rho_- - \bar{\rho}_-) \partial_r \widehat{x}_{nm}^-}{6\rho_-^4}. \quad (\text{K.93})$$

We can rewrite this as

e. x_{nn} in the inner vacuum region

The final equation in the hierarchy may be written schematically as

$$\frac{1}{2}(\rho + \bar{\rho})^2 \left(\frac{x_{nn}}{\rho + \bar{\rho}} \right) = \check{T}_{ln}, \quad (\text{K.94})$$

where

$$\check{T}_{ln} := T_{ln} + \text{Re}[\mathcal{U}x_{m\bar{m}} + \mathcal{V}x_{nm}]. \quad (\text{K.95})$$

Near the left boundary,

$$\check{T}_{ln} = \check{T}_{ln}^{(\delta, -)} \delta(r - R_-) + \check{T}_{ln}^{(\delta', -)} \delta'(r - R_-) + \dots. \quad (\text{K.96})$$

Since the left-hand side is first order in δ , consistency requires

$$\check{T}_{ln}^{(\delta', -)} = 0. \quad (\text{K.97})$$

The δ -coefficient gives

$$[x_{nn}]_- = \frac{2}{\rho_- + \bar{\rho}_-} \check{T}_{ln}^{(\delta, -)}. \quad (\text{K.98})$$

Therefore the inner boundary value is

$$x_{nn}^<(R_-) = X_{nn}^- - \frac{2}{\rho_- + \bar{\rho}_-} \check{T}_{ln}^{(\delta, -)}. \quad (\text{K.99})$$

Write the inner homogeneous solution as

$$x_{nn}^<(r) = 2 \operatorname{Re}[Y_-(r)] + 2(\rho + \bar{\rho})f_-^\circ, \quad (\text{K.100})$$

where $Y_-(r)$ denotes the part fixed by the already-determined coefficients

$$a_-^\circ, \quad b_-^\circ, \quad c_-^\circ, \quad d_-^\circ. \quad (\text{K.101})$$

Evaluating at $r = R_-$, one obtains

$$x_{nn}^<(R_-) = 2 \operatorname{Re}[Y_-(R_-)] + 2(\rho_- + \bar{\rho}_-)f_-^\circ. \quad (\text{K.102})$$

Thus

$$f_-^\circ = \frac{x_{nn}^<(R_-) - 2 \operatorname{Re}[Y_-(R_-)]}{2(\rho_- + \bar{\rho}_-)}. \quad (\text{K.103})$$

Substituting the matching condition for $x_{nn}^<(R_-)$ gives

$$f_-^\circ = \frac{X_{nn}^- - \frac{2}{\rho_- + \bar{\rho}_-} \check{T}_{ln}^{(\delta, -)} - 2 \operatorname{Re}[Y_-(R_-)]}{2(\rho_- + \bar{\rho}_-)}. \quad (\text{K.104})$$

f. Summary of the sign structure

The construction consists of two sign reversals. First, imposing vanishing fields outside the right boundary gives

$$x^{\text{WT}}(R_+^-) = -[x]_+. \quad (\text{K.105})$$

Second, matching the transported worldtube field to the inner vacuum region gives

$$x^<(R_-) = x^{\text{WT}}(R_-^+) - [x]_-. \quad (\text{K.106})$$

Therefore the coefficients

$$a_-^\circ, \quad b_-^\circ, \quad c_-^\circ, \quad d_-^\circ, \quad f_-^\circ \quad (\text{K.107})$$

are obtained by first initializing the worldtube hierarchy at R_+ with the negative of the right-boundary jump data, transporting inward to R_- , and then subtracting the left-boundary jump data before projecting onto the inner homogeneous basis.

5. Transport boundary data for a smooth bump window

We now contrast the preceding box-window construction with the case of a smooth compactly supported bump window. Let

$$R_- := r_{\min}, \quad R_+ := r_{\max}, \quad (\text{K.108})$$

and let

$$W(x^\mu) = W(r; r_p, \lambda) \quad (\text{K.109})$$

be a smooth window centered on the particle and supported inside the worldtube,

$$\text{supp } W \subseteq [R_-, R_+]. \quad (\text{K.110})$$

The puncture is now tapered according to

$$h_{\mu\nu}^{\mathcal{P}} \longrightarrow h_{\mu\nu}^{\mathcal{P}} W. \quad (\text{K.111})$$

In contrast with the box window, there are no Heaviside factors and hence no endpoint distributions if W is a genuine flat bump.

For a standard C^∞ compactly supported bump, one has

$$W(R_\pm) = W'(R_\pm) = W''(R_\pm) = \dots = 0. \quad (\text{K.112})$$

The action of the second-order Einstein operator produces a source inside the worldtube:

$$\mathcal{E}_{\mu\nu}[h^{\mathcal{P}}W] = W \mathcal{E}_{\mu\nu}[h^{\mathcal{P}}] + \mathcal{C}_{\mu\nu}^{(1)}[h^{\mathcal{P}}] \partial_r W + \mathcal{C}_{\mu\nu}^{(2)}[h^{\mathcal{P}}] \partial_r^2 W, \quad (\text{K.113})$$

where $\mathcal{C}_{\mu\nu}^{(1)}$ and $\mathcal{C}_{\mu\nu}^{(2)}$ denote the terms produced by one and two radial derivatives acting on the window. For a circular orbit and a purely radial window, there are no additional $\partial_u W$ terms. Thus the residual source has the schematic form

$$T_{\mu\nu}^{\mathcal{R}} = T_{\mu\nu}^{\text{bump}}(r, z), \quad (\text{K.114})$$

with

$$\text{supp } T_{\mu\nu}^{\text{bump}} \subseteq [R_-, R_+]. \quad (\text{K.115})$$

In particular, the endpoint distributional coefficients vanish:

$$T_{\mu\nu}^{(\delta, \pm)} = T_{\mu\nu}^{(\delta', \pm)} = 0. \quad (\text{K.116})$$

a. Absence of endpoint jump conditions

For the box window, differentiating the Heaviside factors produces δ - and δ' -terms at $R_{\pm}$, which in turn give jump conditions for the corrector scalars. For a smooth flat bump, these endpoint distributions are absent. Consequently the corrector scalars are matched smoothly to the vacuum regions.

Assume, as before, that the corrector scalars vanish in the outer vacuum region,

$$x_{\mu\nu}^>(r) = 0, \quad r > R_+. \quad (\text{K.117})$$

Since the source vanishes smoothly at R_+ , the worldtube fields satisfy ordinary homogeneous matching conditions there. For the second-order fields one has

$$\boxed{x_{m\bar{m}}^{\text{WT}}(R_+) = 0, \quad \partial_r x_{m\bar{m}}^{\text{WT}}(R_+) = 0,} \quad (\text{K.118})$$

and

$$\boxed{x_{nm}^{\text{WT}}(R_+) = 0, \quad \partial_r x_{nm}^{\text{WT}}(R_+) = 0.} \quad (\text{K.119})$$

For the first-order equation for x_{nn} , one imposes

$$\boxed{x_{nn}^{\text{WT}}(R_+) = 0.} \quad (\text{K.120})$$

The hierarchy is then integrated inward through the worldtube with the smooth bump source $T_{\mu\nu}^{\text{bump}}$. Define the transported values at the inner boundary by

$$X_{m\bar{m}}^- := x_{m\bar{m}}^{\text{WT}}(R_-), \quad P_{m\bar{m}}^- := \partial_r x_{m\bar{m}}^{\text{WT}}(R_-), \quad (\text{K.121})$$

$$X_{nm}^- := x_{nm}^{\text{WT}}(R_-), \quad P_{nm}^- := \partial_r x_{nm}^{\text{WT}}(R_-), \quad (\text{K.122})$$

and

$$X_{nn}^- := x_{nn}^{\text{WT}}(R_-). \quad (\text{K.123})$$

At R_- , smooth matching to the inner vacuum region gives

$$x_{\mu\nu}^<(R_-) = x_{\mu\nu}^{\text{WT}}(R_-), \quad (\text{K.124})$$

and, for the second-order fields,

$$\partial_r x_{\mu\nu}^<(R_-) = \partial_r x_{\mu\nu}^{\text{WT}}(R_-). \quad (\text{K.125})$$

Equivalently,

$$[x_{\mu\nu}]_- = 0, \quad [\partial_r x_{\mu\nu}]_- = 0. \quad (\text{K.126})$$

b. $x_{m\bar{m}}$ in the inner vacuum region

Introduce the usual shorthand

$$\rho_- := \rho(R_-), \quad \bar{\rho}_- := \bar{\rho}(R_-), \quad \Sigma_- := R_-^2 + a^2 z^2. \quad (\text{K.127})$$

The inner homogeneous solution is written as

$$x_{m\bar{m}}^<(r) = a_-^\circ \left(\frac{\rho}{\bar{\rho}} + \frac{\bar{\rho}}{\rho} \right) + b_-^\circ (\rho + \bar{\rho}). \quad (\text{K.128})$$

Since there is no jump subtraction at R_- , the boundary data are simply

$$x_{m\bar{m}}^<(R_-) = X_{m\bar{m}}^-, \quad \partial_r x_{m\bar{m}}^<(R_-) = P_{m\bar{m}}^-. \quad (\text{K.129})$$

Therefore

$$a_-^\circ = \frac{R_-^2 - a^2 z^2}{2\Sigma_-} X_{m\bar{m}}^- + \frac{R_-}{2} P_{m\bar{m}}^-, \quad (\text{K.130})$$

and

$$b_-^\circ = -\frac{2a^2 z^2 R_-}{\Sigma_-} X_{m\bar{m}}^- + \frac{R_-^2 - a^2 z^2}{2} P_{m\bar{m}}^-. \quad (\text{K.131})$$

c. x_{nm} in the inner vacuum region

The inner homogeneous solution for x_{nm} takes the form

$$x_{nm}^<(r) = X_-(r) + \frac{2\rho - \bar{\rho}}{\rho^2} c_-^\circ + \rho(\rho + \bar{\rho}) d_-^\circ, \quad (\text{K.132})$$

where

$$\begin{aligned} X_-(r) &:= \bar{\rho} \Omega^\circ \tilde{\partial} a_-^\circ + \frac{(\rho + \bar{\rho})^2}{\rho} \tau^\circ a_-^\circ \\ &\quad + \rho \tilde{\partial} b_-^\circ + (\rho + \bar{\rho})^2 \tau^\circ b_-^\circ. \end{aligned} \quad (\text{K.133})$$

Define the subtracted inner data

$$\hat{X}_{nm}^- := X_{nm}^- - X_-(R_-), \quad (\text{K.134})$$

and

$$\hat{P}_{nm}^- := P_{nm}^- - \partial_r X_-(R_-). \quad (\text{K.135})$$

Using the basis functions

$$u_c(r) := \frac{2\rho - \bar{\rho}}{\rho^2}, \quad u_d(r) := \rho(\rho + \bar{\rho}), \quad (\text{K.136})$$

with Wronskian

$$W = u_c u'_d - u'_c u_d = 6\rho^2, \quad (\text{K.137})$$

we find

$$c_-^\circ = \frac{(2\rho_-^2 + \rho_- \bar{\rho}_- + \bar{\rho}_-^2) \hat{X}_{nm}^- - (\rho_- + \bar{\rho}_-) \hat{P}_{nm}^-}{6\rho_-}. \quad (\text{K.138})$$

Similarly,

$$d_-^\circ = \frac{(2\rho_-^2 - 2\rho_- \bar{\rho}_- + \bar{\rho}_-^2) \hat{X}_{nm}^- + (2\rho_- - \bar{\rho}_-) \hat{P}_{nm}^-}{6\rho_-^4}. \quad (\text{K.139})$$

d. x_{nn} in the inner vacuum region

Finally, write the inner homogeneous solution for x_{nn} as

$$x_{nn}^<(r) = 2 \operatorname{Re}[Y_-(r)] + 2(\rho + \bar{\rho}) f_-^\circ, \quad (\text{K.140})$$

where $Y_-(r)$ denotes the part determined by the previously fixed coefficients

$$a_-^\circ, \quad b_-^\circ, \quad c_-^\circ, \quad d_-^\circ. \quad (\text{K.141})$$

Matching at R_- gives

$$x_{nn}^<(R_-) = X_{nn}^-. \quad (\text{K.142})$$

Therefore

$$X_{nn}^- = 2 \operatorname{Re}[Y_-(R_-)] + 2(\rho_- + \bar{\rho}_-) f_-^\circ, \quad (\text{K.143})$$

and hence

$$f_-^\circ = \frac{X_{nn}^- - 2 \operatorname{Re}[Y_-(R_-)]}{2(\rho_- + \bar{\rho}_-)}. \quad (\text{K.144})$$

e. Endpoint regularity and comparison with the box window

The distinction between a box window and a smooth bump can be summarized as follows. If the window is a Heaviside box, derivatives of the window generate endpoint distributions,

$$\partial_r \Theta \sim \delta, \quad \partial_r^2 \Theta \sim \delta', \quad (\text{K.145})$$

and the corrector scalars satisfy endpoint jump conditions.

If the window is a smooth flat bump, then

$$W(R_\pm) = W'(R_\pm) = W''(R_\pm) = \cdots = 0, \quad (\text{K.146})$$

and the endpoint distributional coefficients vanish:

$$T_{\mu\nu}^{(\delta,\pm)} = T_{\mu\nu}^{(\delta',\pm)} = 0. \quad (\text{K.147})$$

The corrector scalars are then determined by ordinary forced transport through the worldtube with smooth matching to zero in the exterior regions.

Appendix L: Pole-factorized Held operators in outgoing Kerr coordinates

We work in outgoing Kerr coordinates (u, r, θ, φ) and use

$$z = \cos \theta, \quad S(z) := \sqrt{1 - z^2}.$$

The standard Kerr background functions are

$$\Delta(r) = r^2 - 2Mr + a^2, \quad \Sigma(r, z) = r^2 + a^2 z^2. \quad (\text{L.1})$$

For a field of spin weight s and azimuthal mode number m , define

$$\alpha_{m,s} := \frac{|m + s|}{2}, \quad \beta_{m,s} := \frac{|m - s|}{2}, \quad (\text{L.2})$$

and introduce the pole factor

$$W_{m,s}(z) := (1 - z)^{\alpha_{m,s}} (1 + z)^{\beta_{m,s}}. \quad (\text{L.3})$$

We then write the original field as

$$F_s(r, z) = W_{m,s}(z) G_s(r, z). \quad (\text{L.4})$$

If one prefers to use GHP weights (p, q) , the spin is

$$s = \frac{p - q}{2}. \quad (\text{L.5})$$

1. Raw Held operators

With our conventions, the angular Held operators acting on an expression $\text{expr}(r, z)$ are

$$\tilde{\delta} = \frac{1}{\sqrt{2}} \left[S(z) \partial_z + \left(\frac{m + sz}{S(z)} - a\omega S(z) \right) \right], \quad (\text{L.6})$$

$$\tilde{\delta}' = \frac{1}{\sqrt{2}} \left[S(z) \partial_z - \left(\frac{m + sz}{S(z)} - a\omega S(z) \right) \right]. \quad (\text{L.7})$$

The other operators are

$$\mathbf{P} = \partial_r \quad (\text{L.8})$$

$$\tilde{\mathbf{P}}' = -i\omega \quad (\text{L.9})$$

and

$$\tilde{\mathbf{P}}'_r = -i\omega - \frac{\Delta(r)}{2\Sigma(r, z)} \partial_r. \quad (\text{L.10})$$

2. Weighted (pole-factorized) Held operators

The reduced angular operators are defined by conjugation with the pole factor:

$$\widehat{\partial}_s := W_{m,s+1}^{-1} \tilde{\partial} W_{m,s}, \quad \widehat{\tilde{\partial}}_s := W_{m,s-1}^{-1} \tilde{\partial}' W_{m,s}. \quad (\text{L.11})$$

Thus, acting on the reduced field G_s ,

$$\widehat{\partial}_s[G_s] = W_{m,s+1}^{-1} \tilde{\partial}[W_{m,s}G_s], \quad (\text{L.12})$$

$$\widehat{\tilde{\partial}}_s[G_s] = W_{m,s-1}^{-1} \tilde{\partial}'[W_{m,s}G_s]. \quad (\text{L.13})$$

Since $W_{m,s}$ depends only on z , the reduced radial operators are unchanged:

$$\widehat{\mathbf{P}}_s = \mathbf{P}, \quad \widehat{\mathbf{P}}'_s = \tilde{\mathbf{P}}'. \quad (\text{L.14})$$

Equivalently,

$$\widehat{\mathbf{P}}_s[G_s] = \partial_r G_s, \quad (\text{L.15})$$

$$\widehat{\mathbf{P}}'_s[G_s] = -i\omega G_s - \frac{\Delta(r)}{2\Sigma(r, z)} \partial_r G_s. \quad (\text{L.16})$$

3. Direct reduced form

Using

$$\frac{W'_{m,s}(z)}{W_{m,s}(z)} = -\frac{\alpha_{m,s}}{1-z} + \frac{\beta_{m,s}}{1+z}, \quad (\text{L.17})$$

the reduced angular operators may be written directly as

$$\widehat{\partial}_s[G_s] = \frac{1}{\sqrt{2}} \left[S \partial_z G_s + \left(S \frac{W'_{m,s}}{W_{m,s}} + \frac{m + sz}{S} - a\omega S \right) G_s \right] \frac{W_{m,s}}{W_{m,s+1}}, \quad (\text{L.18})$$

$$\widehat{\partial}_s[G_s] = \frac{1}{\sqrt{2}} \left[S \partial_z G_s + \left(S \frac{W'_{m,s}}{W_{m,s}} - \frac{m + sz}{S} + a\omega S \right) G_s \right] \frac{W_{m,s}}{W_{m,s-1}}. \quad (\text{L.19})$$

These forms are useful for symbolic manipulation, but numerically it is preferable to simplify them into endpoint-regular polynomial prefactors.

4. Second-order reduced compositions

The natural second-order reduced angular operators are

$$\widehat{\partial}_{s-1} \widehat{\partial}_s[G_s] = W_{m,s}^{-1} \widetilde{\partial}' \left(\widetilde{\partial}'[W_{m,s} G_s] \right), \quad (\text{L.20})$$

$$\widehat{\partial}_{s+1} \widehat{\partial}_s[G_s] = W_{m,s}^{-1} \widetilde{\partial}' \left(\widetilde{\partial}[W_{m,s} G_s] \right). \quad (\text{L.21})$$

Hence the reduced commutator is

$$[\widehat{\partial}_{s-1}, \widehat{\partial}_s] G_s = \widehat{\partial}_{s-1} \widehat{\partial}_s[G_s] - \widehat{\partial}_{s+1} \widehat{\partial}_s[G_s]. \quad (\text{L.22})$$

With the present normalization, one finds

$$(\widehat{\partial}_{s-1} \widehat{\partial}_s - \widehat{\partial}_{s+1} \widehat{\partial}_s)[G_s] = (-s - 2a\omega z) G_s. \quad (\text{L.23})$$

The same commutator identity holds in the unreduced formulation:

$$(\widetilde{\partial} \widetilde{\partial}' - \widetilde{\partial}' \widetilde{\partial})[F_s] = (-s - 2a\omega z) F_s. \quad (\text{L.24})$$

Appendix M: Zero modes in the “Kerr-Newman” gauge

By definition of the derivative as a one-parameter flow, the zero mode perturbation is given by

$$\dot{g}_{ab} = \frac{d}{d\lambda} g_{ab}(M(\lambda), J(\lambda)) \Big|_{\lambda=0} \quad (\text{M.1})$$

where $M(0)$ and $J(0)$ are the mass and angular momentum of the unperturbed Kerr background. Alternatively, $\dot{g}_{ab}$ may be written as

$$\dot{g}_{ab} = \frac{\delta g_{ab}}{\delta M} \delta M + \frac{\delta g_{ab}}{\delta J} \delta J. \quad (\text{M.2})$$

Here we give the tetrad components of the angular momentum and mass variations $\delta g_{ab}/\delta M$ and $\delta g_{ab}/\delta J$ in the tetrad (A.5a). Using a slight abuse of notation where $(\partial_M g)_u = l^a l^b (\delta g_{ab}/\delta M)$ etc , the results read

$$\begin{aligned}
(\partial_M g)_u &= 0 \\
(\partial_M g)_{ln} &= \frac{a^2(1-z^2)}{M\Sigma}, \\
(\partial_M g)_{lm} &= -\frac{ia\bar{\rho}}{\sqrt{2}M}\sqrt{1-z^2}, \\
(\partial_M g)_{nn} &= \frac{1}{M\Sigma^2} \left\{ a^2(r^2 - a^2)(z^2 - 1) - 2Mr(\Delta + 2Mr + 2a^2z^2) \right\}, \\
(\partial_M g)_{nm} &= \frac{ia\bar{\rho}}{2\sqrt{2}M\Sigma}(\Delta - 4a^2)\sqrt{1-z^2}, \\
(\partial_M g)_{mm} &= \frac{a^2\bar{\rho}^2(z^2 - 1)}{M}, \\
(\partial_M g)_{m\bar{m}} &= \frac{a^2(1+z^2)}{M\Sigma},
\end{aligned} \tag{M.3}$$

for the mass perturbation and

$$\begin{aligned}
(\partial_J g)_u &= 0, \\
(\partial_J g)_{ln} &= -(\partial_M g)_{ln}/a, \\
(\partial_J g)_{lm} &= -(\partial_M g)_{lm}/a, \\
(\partial_J g)_{nn} &= \frac{a^3}{M\Sigma^2} ((z^2 - 1) + ar(r + 2M - z^2(r - 2M))) \\
(\partial_J g)_{nm} &= -(\partial_M g)_{nm}/a \\
(\partial_J g)_{mm} &= -(\partial_M g)_{mm}/a, \\
(\partial_J g)_{m\bar{m}} &= -(\partial_M g)_{m\bar{m}}/a.
\end{aligned} \tag{M.4}$$

for the angular momentum perturbation. As the zero-mode perturbation above contains $m\bar{m}$ and l components, evidently the Kerr-Newman gauge is not a radiation gauge.

As in [6] transform to the IRG, defining

$$\dot{g}_{ab} = \dot{g}_{ab}^{\text{KN}} + \mathcal{L}_\zeta g_{ab}, \tag{M.5}$$

with a gauge vector ζ^a chosen such that $\dot{g}_{ab}l^b = 0 = \dot{g}_{ab}m^a\bar{m}^b$. Solving the gauge transformation equations one finds

$$\zeta^b = -\frac{\dot{a}}{\rho^{\circ}a} \text{Re} \left\{ \tau^{\circ}\bar{\tau}^{\circ}\rho l^b - \bar{\tau}^{\circ}m^b \right\}. \tag{M.6}$$

The components of $\dot{g}_{ab}$ are then given by

$$\dot{g}_{nn} = (\rho + \bar{\rho}) g_M^\circ + 2\rho\bar{\rho}(\rho + \bar{\rho}) \operatorname{Re} \left\{ \frac{1}{2} \Omega^\circ \tilde{\delta}' g_a^\circ + \bar{\tau}^\circ g_a^\circ \right\}, \quad (\text{M.7a})$$

$$\dot{g}_{nm} = \rho(\rho + \bar{\rho}) g_a^\circ, \quad (\text{M.7b})$$

$$\dot{g}_{mm} = \dot{g}_{nl} = \dot{g}_{ml} = \dot{g}_{il} = \dot{g}_{\bar{m}m} = 0, \quad (\text{M.7c})$$

where

$$g_M^\circ = \Psi^\circ \frac{\dot{M}}{M} \doteq \{-3, -3\}, \quad (\text{M.8})$$

$$g_a^\circ = \frac{\tau^\circ}{2\rho^{\circ'}} \Psi^\circ \frac{\dot{a}}{a} \doteq \{-2, -4\}. \quad (\text{M.9})$$

In the Kinnersley tetrad, the Held scalars reduce to $g_M^\circ = \dot{M}$ and $g_a^\circ = \frac{i}{\sqrt{2}} M \dot{a} \sin \theta$.

Appendix N: Zero mode charges

Abbott and Desser [] introduced gauge invariant definitions for spacetime charges that are conserved at the linearized level. Given a killing field of the background spacetime ζ^a , the charges may be derived from the 2-form

$$F_{ab} = \zeta^c \nabla_{[a} \hat{h}_{b]c} + \hat{h}_{c[a} \nabla_{b]} \zeta^c - \zeta_{[a} \nabla^c \hat{h}_{b]c} \quad (\text{N.1})$$

where

$$\hat{h}_{ab} = h_{ab} - \frac{1}{2} g_{ab} g^{cd} h_{cd}. \quad (\text{N.2})$$

The divergence of F_{ab} defines a conserved current

$$j_a = \nabla^c F_{ac} = T_{ac} \zeta^c \quad (\text{N.3})$$

through the Einstein equations and Killing's equation. It follows from Gauss' law that the integral

$$Q_\zeta \equiv \frac{1}{6} \int_\Sigma j^a \epsilon_{abcd} \quad (\text{N.4})$$

is independent of spatial hypersurface Σ . We may think of it as a conserved charge of the linearized system. It follows from Stokes' theorem and (N.3) that the charge may be written as a closed-surface integral

$$Q_\zeta = \int_{i^0} \star F_{ab} - \int_{\mathcal{B}} \star F_{ab}, \quad (\text{N.5})$$

where $\mathcal{B}$ is the bifurcation surface (assuming a black hole spacetime with $a < M$) and i^0 is spatial infinity. The integral itself, which we denote

$$\mathcal{F}_S[\zeta, h] = \int_S \star F_{ab}[\zeta, h], \quad (\text{N.6})$$

is gauge invariant, as the flux for a gauge perturbation can be written as $\mathbf{F}[\zeta, \mathcal{L}_\xi g] = d \star \boldsymbol{\eta}$ for a certain 3-form $\boldsymbol{\eta}$ given in [12]. It then follows from Poincaré's Lemma that $Q = 0$ for a pure gauge perturbation.

1. Mass perturbation in GHP form

For the time-translational killing field (added $M^{1/3}$ so it has the right flat spacetime limit),

$$t^a = M^{1/3} \Psi_2^{-1/3} (\rho n^a - \rho' l^a + \tau' m^a - \tau \bar{m}^a) \quad (\text{N.7})$$

the mass perturbation can be written in terms of the metric as

$$\begin{aligned} Q_t = M^{1/3} \text{Re} \int_S \Psi_2^{-1/3} \Big\{ & (\rho^2 - \rho \mathbb{P}) h_{nn} + (\rho'^2 - \rho' \mathbb{P}') h_{ll} + 2(\rho \rho' - \tau' \bar{\tau}') h_{nl} \\ & + (\tau \mathbb{P} + \rho \bar{\partial} - 4\rho \bar{\tau}') h_{n\bar{m}} + (\tau \mathbb{P}' - \rho' \bar{\partial} - 4\rho' \tau) h_{l\bar{m}} \\ & + (2\rho \rho' - 2\tau' \bar{\tau}' - \Psi_2) h_{m\bar{m}} - \tau'^2 h_{mm} \Big\} m \wedge \bar{m}. \end{aligned} \quad (\text{N.8})$$

In Kerr-Newman coordinates $m \wedge \bar{m} = \Sigma dz \wedge d\varphi$.

2. Point-particle

Consider a particle on a bound orbit with $r_{\min} < r_p < r_{\max}$. The source T_{ab} then vanishes on each Σ_t outside this region. Outside the support of the source let us choose boundary surfaces with radii r_1 and r_2 with $r_1 < r_{\min} < r(t) < r_{\max} < r_2$. Let the corresponding two-surfaces be called S_1 and S_2 . We start with the metric in GHZ form

$$h_{ab}^\pm = \hat{h}_{ab}^\pm + x_{ab}^\pm, \quad \hat{h}_{ab}^\pm = 2 \text{Re} \mathcal{S}_{ab}^\dagger \Phi^\pm \quad (\text{N.9})$$

where Φ^+ satisfies advanced boundary conditions at infinity and obeys $\mathcal{O}^\dagger \Phi^+ = 0$ in the region $r > r_{\max}$ and Φ^- satisfies retarded boundary conditions at infinity at solves $\mathcal{O}^\dagger \Phi^- = 0$ in the region $r < r_{\max}$. Similarly, x^- vanishes for $r < r_-$ and x^+ vanishes for $r > r_+$. As $\mathcal{T}x^\pm = 0$, both metrics have the same Weyl scalars in the absence of algebraically special

modes and therefore can only differ by a gauge transformation and perturbation to a nearby Kerr solution. Since Φ^+ is regular at infinity and Φ^- is regular the horizon the gauge transformation can only be singular on the support of the source. Taking both solutions, we put the metric in the so-called no-string gauge, where the perturbation may be written as

$$h_{ab} = \hat{h}_{ab}^- \Theta^- + (\hat{h}_{ab}^+ + \dot{g}_{ab}) \Theta^+ + \mathcal{L}_\xi g_{ab} \quad (\text{N.10})$$

$$= \hat{h}_{ab}^- \Theta^- + (\hat{h}_{ab}^+ + \dot{g}_{ab}) \Theta^+ + D_{ab} \delta(\mathcal{S}_p), \quad (\text{N.11})$$

where $\mathcal{S}_p$ is the sphere piercing the particle at $r = r_p(u)$ and $\Theta^\pm = \Theta(\pm[r - r_p(u)])$ are Heaviside functions. Here, $\hat{h}^\pm = 2 \text{Re} \mathcal{S}_{ab}^\dagger \Phi_{vac}^\pm$ correspond to the smooth vacuum solutions of the Teukolsky equation $\Phi_{vac}^\pm$ which are regular at the future horizon $+$ and future null infinity respectively. As such solutions generate vacuum metrics [13, 14], the contribution from Φ_{vac}^- gives $\mathcal{F}_{S_1} = 0$ and therefore gives vanishing Abbott-Desser charges. To evaluate the contribution from the outer boundary S_2 , with $r_2 > r_{\max}$, we use that

$$\dot{g}_{ab} + \mathcal{L}_\xi g_{ab} = \hat{h}_{ab}^S + x_{ab}, \quad r > r_{\max}, \quad (\text{N.12})$$

where

$$\hat{h}_{ab}^S = 2 \text{Re} \mathcal{S}_{ab}^\dagger \Phi^S$$

and

$$\Phi^S = d_0^c + d_1^c \rho^{-1} + d_2^c \rho^{-2} + d_3^c \rho^{-3} \quad (\text{N.13})$$

is determined from the source via jump conditions on the inversion relation $\mathbb{P}^4 \bar{\Phi} = -2\psi_0$. Therefore we have that

$$\begin{aligned} Q_\zeta &= \oint_S \star F_{ab}[\zeta, \dot{g}] \\ &= \oint_S \star F_{ab}[\zeta, \hat{h}^S] + \oint_S \star F_{ab}[\zeta, x] \end{aligned} \quad (\text{N.14})$$

where S is any bounding two-surface at $r > r_{\max}$.

For the Kerr Killing fields t^a and ϕ^a the respective mass and spin perturbations due to the particle are [12, 14, 15]

$$\begin{aligned} \delta M &= p_a t^a \\ \delta J &= -p_a \phi^a, \end{aligned} \quad (\text{N.15})$$

where p_a is the four momentum of the particle. Therefore by (N.15) and (N.14) we find

$$p_a t^a = \oint_S \star F_{ab}[t, x] + \oint_S \star F_{ab}[\zeta, \hat{h}^S] \quad (\text{N.16})$$

$$p_a \phi^a = - \oint_S \star F_{ab}[\phi, x] + \oint_S \star F_{ab}[\zeta, \hat{h}^S] \quad (\text{N.17})$$

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